

GRID CONNECTION CODE FOR *RENEWABLE POWER PLANTS (RPPs)* CONNECTED TO THE ELECTRICITY TRANSMISSION SYSTEM (TS) OR THE *DISTRIBUTION SYSTEM (DS)* IN SOUTH AFRICA

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**This document is approved by the National Energy Regulator
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Administered by:

The SA Grid Code Secretariat

Contact: Mr. T. Mchunu

System Operator, Eskom Transmission Division

P.O Box 103, Germiston 1400

Tell: +27 (0)11 871 3076

Email: mchunut@eskom.co.za

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Grid Connection Code Basis

1.1 Legislation

(1) The legal basis for this renewable power plants grid connection code is specified in terms of the Electricity Regulation Act (Act 4 of 2006), as amended.

(2) This *Grid Connection Code for Renewable Power Plants (RPPs) connected to the electricity Transmission System (TS) or the Distribution System (DS) in South Africa* has, on the date of approval by NERSA, superseded the *Grid Code Requirements for Wind Energy Facilities Connected to the Distribution or Transmission Systems in South Africa*.

1.2 Handling of Non-compliances and Deviations

(1) Amendments, derogations or exemptions shall be processed as specified in the RSA Grid Code, as amended.

2. Objectives

(1) The primary objective of this grid connection code is to specify minimum technical and design grid connection requirements for *Renewable Power Plants (RPPs)* connected to or seeking connection to the South African electricity *transmission system (TS)* or *distribution system (DS)*.

(2) This document shall be used together with other applicable requirements of the *code* (i.e. the Grid Code, the Distribution Code and the *Scheduling and Dispatch Rules*), as compliance criteria for *RPPs* connected to the *TS* and the *DS*.

3. Scope

(1) The grid connection requirements in this code shall apply to all *RPPs* connected or seeking connection to the *TS* or *DS*, *the SO*, as well as to the respective electrical *Network Service Providers (NSPs)*.

(2) This grid connection code shall, at minimum, apply to the following *RPP* technologies:

- (a) Photovoltaic
- (b) Concentrated Solar Power
- (c) Small Hydro
- (d) Landfill gas
- (e) Biomass
- (f) Biogas
- (g) Wind

(3) All thermal *RPPs* and hydro *units* of category C (as defined in paragraph 7 below) shall also comply with the design requirements specified in the SA Grid Code (specifically section 3.1. of the Network Code). This *RPP* grid connection code shall take precedence whenever there is a conflict between this code and other *codes*.

(4) Unless otherwise stated, the requirements in this grid connection code shall apply equally to all *RPP* technologies and *categories*.

(5) The *RPP* shall, for duration of its generation licence issued by *National Energy Regulator of South Africa (NERSA)*, comply with the provisions of this grid connection code and all other applicable codes, rules and regulations approved by *NERSA*.

(6) Where there has been a replacement of or a major modification to an existing *RPP*, the *RPP* shall be required to demonstrate compliance to these requirements before being allowed to operate commercially.

(7) Compliance with this grid connection code shall be applicable to the *RPP* depending on its *rated power* and, where indicated, the nominal voltage at the *POC*. Accordingly, *RPPs* are grouped into the following three categories:

(a) Category A: 0 – 1 MVA

This category includes *RPPs* with *rated power* of less than 1 MVA and connected to the *LV* voltage (typically called 'small or micro turbines'). This category shall further be divided into 3 sub-categories:

(i) Category A1: 0 - 13.8 kVA

This sub-category includes *RPPs* of *Category A* with *rated power* in the range of 0 to 13.8 kVA.

(ii) Category A2: 13.8 kVA – 100 kVA

This sub-category includes *RPPs* of *Category A* with *rated power* in the range greater than 13.8 kVA but less than 100 kVA.

(iii) Category A3: 100 kVA – 1 MVA

This sub-category includes *RPPs* of *Category A* with *rated power* in the range 100 kVA but less than 1 MVA.

Note: For *RPPs* connected to multi-phase supplies (two- or three-phase connection at the *POC*), the difference in installed capacity between phases may not exceed 4.6 kVA per phase.

(b) Category B: 1 MVA – 20 MVA

This category includes *RPPs* with *rated power* in the range equal or greater than 1 MVA but less 20 MVA.

(c) Category C: 20 MVA or higher

This category includes *RPPs* with *rated power* equal to or greater than 20 MVA.

(8) The requirements of this *grid connection code* are organized according to above defined categories. Unless otherwise stated, requirements in this *grid connection code* shall apply equally to all *categories* of *RPPs*.

(9) Compliance with this and other *codes* requirements will depend on the interaction between the *RPP* and the grid to which it is connected. The *NSP* shall supply the *RPP Generator* with a reasonable detail of their *TS* or *DS* that is sufficient to allow an accurate analysis of the interaction between the *RPP* and the *NIPS*, including other *generation facilities*. Network technical data shall be provided in line with the requirements of **Appendix 12** - Provision of Technical Network/Grid Data to *RPP Generators*.

4. Definitions and Abbreviations

(1) Unless otherwise indicated, words and terminology in this document shall have the same meaning as those in the *codes*. The following definitions and abbreviations are used in this document.

Active Power Curtailment Set-point

The limit set by the *SO*, *NSP* or their agent for the amount of active power that the *RPP* is permitted to generate. This instruction may be issued manually or automatically via a tele-control facility. The manner of applying the limitation shall be agreed between the parties.

Available Active Power

The amount of active power (MW), measured at the *POC*, that the *RPP* could produce based on plant availability as well as current renewable primary energy conditions (e.g. wind speed, solar radiation).

Codes

The Distribution Code, the Transmission Grid Code or any other Code approved by *NERSA*.

Connection Agreement

As defined in the *Code*,

Communication Gateway Equipment

As defined in the *Code*,

Curtailed Active Power

The amount of Active Power that the *RPP* is permitted to generate by the *SO*, *NSP* or their agent subject to network or system constraints.

Distribution System (DS)

As defined in the *Code*

Distributor

As defined in the *Code*

Droop

A percentage of the frequency change required for an *RPP* to move from no-load to *rated power* or from *rated power* to no-load.

Extra High Voltage (EHV)

The set of nominal voltage levels greater than 220 kV.

Frequency control

The control of active power with a view to stabilising frequency of the *NIPS*.

Generator

As defined in the *Code*

High voltage (HV)

The set of nominal voltage levels greater than 33 KV up to and including 220 kV.

Low voltage (LV)

Nominal voltage levels up to and including 1 kV.

Maximum voltages (U_{max})

Maximum continuous operating voltage

Medium voltage (MV)

The set of nominal voltage levels greater than 1 kV up to and including 33 kV.

Minimum voltages (U_{min})

Minimum continuous operating voltage

National Energy Regulator of South Africa (NERSA)

The legal entity established in terms of the National Energy Regulator Act, 2004 (Act 40 of 2004), as amended.

National Interconnected Power Systems (NIPS)

The electrical network comprising components that have a measurable influence on each other as they are operating as one system, this includes:

- the *TS*;
- the *DS*;
- assets connected to the *TS* and *DS*;
- *power stations* connected to the *TS* and *DS*;
- international *interconnectors*;
- the *control area* for which the *SO* is responsible.

National Transmission Company (NTC)

As defined in the *Code*

Network Service Provider (NSP)

As defined in the *Code*

Nominal voltage

The voltage for which a network is defined and to which operational measurements are referred.

Participants

As defined in the *Code*,

Point of Common Coupling (PCC)

As defined in the *Code*,

Point of Connection (POC)

As defined in the *Code*

Power Quality

Characteristics of the electricity at a given point on an electrical system, evaluated against a set of reference technical parameters. These characteristics include:

- voltage or current quality, i.e. regulation (magnitude), harmonic distortions, flicker, unbalance;
- voltage events, i.e. voltage dips, voltage swells, voltage transients;
- (supply) interruptions;
- frequency of supply.

Rated power (of the *RPP*)

The highest active power measured at the *POC*, which the *RPP* is designed to continuously supply.

Rated wind speed

The average wind speed at which a *wind power plant* achieves its *rated power*. The average renewable speed is calculated as the average value of renewable speeds measured at hub height over a period of 10 minutes.

Renewable Power Plant (RPP)

One or more unit(s) and associated equipment, with a stated *rated power*, which has been connected to the same *POC* and operating as a single power plant.

Notes:

- (i) It is therefore the entire *RPP* that shall be designed to achieve requirements of this code at the *POC*. A *RPP* has only one *POC*.
- (ii) In this code, the term *RPP* is used as the umbrella term for a unit or a system of generating units producing electricity based on a primary renewable energy source (e.g. wind, sun, water, biomass etc.). A *RPP* can use different kinds of primary energy source. If a *RPP* consists of a homogeneous type of generating units it can be named as follows:

- **PV Power Plant (PVPP)**

A single photovoltaic panel or a group of several photovoltaic panels with associated equipment operating as a power plant.

- **Concentrated Solar Power Plant (CSPP)**

A group of aggregates to concentrate the solar radiation and convert the concentrated power to drive a turbine or a group of several turbines with associated equipment operating as a power plant.

- **Small Hydro Power Plant (SHPP)**

A single hydraulic driven turbine or a group of several hydraulic driven turbines with associated equipment operating as a power plant with rated capacity of less than 20MVA.

- **Landfill Gas Power Plant (LGPP)**

A single turbine or a group of several turbines driven by landfill gas with associated equipment operating as a power plant.

- **Biomass Power Plant (BMPP)**

A single turbine or a group of several turbines driven by biomass as fuel with associated equipment operating as a power plant.

- **Biogas Power Plant (BGPP)**

A single turbine or a group of several turbines driven by biogas as fuel with associated equipment operating as a power plant.

- **Wind Power Plant (WPP)**

A single turbine or a group of several turbines driven by wind as fuel with associated equipment operating as a power plant. This is also referred to as a wind energy facility (WEF)

Renewable Power Plant (RPP) Controller

A set of control functions that make it possible to control the *RPP* at the *POC*. The set of control functions shall form a part of the *RPP*.

RPP Generator

Means a legal entity that is licensed to develop and operate a *RPP*.

System Operator (SO)

As defined in the *Code*

Transmission Network Service Provider (TNSP)

As defined in the *Code*

Transmission System (TS)

As defined in the *Code*

Unit / Generation facility

As defined in the *Code*

Voltage Quality

Subset of *power quality* referring to steady-state voltage quality, i.e. voltage regulation (magnitude), voltage harmonics, voltage flicker, voltage unbalance, voltage dips. The current drawn from or injected into the *POC* is the driving factor for voltage quality deviations.

Voltage Ride Through (VRT) Capability

Capability of the *RPP* to stay connected to the network and keep operating following voltage dips or surges caused by short-circuits or disturbances on any or all phases in the *TS* or *DS*.

5. Tolerance of Frequency and Voltage Deviations

(1) The *RPP* shall be able to withstand frequency and voltage deviations at the *POC* under normal and abnormal operating conditions described in this grid connection code while reducing the active power as little as possible.

(2) The *RPP* shall be able to support network frequency and voltage stability in line with the requirements of this grid connection code.

(3) Normal operating conditions and abnormal operating conditions are described in section 5.1 and section 5.2, respectively.

5.1 Normal Operating Conditions

(1) Unless otherwise stated, requirements in this section shall apply to all *categories* of *RPPs*.

(2) *RPPs* of *Category A* shall be designed to be capable of operating continuously within the voltage range of -15% to +10% around the nominal voltage at the *POC*. The actual operating voltage differs from location to location, and this shall be decided by the *NSP* in consultation with the affected *customers* (including the *RPP generator*), and implemented by the *RPP generator*.

(3) *RPPs* of *Category B and C* shall be designed to be capable of operating continuously within the *POC* voltage range specified by U_{min} and U_{max} as shown in table 1 below, measured at the *POC*. The actual operating voltage differs from location to location, and this shall be decided by the *NSP* in consultation with the affected *customers* (including the *RPP generator*), and implemented by the *RPP generator*.

Table 1: Minimum and maximum operating voltages at POC

Nominal (U_n) [kV]	U_{min} [pu]	U_{max} [pu]
132	0.90	1.0985
88	0.90	1.0985
66	0.90	1.0985
44	0.90	1.08
33	0.90	1.08
22	0.90	1.08
11	0.90	1.08

(4) The nominal frequency of the *National Integrated Power System (NIPS)* is 50 Hz and is normally controlled within the limits as defined in the *Grid Code*. The *RPP* shall be designed to be capable of operating for the minimum operating range illustrated in Figures 1 (total cumulative over the life of the *RPP*) and Figure 2 (during a system frequency disturbance).

(5) When the frequency on the *NIPS* is higher than 51.5 Hz for longer than 4 seconds, the *RPP* shall be disconnected from the grid.

(6) When the frequency on the *NIPS* is less than 47.0 Hz for longer than 200ms, the *RPP* may be disconnected.

(7) The *RPP* shall remain connected to the *NIPS* during rate of change of frequency of values up to and including 1.5 Hz per second, provided the network frequency is still within the minimum operating range indicated in Figures 1 and 2.

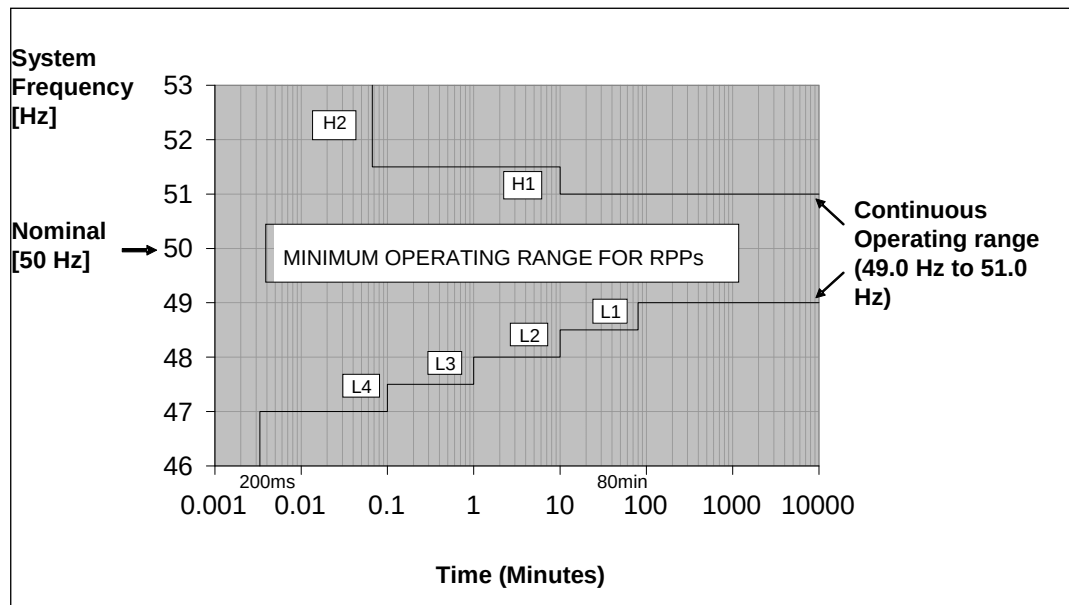


Figure 1: Minimum frequency operating range for *RPP* (Cumulative over the life of the *RPP*)

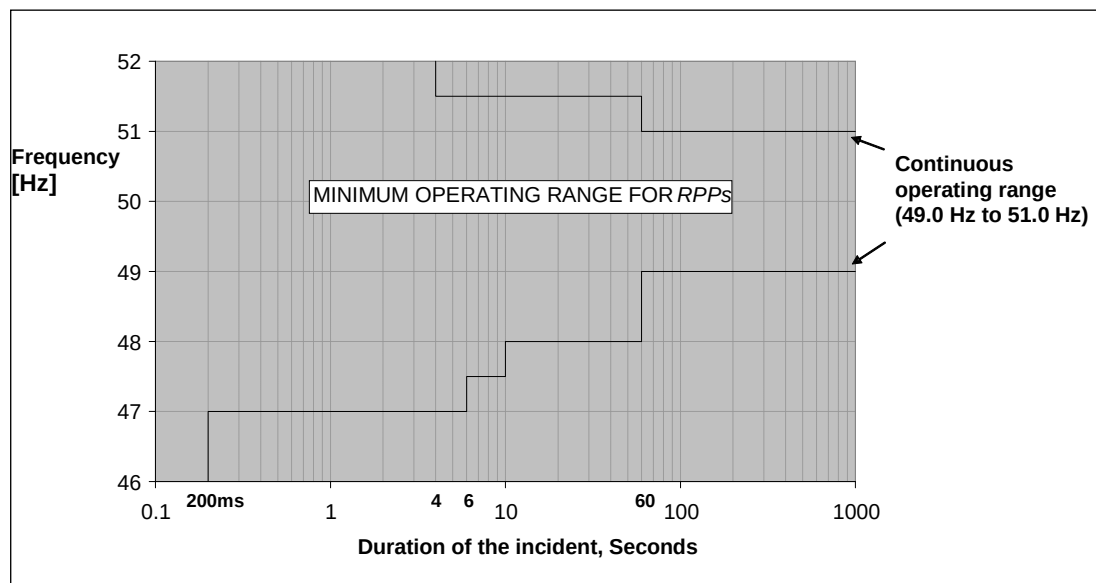


Figure 2: Minimum *frequency* operating range of a *RPP* (during a system *frequency* disturbance)

5.1.1.1 Synchronising to the NIPS

(1) *RPPs* of *Category A* shall only be allowed to connect to the *NIPS*, at the earliest, 60 seconds after the following conditions have been satisfied:

- (a) the voltage at the *POC* is in the range -15% to +10% around the *nominal voltage*,
- (b) frequency in the *NIPS* is within the range of 49.0Hz and 50.2Hz, or otherwise as agreed with the SO.

(2) *RPPs* of *Category B* and *C* shall only be allowed to connect to the *NIPS*, at the earliest, 3 seconds after the following conditions have been satisfied:

- (a) (for *TS* connected *RPPs*), the voltage at the *POC* is within $\pm 5\%$ around the *nominal voltage*,
- (b) (for *DS* connected *RPPs*), the voltage at the *POC* is within U_{max} and U_{min} , as specified in Table 1, around the *nominal voltage*,
- (c) frequency in the *NIPS* is within the range of 49.0Hz and 50.2Hz, or otherwise as agreed with the SO.

5.2 Abnormal Operating Conditions

(1) The *RPP* shall be designed to withstand sudden phase jumps of up to 20° at the *POC* without disconnecting or reducing its output. The *RPP* shall after a settling period resume normal production not later than 5 sec after the operating conditions in the *POC* have reverted to the normal operating conditions.

5.2.1 Tolerance to sudden voltage drops and peaks

(a) *RPPs* of *Category A1* and *A2*

(1) *RPPs* of *Categories A1* and *A2* shall be designed to withstand and fulfil, at the *POC*, voltage ride through conditions illustrated in Figure 3 below.

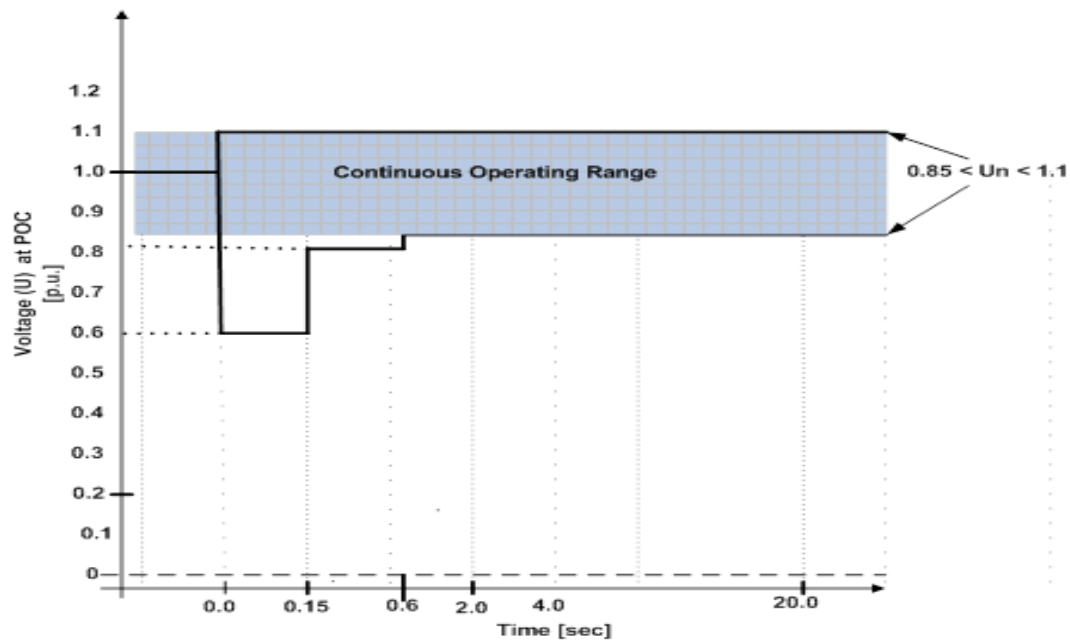


Figure 3: Voltage Ride Through Capability for the RPPs of Category A1 and A2

(2) In addition, the maximum disconnection times for RPPs of Category A1 and A2 is given in **Table 2** below.

Table 2: Maximum disconnection times for RPPs of Categories A1 and A2

Voltage range (at point of connection)	Maximum trip time S
$V < 50 \%$	0,2 s
$50 \% \leq V < 85 \%$	10 s
$85 \% \leq V \leq 110 \%$	Continuous operation
$110 \% < V < 115 \%$	40 s
$115 \% \leq V < 120 \%$	2 s
$120 \% \leq V$	0,16 s
NOTE If multi-voltage control settings are not possible, the more stringent trip time should be implemented, e.g. 2 s between 110% and 120% of voltage.	

(b) RPPs of Categories A3, B and C

(1) RPPs of Categories A3, B and C shall be designed to withstand and fulfil, at the POC, voltage conditions described in this section and illustrated in Figures 4, 4a, 4b and 5 below. The Area D is only applicable to category C RPPs.

(2) The *RPP* shall be designed to withstand voltage drops and peaks, as illustrated in Figure 4, 4a and 4b, and supply or absorb reactive current as illustrated in Figure 5 without disconnecting.

(3) The *RPP* shall be able to withstand voltage drops to zero, measured at the POC, for a minimum period of 0.150 seconds without disconnecting, as shown in Figures 4, 4a and 4b with exception for synchronous generators category B during symmetrical 3 phase faults.

(4) The *RPP* of category C shall be able to withstand voltage peaks up to 120% of the nominal voltage, measured at the POC, for a minimum period of 2 seconds without disconnecting, as shown in Figures 4, 4a and 4b.

(5) Figures 4, 4a and 4b shall apply to all types of faults (symmetrical and asymmetrical i.e. one-, two- or three-phase faults) and the bold line shall represent the minimum voltage of all the phases.

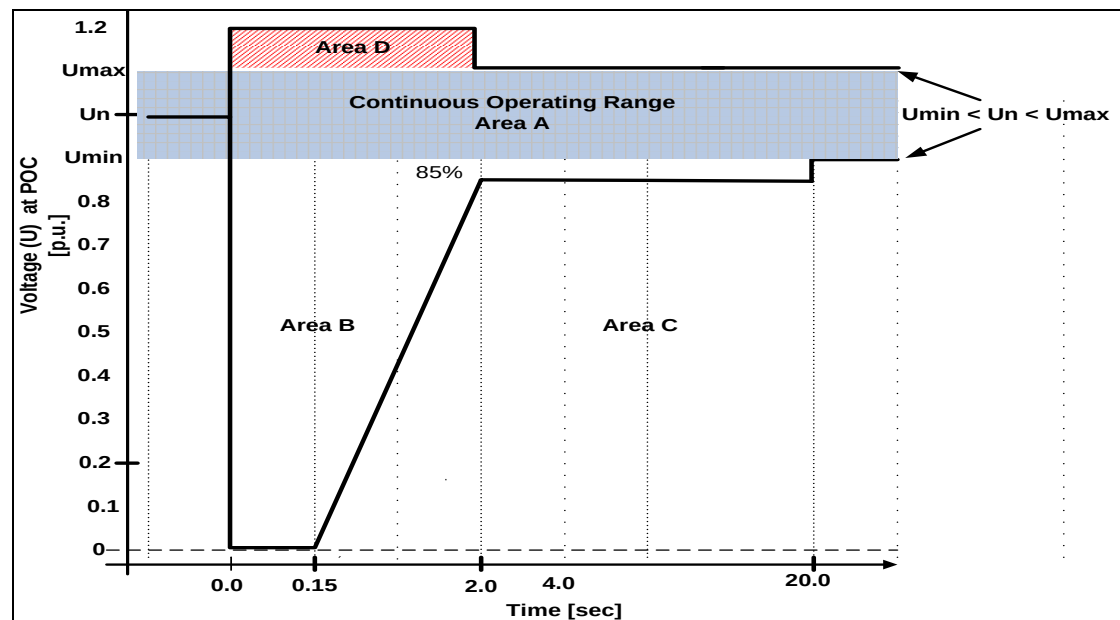


Figure 4: Voltage Ride through Capability for the RPPs of Category A3, B and C utilising non-synchronous machines

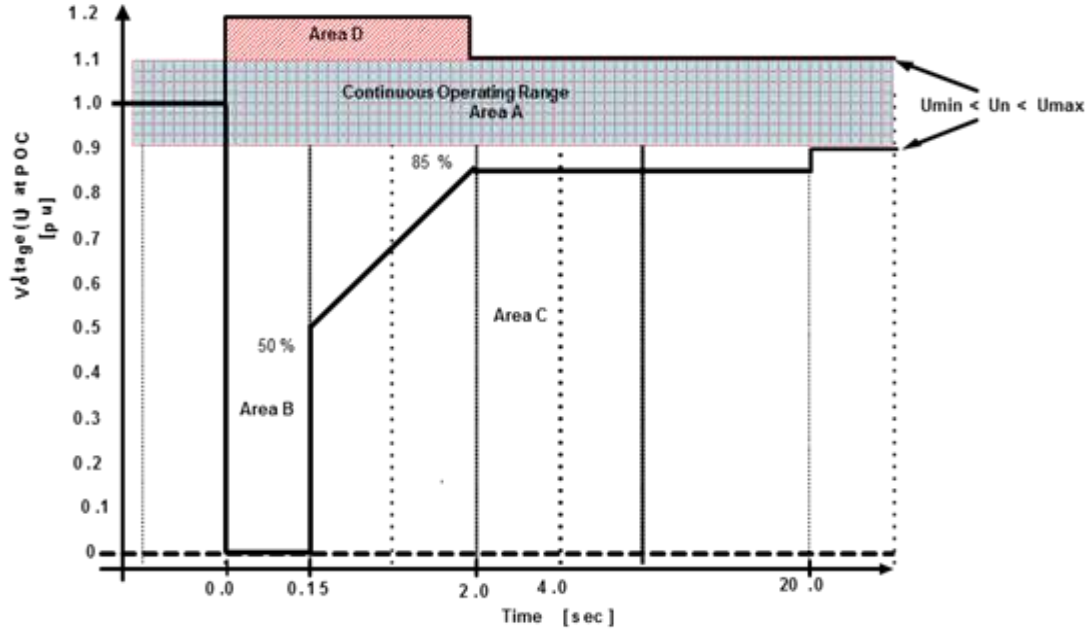


Figure 4a: Voltage Ride Through Capability for the RPPs of Category A3, B and C utilising synchronous machines

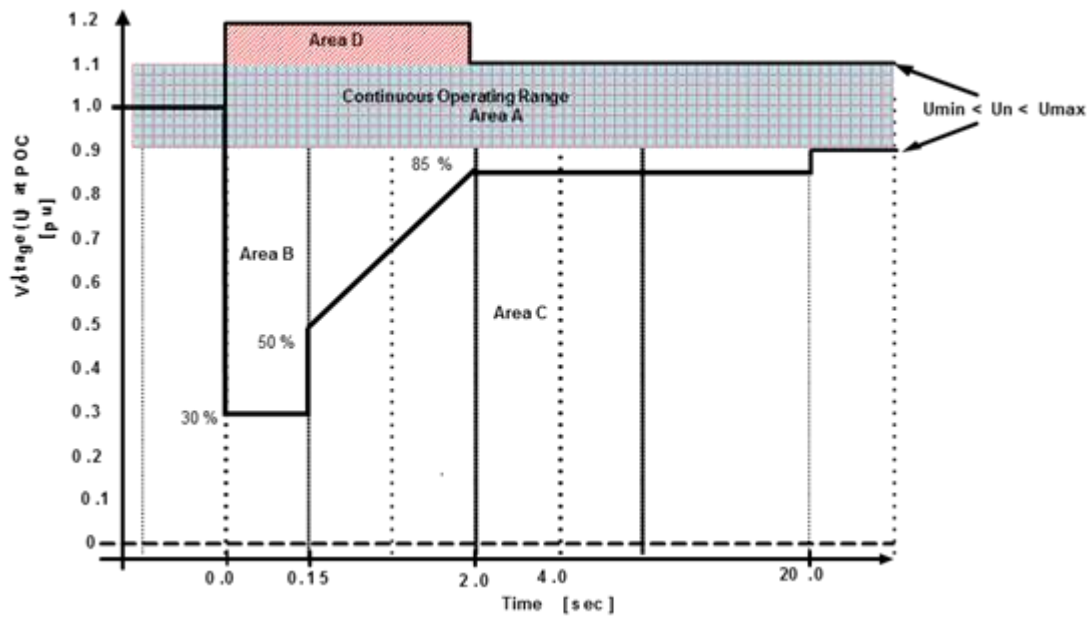


Figure 4b: Voltage Ride Through Capability for the RPPs of Category A3 and B synchronous machine, for 3 phase faults only

(6) If the voltage (U) reverts to area A during a fault sequence, subsequent voltage drops shall be regarded as a new fault condition. If several successive fault sequences occur within area B and evolve into area C, disconnection is allowed, see Figures 4, 4a and 4b.

(7) In the area C (Figures 4, 4a and 4b): disconnection of the *RPP* is allowed.

(8) In connection with symmetrical fault sequences in areas B and D of Figures 4, 4a and 4b, the *RPP* (other than synchronous generating units) shall have the capability of controlling the reactive current, as illustrated in Figure 5. The following requirements shall be complied with:

(a) **Area A:** The *RPP* shall stay connected to the network and uphold normal production.

(b) **Area B:** The *RPP* shall stay connected to the network and in addition:

- (i) *RPPs* of category A3 shall not inject any reactive current into the network;
- (ii) *RPPs* of category B and category C shall provide maximum voltage support by supplying a controlled amount of reactive current so as to ensure that the *RPP* assists in stabilising the voltage as shown in Figure 5;
- (iii) Inverter driven *RPPs* of category B and category C shall be able to disable reactive current support functionality at the request of SO or local network operator.

(c) **Area D:** The *RPP* shall stay connected to the network and provide maximum voltage support by absorbing a controlled amount of reactive current so as to ensure that the *RPP* helps to stabilise the voltage within the design capability offered by the *RPP*, see Figure 5.

(d) **Area E (Figure 5):** Once the voltage at the *POC* is below 20%, the *RPP* shall continue to supply reactive current within its technical design limitations so as to ensure that the *RPP* helps to stabilise the voltage. Disconnection is only allowed after conditions of Figures 4, 4a and 4b have been fulfilled.

(9) Control shall follow Figure 5 so that the reactive current follows the control characteristic with a tolerance of $\pm 20\%$ after 60 ms.

(10) The supply of reactive power has first priority in area B, while the supply of active power has second priority. Active power shall be maintained during voltage drops, but a reduction in active power within the *RPP*'s design specifications is required in proportion to voltage drop for voltages below 85%.

(11) Upon clearance of fault each *RPP* shall restore active power production to at least 90% of the level available immediately prior to the fault within 1 second.

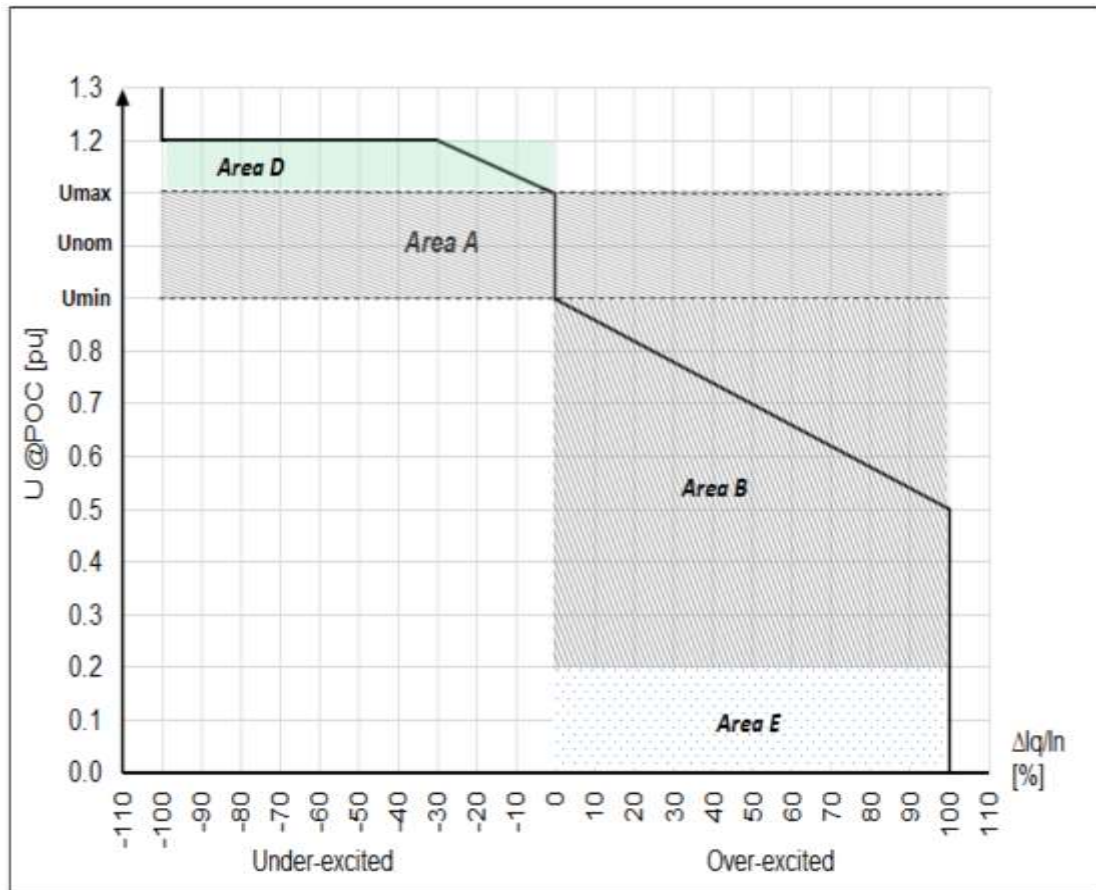


Figure 5: Requirements for Reactive Power Support, I_Q , during voltage drops or peaks at the POC

6. Frequency Response

(1) In case of frequency deviations in the NIPS, RPPs shall be designed to be capable to provide power-frequency response in order to stabilise the grid frequency.

6.1 Power-frequency response curve for RPPs

(1) During high frequency operating conditions, RPPs shall be able to provide mandatory active power reduction requirement in order to stabilise the frequency in accordance with Figure 6 below. The metering accuracy for the grid frequency shall be ± 10 mHz or better.

(2) When the frequency on the NIPS exceeds 50.5 Hz, the RPP shall reduce the active power as a function of the change in frequency as illustrated in Figure 6 below.

(3) Once the frequency exceed 51.5Hz for longer than 4 seconds the RPP shall be tripped to protect the NIPS.

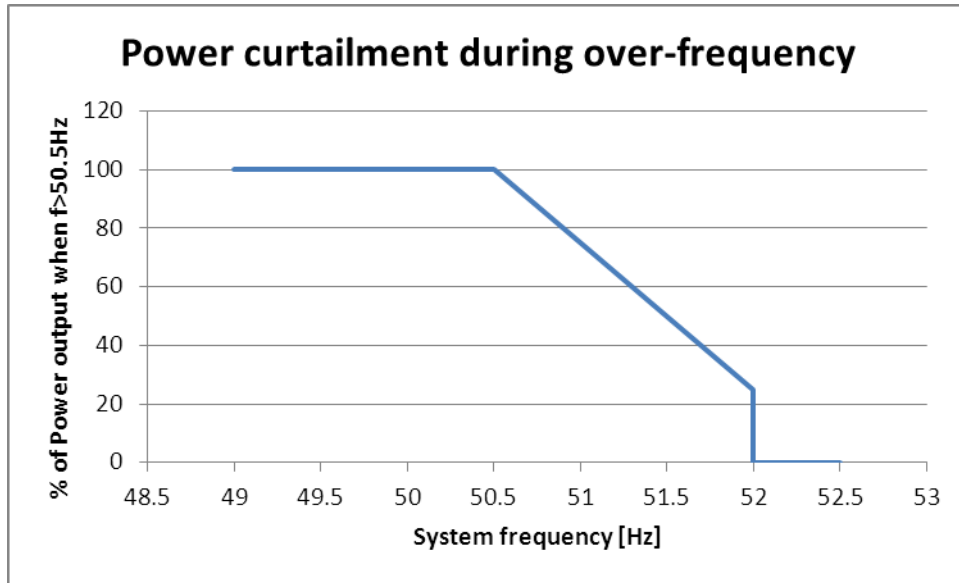


Figure 6: Power curtailment during over-frequency for *RPPs*

6.2 Power-frequency response curve for *RPPs* of Category C

(1) *RPPs* shall be designed to be capable to provide power-frequency response as illustrated in Figure 7.

(2) Except for the mandatory high frequency response (above 50.5 Hz), the *RPP* shall not perform any frequency response function (i.e. there shall be no P_{Δ} , dead-band and control-band functions implement) without having entered into a specific agreement with the *SO*.

(3) It shall be possible to set the frequency response control function for all frequency points shown in Figure 7. It shall be possible to set the frequencies $f_{\min}$, $f_{\max}$, as well as f_1 to f_6 to any value in the range of 47 - 52 Hz with a minimum accuracy of 10 mHz.

(4) The purpose of frequency points f_1 to f_4 is to form a dead band and a control band for for *RPPs* contracted for primary frequency response. The purpose of frequency points f_4 to f_6 is to supply mandatory critical power/frequency response.

(5) The *RPP* shall be equipped with the frequency control *droop* settings as illustrated in figure 7. Each *droop* setting shall be adjustable between 0% and 10%. The actual *droop* setting shall be as agreed with the *SO*.

(6) The *SO* shall decide and advise the *RPP generator* (directly or through its agent) on the *droop* settings required to perform control between the various frequency points.

(7) If the active power from the *RPP* is regulated downward below the unit's design limit $P_{\min}$, shutting-down of individual *RPP units* is allowed.

(8) The *RPP* (with the exception of *RPPPV*) shall be designed with the capability of providing a P_{Δ} of not less than 3% of $P_{\text{available}}$. P_{Δ} is the amount of active power by which the available active power has been reduced in order to provide reserves for frequency stabilisation.

(9) It shall be possible to activate and deactivate the frequency response control function in the interval from $f_{\min}$ to $f_{\max}$.

(10) If the frequency control setpoint (P_{Δ}) is to be changed, such change shall be commenced within two seconds and completed no later than 10 seconds after receipt of an order to change the setpoint.

(11) The accuracy of the control performed (i.e. change in active power output) and of the setpoint shall not deviate by more than $\pm 2\%$ of the setpoint value or by $\pm 0.5\%$ of the rated power, depending on which yields the highest tolerance.

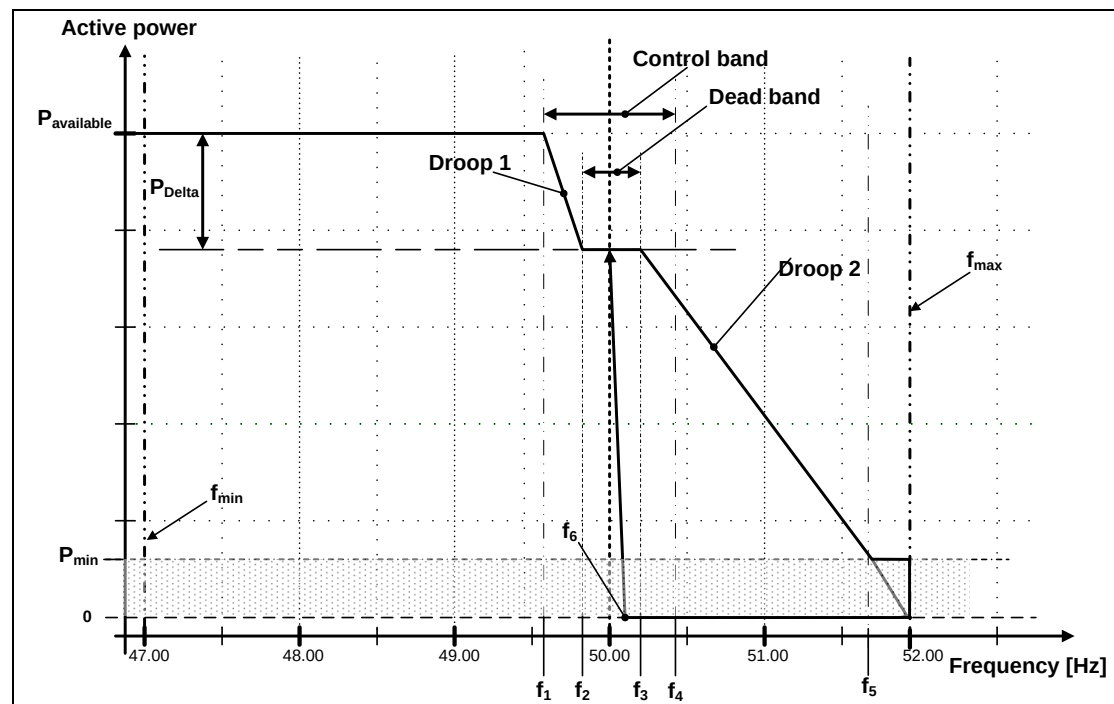


Figure 7: Frequency response requirement for *RPPs* of category *C*

(12) The default settings for $f_{\min}$, $f_{\max}$, f_4 , f_5 and f_6 shall be as shown in **Table 3**, unless otherwise agreed upon between the *SO* and the *RPP generator*. Settings for f_1 , f_2 and f_3 shall be as agreed with the *SO*.

Table 3: Frequency Default Settings

Parameter	Magnitude (Hz.)
$f_{\min}$	47
$f_{\max}$	52
f_1	As agreed with SO
f_2	As agreed with SO
f_3	As agreed with SO
f_4	50.5
f_5	51.5
f_6	50.2

6.3 Procedure for setting and changing the power-frequency response curves for RPPs of Categories C

(1) The SO or its agent shall give the RPP generator a minimum of 2 weeks if changes to any of the frequency response parameters (i.e. f_1 to f_6) are required. The RPP generator shall confirm with the SO or its agent that requested changes have been implemented within two weeks of receiving the SO's request.

7. Reactive Power Capabilities

7.1 RPPs of Category A

(1) RPPs of category A1 & A2 shall operate at better than 0.98 power factor (leading or lagging) measured at the generator terminals, unless otherwise specified by the NSP or the SO.

(2) RPPs of category A3 shall be designed with the capability to supply *rated power* (P_n) (MW) for power factors ranging between 0.95 lagging and 0.95 leading, measured at the POC available from 20% to 100% of *rated power* (P_n).

(3) The RPP shall be designed to operate according to a power factor characteristic curve, which will be determined by the NSP or the SO.

(4) The default power factor setting shall be **unity** power factor, unless otherwise specified by the NSP or the SO.

7.2 RPPs of Category B

(1) RPPs of category B shall be designed with the capability to operate in a voltage (V), power factor or reactive power (Q or Mvar) control modes as described in section 8 below.

The actual operating mode (V, power factor or Q control) as well as the operating point shall be agreed with the *NSP*.

(2) When operating between 5% and 100% of *rated power P_n* (MW) the *RPP* of category *B* shall have the capability of varying reactive power (Mvar) support at the *POC* within the reactive power capability ranges as defined by Figure 8a, where *Q_{min}* and *Q_{max}* are voltage dependent as defined by Figure 9.

(3) At nominal voltage, the required *RPP* reactive power capability (measured at the *POC*) shall be as shown in Figure 8b.

(4) When operating below 5% of *rated power P_n* (MW), there is no reactive power capability requirement, however the *RPP* can only operate within the reactive power tolerance range not exceeding $\pm 5\%$ of *rated power*; that is within Area A,B,C and D in Figure 8b.

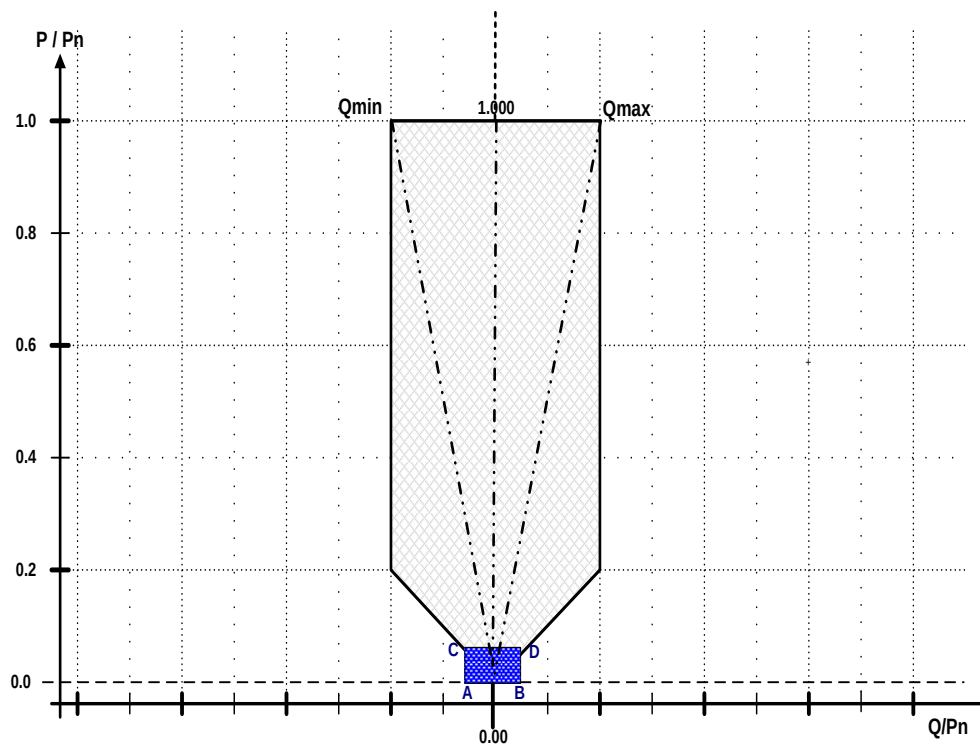


Figure 8a: Reactive power requirements for *RPPs* of category *B* at the *POC* (*Q_{min}* and *Q_{max}* are voltage dependent as defined by Figure 9)

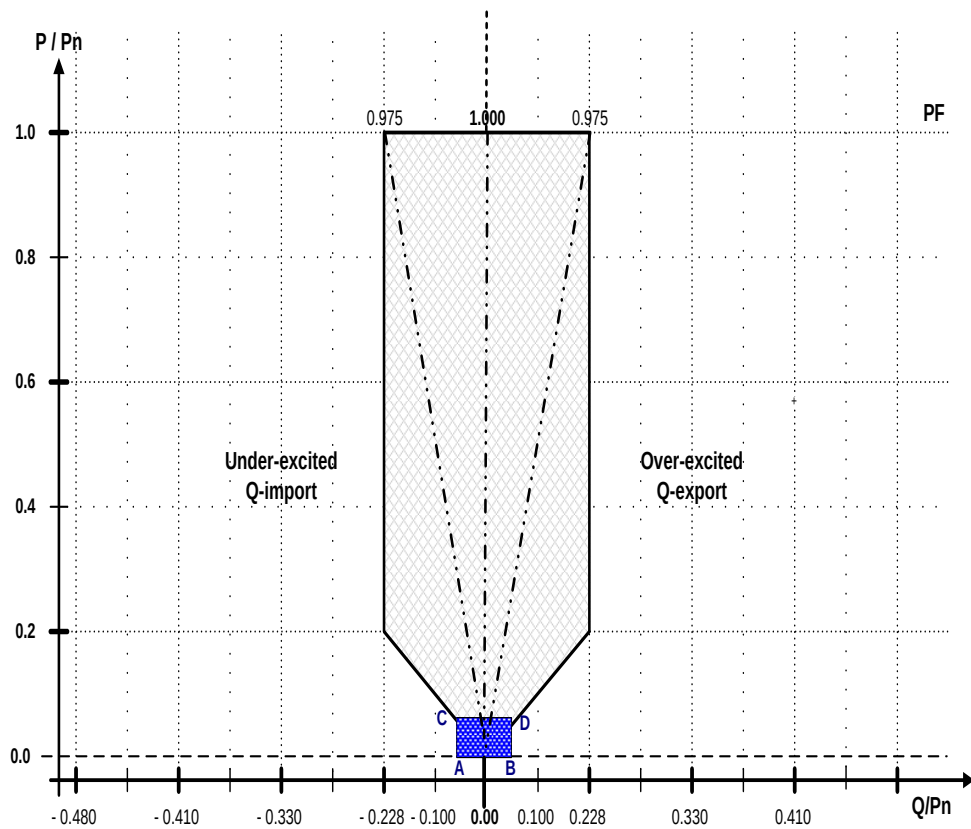


Figure 8b: Reactive power requirements for RPPs of category B (at nominal voltage at POC)

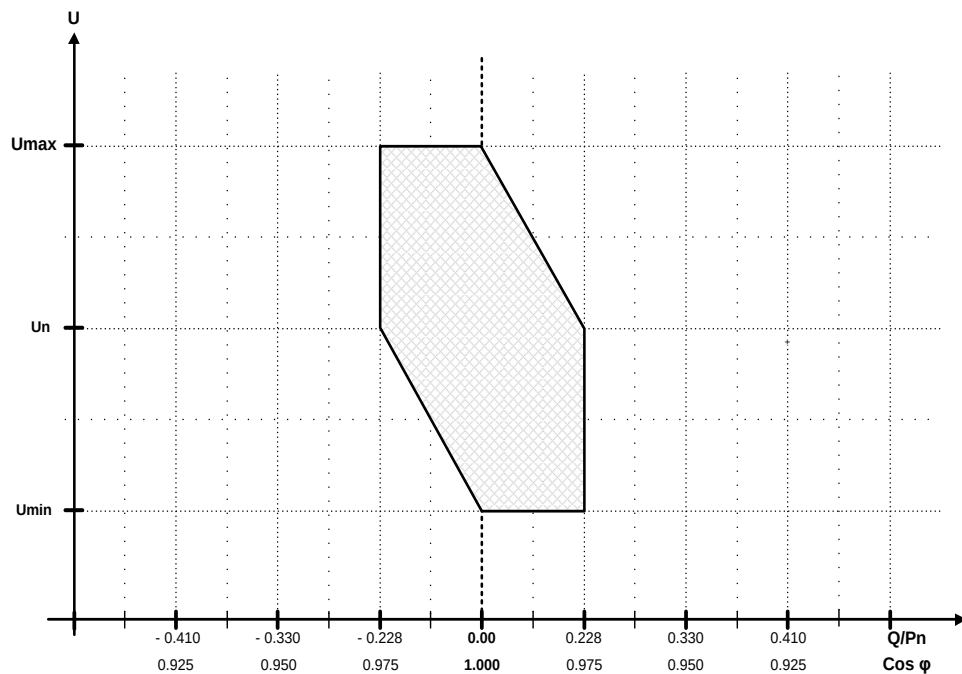


Figure 9: Requirements for reactive power and voltage control range for RPPs of category B.

7.3 RPPs of Category C

(1) *RPPs* of category C shall be designed with the capability to operate in a voltage, power factor or, reactive power (Q or $Mvar$) control modes. The actual control operating mode (V , power factor or Q control) as well as operating point shall be agreed with the *NSP*.

(2) When operating between 5% and 100% of *rated power* P_n (MW) the *RPP* of category C shall have the capability of varying reactive power ($Mvar$) support at the *POC* within the reactive power capability ranges as defined by Figure 10a, where Q_{min} and Q_{max} are voltage dependent as defined by Figure 11.

(3) At nominal voltage, the required *RPP* reactive power capability (measured at the *POC*) shall be as shown in Figure 10b.

(4) When operating below 5% of *rated power* P_n (MW), there is no reactive power capability requirement, however the *RPP* can only operate within the reactive power tolerance range not exceeding $\pm 5\%$ of *rated power*; that is within Area A,B,C and D in Figure 10b.

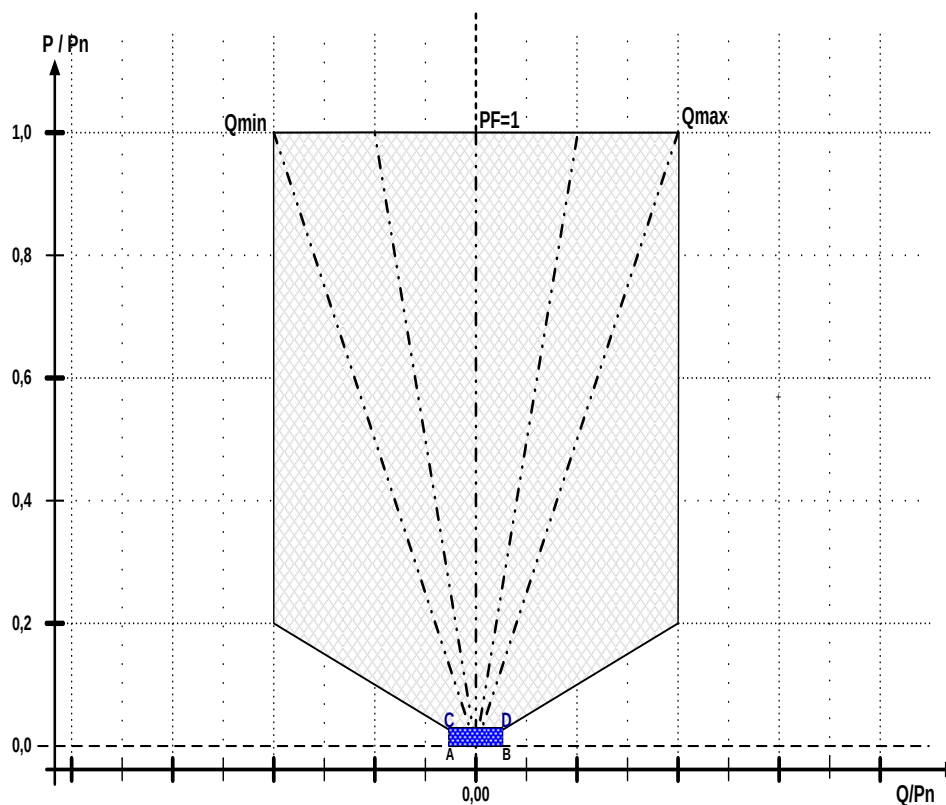


Figure 10a: Reactive power requirements for RPPs of category C at the POC (Q_{min} and Q_{max} are voltage dependent as defined by Figure 11)

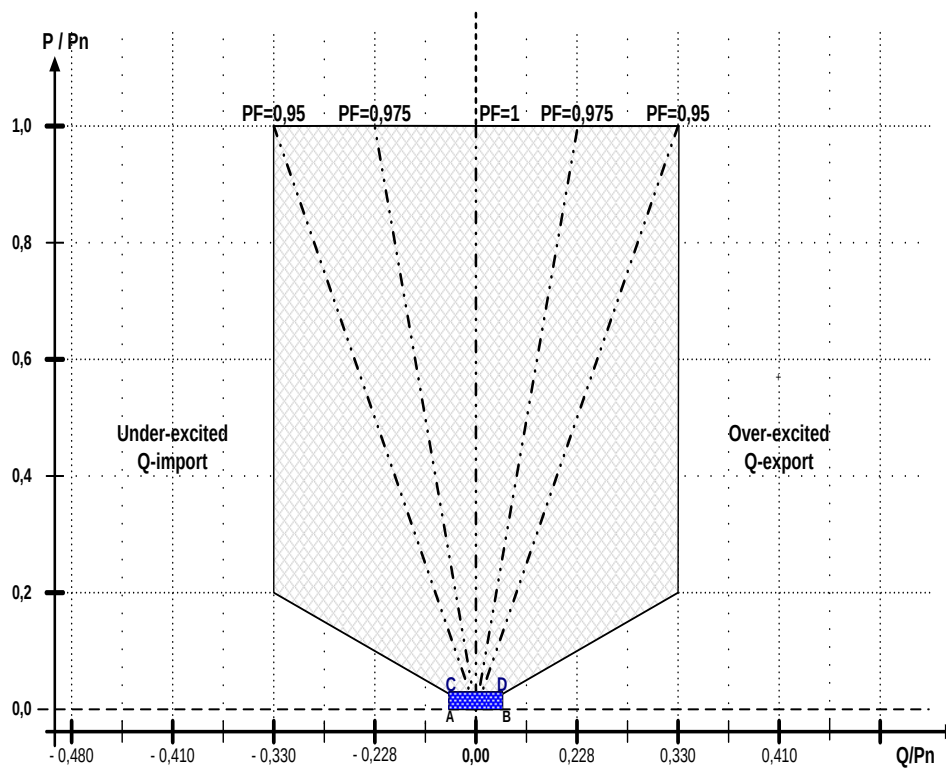


Figure 10b: Reactive power requirements for RPPs of category C (at nominal voltage at POC)

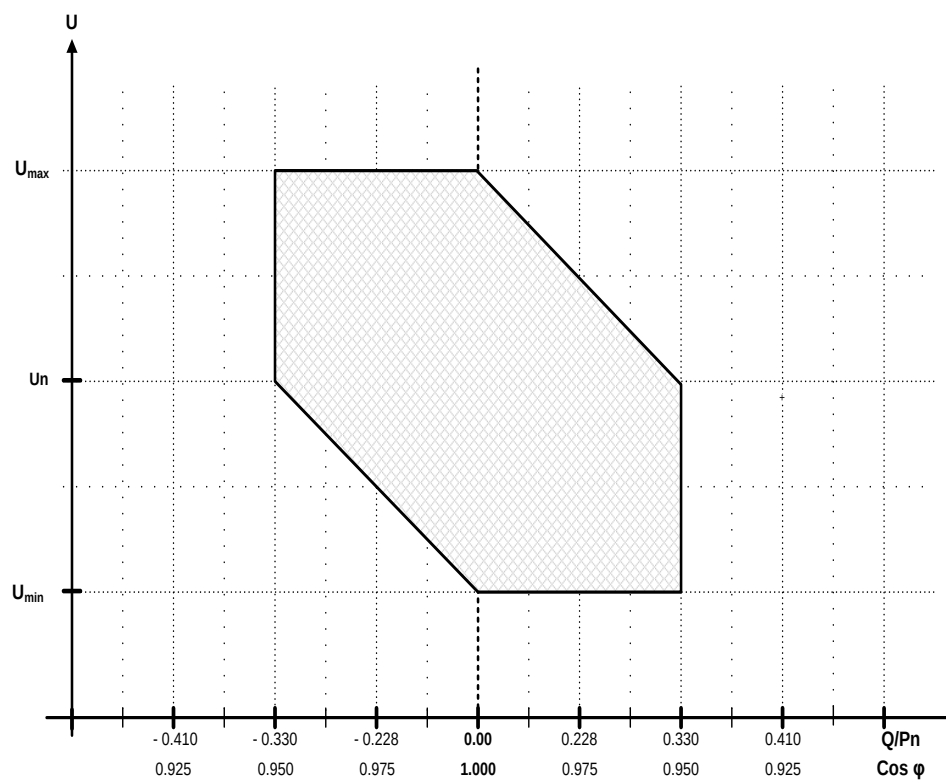


Figure 11: Requirements for reactive power and voltage control range for RPPs of category C

8. Reactive Power and Voltage Control Functions

- (1) The following requirements shall apply to *RPPs* of *category B and C*.
- (2) The *RPP* shall be equipped with reactive power control functions capable of controlling the reactive power supplied by the *RPP* at the *POC* as well as a voltage control function capable of controlling the voltage at the *POC* via orders using setpoints and gradients.
- (3) The reactive power and voltage control functions are mutually exclusive, which means that only one of the three functions mentioned below can be activated at a time.
 - (a) Voltage control
 - (b) Power Factor control
 - (c) Q control
- (4) The control function and applied parameter settings for reactive power and voltage control functions shall be determined by the *NSP* in collaboration with the *SO*, and implemented by the *RPP generator*. The agreed control functions shall be documented in the *operating agreement*.

8.1 Reactive power (Q) Control

- (1) Q control is a control function controlling the reactive power supply and absorption at the *POC* independently of the active power and the voltage. This control function is illustrated in **Figure 12** as a vertical line.
- (2) If the Q control setpoint is to be changed by the *NSP*, *SO* or their agent, the *RPP generator* shall update its echo analog set point value in response to the new value within two seconds. The *RPP* shall respond to the new set point within 30 seconds after receipt of an order to change the setpoint.
- (3) The accuracy of the control performed and of the setpoint shall not deviate by more than $\pm 2\%$ of the setpoint value or by $\pm 0.5\%$ of maximum reactive power, depending on which yields the highest tolerance.
- (4) The *RPP* shall be able to receive a Q setpoint with an accuracy of at least 1kVar.

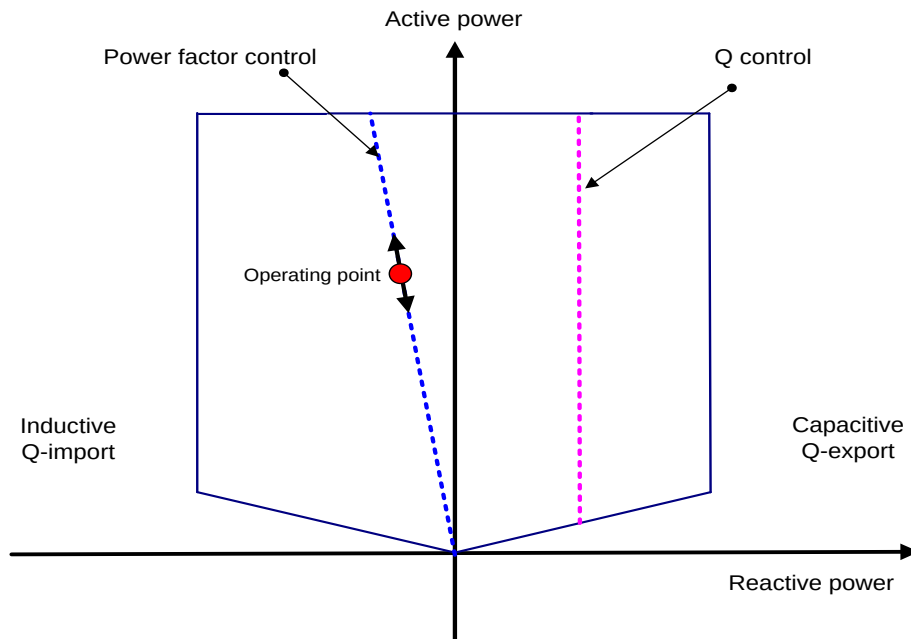


Figure 12: Reactive power control functions for the RPP

8.2 Power Factor Control

- (1) Power Factor Control is a control function controlling the reactive power proportionally to the active power at the POC. This is illustrated in **Figure 12** by a line with a constant gradient.
- (2) If the power factor setpoint is to be changed by the NSP, SO or their agent, the RPP shall update its echo analog set point value to in response to the new value within two seconds. The RPP shall respond to the new set point within 30 seconds after receipt of an order to change the setpoint.
- (3) The accuracy of the control performed and of the setpoint shall not deviate by more than ± 0.02 .

8.3 Voltage Control

- (1) Voltage control is a control function controlling the voltage at the POC.
- (2) If the voltage setpoint is to be changed, such change shall be commenced within two seconds and completed no later than 30 seconds after receipt of an order to change the setpoint.
- (3) The accuracy of the voltage setpoint shall be within $\pm 0.5\%$ of nominal voltage, and the accuracy of the control performed shall not deviate by more than $\pm 2\%$ of the required injection or absorption of reactive power according to *droop* characteristics as defined in **Figure 13**.

(4) The individual *RPP* shall be able to perform the control within its dynamic range and voltage limit with the *droop* configured as shown in **Figure 13**. In this context, *droop* is the voltage change (p.u.) caused by a change in reactive power (p.u.).

(5) When the voltage control has reached the *RPP*'s dynamic design limits, the control function shall await possible overall control from the tap changer or other voltage control functions.

(6) Overall voltage coordination shall be handled by the *NSP* in collaboration with the *SO*.

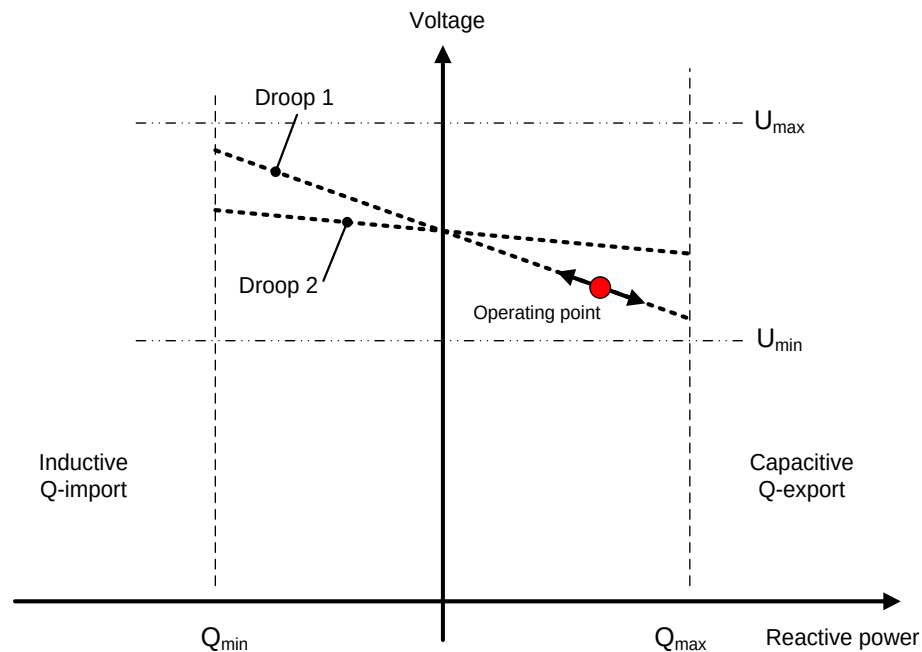


Figure 13: Voltage control for the *RPP*

9. Power Quality

(1) *RPPs* of categories A1, A2, A3, and category B with rated power ≤ 5 MVA shall comply with the requirements as detailed in **Appendix 13** - *RPP* Power Quality Compliance Guideline.

(2) The following requirements shall apply to the following *RPP* categories:

- (a) Category B with rated power above 5 MVA,
- (b) All category C *RPPs*.

(3) For grid code compliance the *RPP* shall monitor and report on power quality using an IEC 61000-4-30 Class A power quality monitoring device. The reporting will be done to prove compliance at the *POC* against the *NSP* requirements specified in the system supply agreements. The power quality parameters to be reported on include:

- (a) flicker

- (b) harmonics
- (c) unbalanced voltages

(4) Voltage and current quality distortion levels emitted by the *RPP* at the *POC* may not exceed the apportioned limits as determined by the relevant *NSP*. The calculation of these emission levels shall be based on international and local specifications. The allocation shall be fair and transparent. For current harmonics, allowance is made for individual harmonics to exceed specified limits by up to 50%, provided the Group Harmonic Distortion limits for the four bandwidth groupings (specified in Appendix 13: *RPP* Power Quality Compliance Guideline) are met.

(5) Appendix 13 provides guidance to *RPP generators* on a method to prove compliance to the Power Quality requirements as set out by the *NSP*.

(6) The *RPP generator* shall ensure that the *RPP* is designed, configured and implemented in such a way that the specified emission limit values are not exceeded.

(7) The *NSP* will manage any voltage harmonic compatibility level exceedances due to the network harmonic impedance at the *POC* being more than 3 times the base harmonic impedance for the range of reference fault levels at the *POC* in line with its license conditions. The 3 times base harmonic impedance is calculated using the following equation:

$$Z(h) = 3 * \frac{V^2}{S} * h$$

where *h* is the harmonic number, *V* is the nominal voltage (line-to-line) in kV, and *S* is the fault level in MVA. The angle of the network harmonic impedance is such that the impedance may range from fully inductive to fully capacitive.

(8) In order to assist with the maximum resonance of 3 times as per clause (6) above, no *RPP* may connect equipment, e.g. shunt capacitor banks, that will cause a resonance of more than 3 times at the *POC* at any frequency.

(9) The *Transmission and Distribution network service providers* shall use reasonable endeavours to furnish the *RPP* with a reliable and continuous connection for the delivery of electrical energy up to the *POC*. The network operators do not guarantee that the continuity and voltage quality of the connection will always be maintained under all contingencies. It is therefore incumbent upon the *RPP* to take adequate measures to protect the *RPP* facility against any losses and/or damage arising from frequency deviations, connection/supply interruptions, voltage variations (including voltage dips), voltage harmonics, voltage flicker, voltage unbalance, voltage swells and transients, undervoltages and overvoltages in the connection. It is also incumbent upon the *RPP* to take such necessary measures so as not to cause any damage to the *TS* and *DS*.

10. Protection and Fault levels

- (1) Unless otherwise stated, requirements in this section apply to all *categories* of *RPPs*.
- (2) Protection functions shall be available to protect the *RPP* and to ensure a stable *TS* and *DS*.
- (3) The *RPP generator* shall ensure that a *RPP* is dimensioned and equipped with the necessary protection functions so that the *RPP* is protected against damage due to faults and incidents in the *TS* and *DS*.
- (4) The *RPP* of *category A* shall be equipped with effective detection of islanded operation in all system configurations and capability to shut down generation of power in such condition within 2 seconds. Islanded operation with part of the *TS* or *DS* is not permitted unless specifically agreed with the *NSP*.
- (5) The *RPP* of *category B* and *C* shall be equipped with effective detection of islanded operation in all system configurations and capability to shut down generation of power in such condition within 2 seconds. Islanded operation with part of the *TS* or *DS* is not permitted unless specifically agreed with the *NSP*.
- (6) The above requirement does not apply to *RPP* of category *B* and *C* that is able to control voltage and frequency and which *POC* is at *TS*.
- (7) The *NSP* or the *SO* may request that the set values for protection functions be changed following commissioning if it is deemed to be of importance to the operation of the *TS* and *DS*. However, such change shall not result in the *RPP* being exposed to negative impacts from the *TS* and *DS* lying outside of the design requirements.
- (8) The *NSP* shall inform the *RPP generator* of the highest and lowest short-circuit current that can be expected at the *POC* as well as any other information about the *TS* and *DS* as may be necessary to define the *RPP's* protection functions.

11. Active Power Constraint Functions

- (1) This section shall apply to *RPPs* of *categories A3, B & C*
- (2) For system security reasons it may be necessary for the *SO, NSP* or their agent to curtail the *RPP* active power output.
- (3) The *RPP generator* shall be capable of:

- (a) operating the *RPP* at a reduced level if active power has been curtailed by the *SO*, *NSP* or their agent for network or system security reasons,
 - (b) receiving a telemetered *MW Curtailment set-point* sent from the *SO*, *NSP* or their agent. If another operator is implementing power curtailment, this shall be in agreement with all the parties involved.
- (4) The *RPP* shall be equipped with constraint functions, i.e. supplementary active power control functions. The constraint functions are used to avoid imbalances in the *NIPS* or overloading of the *TS* and *DS* in connection with the reconfiguration of the *TS* and *DS* in critical or unstable situations or the like, as illustrated in **Figure 14**.
- (5) Activation of the active power constraint functions shall be agreed with the *SO* or *NSP*. The required constraint functions are as follows:
- (a) Absolute production constraint
 - (b) Delta production constraint
 - (c) Power gradient constraint.
- (6) Delta productions constraint and power gradient constraint functions may not be required for CSP, SHPP, LGPP, BMPP and BGP 's subject to agreement by the *SO* or *NSP*.
- (7) The required constraint functions are described in the following sections.

11.1 Absolute Production Constraint

- (1) An Absolute Production Constraint is used to constrain the output active power from the *RPP* to a predefined power MW limit at the *POC*. This is typically used to protect the *TS* and *DS* against overloading.
- (2) If the setpoint for the Absolute Production Constraint is to be changed, the *RPPs* (*PVPP* and *WPP* s) shall commence such change within two seconds and the change shall be completed not later than 30 seconds after receipt of an order to change the setpoint.
- (3) For *RPPs* (*PVPP* and *WPP*), the accuracy of the control performed and of the setpoint shall not deviate by more than $\pm 2\%$ of the setpoint value or by $\pm 0.5\%$ of the *rated power*, depending on which yields the highest tolerance.
- (4) For *RPPs*, if the setpoint for the Absolute Production Constraint is to be changed, such change shall be as stipulated for each renewable technology in the Appendices to this code; Appendix 3 (CSP), Appendix 4 (SHPP), Appendix 5 (LGPP), Appendix 6 (BMPP) and

Appendix 7 (BGPP). . The RPP shall complete the change in the minimum time and to the highest accuracy achievable within its technical design limitations.

11.2 Delta Production Constraint

(1) A Delta Production Constraint is used to constrain the active power from the *RPP* to a required constant value in proportion to the possible active power.

(2) A Delta Production Constraint is typically used to establish a control reserve for control purposes in connection with *frequency control*.

(3) If the setpoint for the Delta Production Constraint is to be changed, such change shall be commenced within two seconds and completed no later than 30 seconds after receipt of an order to change the setpoint.

(4) The accuracy of the control performed and of the setpoint shall not deviate by more than $\pm 2\%$ of the setpoint value or by $\pm 0.5\%$ of the *rated power*, depending on which yields the highest tolerance.

11.3 Power Gradient Constraint

(1) A Power Gradient Constraint is used to limit the maximum ramp rates by which the active power can be changed in the event of changes in primary renewable energy supply or the setpoints for the *RPP*, taking into account the availability of primary energy to support these gradients. A Power Gradient Constraint is typically used for reasons of system operation to prevent changes in active power from impacting the stability of the *TS* or the *DS*.

(2) If the setpoint for the Power Gradient Constraint is to be changed, such change shall be commenced within two seconds and completed no later than 30 seconds after receipt of an order to change the setpoint.

(3) The accuracy of the control performed and of the setpoint shall not deviate by more than $\pm 2\%$ of the setpoint value or by $\pm 0.5\%$ of the *rated power*, depending on which yields the highest tolerance.

(4) The active power constraint functions are illustrated on **Figure 14**.

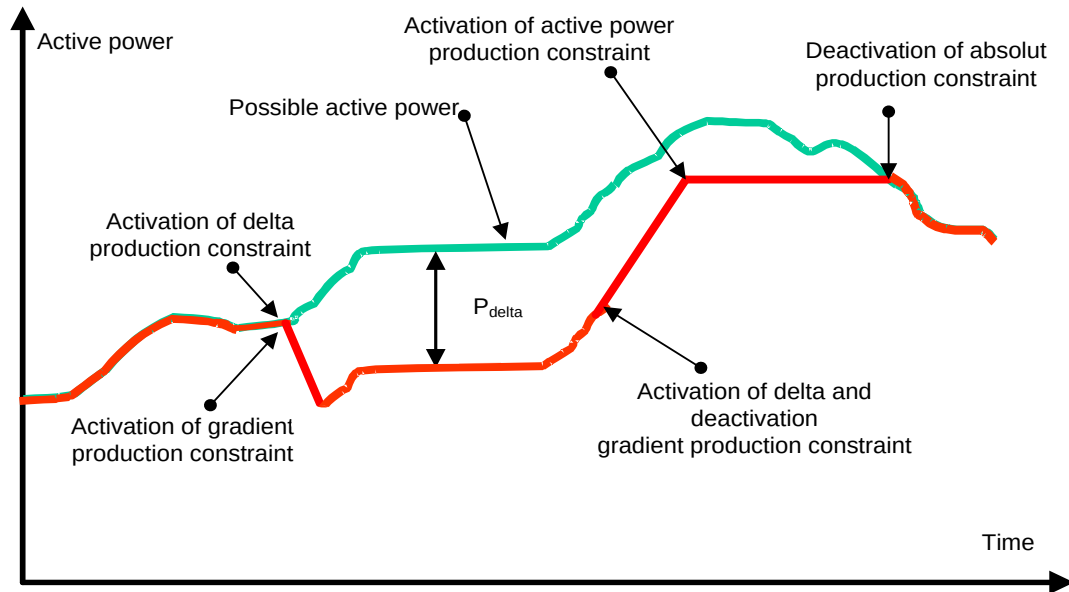


Figure 14: Active power control functions for a *Renewable Power Plant*

12. Control Function Requirements

- (1) *RPPs* shall be equipped with the control functions specified in **Table 4**. The purpose of the various control functions is to ensure overall control and monitoring of the *RPP's* generation.
- (2) The *RPP* control system shall be capable of controlling the ramp rate of its active power output with a maximum MW per minute ramp rate set by *SO* or *NSP*.
- (3) These ramp rate settings shall be applicable for all ranges of operation including positive ramp rate during start up, positive ramp rate only during normal operation and negative ramp rate during controlled shut down. They shall not apply to frequency regulation.
- (4) The *RPP generator* shall not perform any frequency response or voltage control functions without having entered into a specific agreement to this effect with the *NSP*.

Table 4: Control functions required for *RPPs*

Control function	Category A3	Category B	Category C
Frequency control	-	-	X
Absolute production constraint	X	X	X
Delta production constraint	-	-	X

Power gradient constraint	X	X	X
Q control	-	X	X
Power factor control	-	X	X
Voltage control	-	X	X

13. RPP Availability, Forecast, and Supervisory Control and Data Acquisition

- (1) All specified signals shall be made available at the *POC* by the *RPP generator*.
- (2) Requirements for the exchange of signals between *RPPs* of *category B and C*, and the *NSP*, *SO* or their agent are described below. A detailed description on the implementation of the signals lists is provided in Appendix 10.
- (3) The signals indicated in clauses 13.1, 13.2 and 13.3 below are the minimum requirement that the *RPP generator* must provide; additional site specific signals may also be accommodated in agreement with the relevant *SO/NSP*.
- (4) All digital signals reported to the *Gateway* shall be time-stamped (UTC +2:00) to an accuracy of +/-1 millisecond and shall be reported within 1 second of any change as shown in Figure 15.
- (5) Timestamps on analogue changes are not required.
- (6) All meteorological data in clause 13.2.2 shall be reported every minute.
- (7) All analogue input changes shall be reported to the *Gateway* within 1 second (see Figure 15) of any change greater than or equal to the following:
 - (a) Frequency shall be updated when the value changes by 0.01 Hz or more.
 - (b) Power factor values shall be updated when the value changes by 0.01 or more.
 - (c) Active power values shall be updated when the value changes by 1% or more of *rated power*. If *rated power* is 50 MW or more then values shall be updated when the value changes by 0.5 MW or more.
 - (d) Actual Ramp Rate shall be updated when the value changes by 1 MW/min or more.
 - (e) All other analogues shall be updated when the value changes by 1% or more of the full-scale or nominal value.

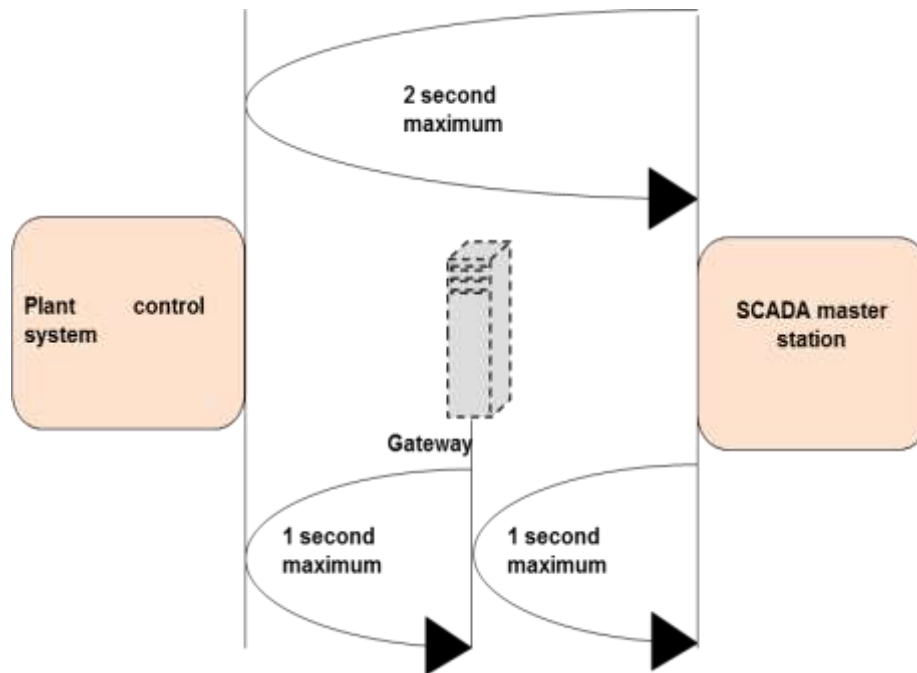


Figure 15: Data collection time frame

- (8) To protect the communications bandwidth especially on Area Radio channels the values used by the *Gateway* to report analogue changes to the *NSP* may be set higher than the values listed in clause 13 (7) above.
- (9) Support for both direct-operate and select-before-operate Commands shall be provided between the *Gateway* and the *RPP* plant control system.

13.1 Generator Availability and Forecast Production Values for RPPs of Category B and C

- (1) This section shall apply to *RPPs* of categories *B* and *C*.
- (2) *RPP* generator shall make available the following signals to the *NSP/SO* in the format and method specified by the *NSP/SO*:
 - (a) Available MW and forecast MW day-ahead for a week for each hour before 10a.m,
 - (b) Available MW and forecast MW for the next 6 hours updated hourly 10 to 20 minutes before the hour,
 - (c) Available MVar for the next 6 hours updated hourly 10 to 20 minutes before the hour.
- (3) The content of each forecast will be structured using XML tags. Examples are available on request. The format is subject to change.

13.2 Signals from the RPP, to be made available at the POC

- (1) This section shall apply to *RPPs* of category *B* and *C*.

(2) Signals from the *RPP* to the *SO* or *NSP* or their agent shall be broken up into a number of logical groups depending on functionality.

(3) The following signal list groups shall apply:

(a) Signals List #1 – General.

In addition, the *RPP generator* shall be required to provide certain signals from Signals Lists 2, 3, 4 and 5. These lists relate to:

(b) Signals List #2 - RPP Meteorological Data

(c) Signals List #3 - Frequency Response System Settings

(d) Signals List #4 - Active Power Constraints

(e) Signals List #5 - Reactive power and voltage control.

13.2.1 Signals List #1 – General

(1) The *RPP generator* shall make the following signals available at a *SO* or *NSP* designated *communication gateway equipment* located at the *RPP* site. These signals should be provided for each Bay. The concept of a Bay and further details on the signals below is explained in Appendix 10.

(a) Actual sent-out (MW) at the POC

(b) Actual Ramp rate of the entire RPP

(c) Reactive Power Import/Export (-/+Mvar) at the POC

(d) Current sent out

(e) Power Factor

(f) Voltage output

(g) Frequency

(h) Breaker status

(i) Isolator status

(j) Supervisory switch status

(k) Plant islanded

(l) Plant shutdown

(m) Plant trip on loss of grid detection.

13.2.2 Signals List #2 – RPP Meteorological Data.

(1) *RPP generator* (with the exception of *SHPP generators*) shall make the following signals available at a *SO* or *NSP* designated *Communication Gateway equipment* located at the *RPP* site:

- (a) Wind speed (within 75% of the hub height) – measured signal in meters/second (for *WPP* only)
- (b) Wind direction (within 75% of the hub height) – measured signal in degrees from true North (0-359) (for *WPP* only)
- (c) Air temperature- measured signal in degrees centigrade (-20 to 50)
- (d) Air pressure- measured signal in millibar (800 to 1400).
- (e) Air density (for *WPP* only)
- (f) Diffuse Solar radiation (for solar power plants only)
- (g) Direct Solar radiation (for solar power plants only).

(2) The meteorological data signals shall be provided by a dedicated Meteorological Mast located at the *RPP* site or, where possible and preferable to do so, data from a means of the same or better accuracy. In the case of *WPPs*, the Meteorological Mast shall be located at a position that optimizes the predictability of the output of the plant.

(3) Energy resource conversion data for the facility (e.g. MW / wind speed) for the various resource inputs shall be provided to enable the *SO* to derive a graph of the full range of the facilities output capabilities. An update shall be sent to the *SO* following any changes in the output capability of the facility.

(4) For *RPPs* where the wind turbines are widely dispersed over a large geographical area and rather different weather patterns are expected for different sections of the *RPP*, the meteorological data shall be provided from a number of individual Meteorological Masts, or where possible and preferable to do so, data from a source of the same or better reliability for groups of wind turbines. It is expected that wind turbines within an individual group shall demonstrate a high degree of correlation in Active Power output at any given time. The actual signals required shall be specified by the *SO*. There shall be at least one Meteorological Mast for every 10x10 square km area of the facility.

13.2.3 Signals List #3 – Frequency Response System Settings

(1) The *RPP generator* shall make the following signals available at a *SO* or *NSP* designated *communication gateway equipment* located at the *RPP* site:

- (a) *Frequency Response System* mode status indication (ON/OFF) as a single bit point.

13.2.4 Signals List #4 –Active Power Constraints

(1) The *RPP generator* shall make the following signals available at a *SO or NSP* designated *communication gateway equipment* located at the *RPP* site:

- (a) Curtailment mode status (ON/OFF) as a single bit point
- (b) Curtailment in progress as a single bit point
- (c) Curtailment setpoint feedback
- (d) Curtailment mode state (Not ready/Ready) as a single bit point
- (e) P-delta constraint mode (ON/OFF) as a single bit point
- (f) P-delta setpoint feedback
- (g) P-delta mode (Not ready/Ready) as a single bit point
- (h) Power gradient constraint mode (ON/OFF) as a single bit point
- (i) Up Ramp rate setpoint feedback
- (j) Down Ramp rate setpoint feedback
- (k) Power gradient constraint mode (Not ready/Ready) as a single bit point.

13.2.5 Signal List #5- Reactive power and voltage control

(1) The *RPP generator* shall make the following signals available at a *SO or NSP* designated *communication gateway equipment* located at the *RPP* site:

- (a) Reactive power control mode (ON/OFF) as a single bit point
- (b) Reactive power control setpoint/raise or lower command
- (c) Reactive power lower limit
- (d) Reactive power upper limit
- (e) Reactive power control mode (Not ready/Ready) as a single bit point
- (f) Power factor control mode (ON/OFF) as a single bit point
- (g) Power factor control setpoint/raise or lower command
- (h) Power factor control setpoint feedback
- (i) Power factor control mode (Not ready/Ready) as a single bit point
- (j) Voltage control mode (ON/OFF) as a single bit point
- (k) Voltage control setpoint/raise or lower command
- (l) Voltage control mode setpoint feedback
- (m) Voltage control mode (Not ready/Ready) as a single bit point.

13.3Control Signals Sent from SO to the RPPs

(1) The control signals described below shall be sent from *SO* to the *RPP*. The *RPP* shall be capable of receiving these signals and shall act accordingly.

- (a) Connection Point CB Trip facility. A facility shall be provided by the *NSP* to facilitate the disconnection of the *RPP*. It shall be possible for *SO*, *NSP* or their agent to send a trip signal to the circuit breaker at the *HV* side of the *POC*. This is currently implemented via the breaker shown as (a) in the example in Figure 16 below.
- (b) Primary frequency control ON/OFF
- (c) Curtailment mode ON/OFF
- (d) Curtailment setpoint command
- (e) Stop command
- (f) P-delta mode ON/OFF
- (g) P-delta setpoint command
- (h) Power gradient constraint ON/OFF
- (i) Up ramp rate setpoint command
- (j) Down ramp rate setpoint command
- (k) Reactive power control mode ON/OFF
- (l) Reactive power control setpoint command
- (m) Power factor mode ON/OFF
- (n) Power factor setpoint command
- (o) Voltage control mode ON/OFF
- (p) Voltage control setpoint

13.4 Communications Specifications

13.4.1 Gateway

- (1) The *Gateway* shall be located in the *RPP* control room. The *Gateway* shall be owned, operated and maintained by the *RPP generator*.
- (2) The *Gateway* shall be compatible with the *SCADA* Master Station protocols as per 13.4.2.1 and shall be tested as per the process described in Appendix 11.
- (3) In the event of any proposed modification to the *RPP* plant control system which may affect the operation and functionality of the *Gateway*, the *RPP generator* shall follow the test process described in Appendix 11.
- (4) The *Gateway* shall have at least three communication ports available exclusively for *NSP* and *SO SCADA* Master station communications.

13.4.2 Protocols for Information Exchange

(1) Only the IEC/IEEE protocols as specified below in 13.4.2.1 shall be used for SCADA information exchange between the *RPP* and the Gateway and the *NSP* or *SO*.

(2) The *RPP* control system shall be capable of reliably exchanging system status and data with the *NSP* or *SO* via the Gateway. The *RPP* shall cater for the communication infrastructure to connect the required number of Gateway ports to the *NSP* or *TNSP* telecommunications infrastructure in the *NSP* or *TNSP* substation.

13.4.2.1 SCADA Protocol between the Gateway and the NSP or SO

(1) The IEC 60870-5-101 protocol shall be implemented on the Gateway for communication between the *RPP* and *SO* when the *RPP* connects to the *Transmission System*.

(2) The DNP3 protocol shall be implemented on the Gateway for communication between the *RPP* and *NSP* when the *RPP* connects to the *Distribution System*.

13.4.2.2 SCADA Protocols within the RPP system

(1) The *RPP generator* may select protocols for communication within the *RPP* plant control system which shall use GPS based time stamping for all digital signals at the Intelligent Electronic Device (IED) or plant controller level (if IED is not applicable) which will be propagated to the SCADA master station with date and time to 1 millisecond accuracy in the defined protocol as per 13.4.2.1

(2) The *RPP generator* shall ensure that the data propagated to the SCADA master station has the data quality flags set by the IED or plant controller (if IED is not applicable) from which the data originated.

13.4.3 Telecommunication Interface Requirements

(1) The telecommunication interface to the *NSP* or *SO* may vary from substation to substation of the *NSP*. The *RPP generator* shall provide the equipment to interface with the *NSP* telecommunication infrastructure.

(2) The preferred telecommunication interface is X.21. Other telecommunication interfaces if necessary shall be agreed upon by the parties.

(3) The communication link from the *RPP* up to the *NSP's* telecommunication interface shall be owned and maintained by the *RPP generator*.

13.4.4 Operational Communication

(1) Each *RPP generator* shall be responsible for providing the following to the *NSP* or *SO* for operational purposes:

- (a) one South African landline contact number,
- (b) one mobile telephone contact number and
- (c) one email address that shall be continuously attended to and responded by authorised personnel within 5 minutes.

13.5 Data Communications Requirements

(1) The necessary communications links, communications protocol and the requirement for analogue or digital signals shall be specified by the *SO* as appropriate before a *connection agreement* is signed between the *RPP Generator* and the *Distributor* or *TNSP*.

(2) Active Power Curtailment or Voltage Regulation facilities at the *RPP* shall be tested once a month. It is essential that facilities exist to allow the testing of the functionality without tripping the actual equipment.

(3) Test procedures shall involve a setpoint/raise or lower command being sent by the *NSP/SO*, the setpoint feedback observed and the *RPP* responding to the setpoint. This shall be followed by return to normal setpoint/raise or lower command being sent to the *RPP* and the appropriate responses being measured

(4) Where signals or indications required to be provided by the *RPP generator* become unavailable or do not comply with applicable standards due to failure of the *RPP* equipment or any other reason under the control of the *RPP*, the *RPP generator* shall restore or correct the signals and/or indications within 24 hours

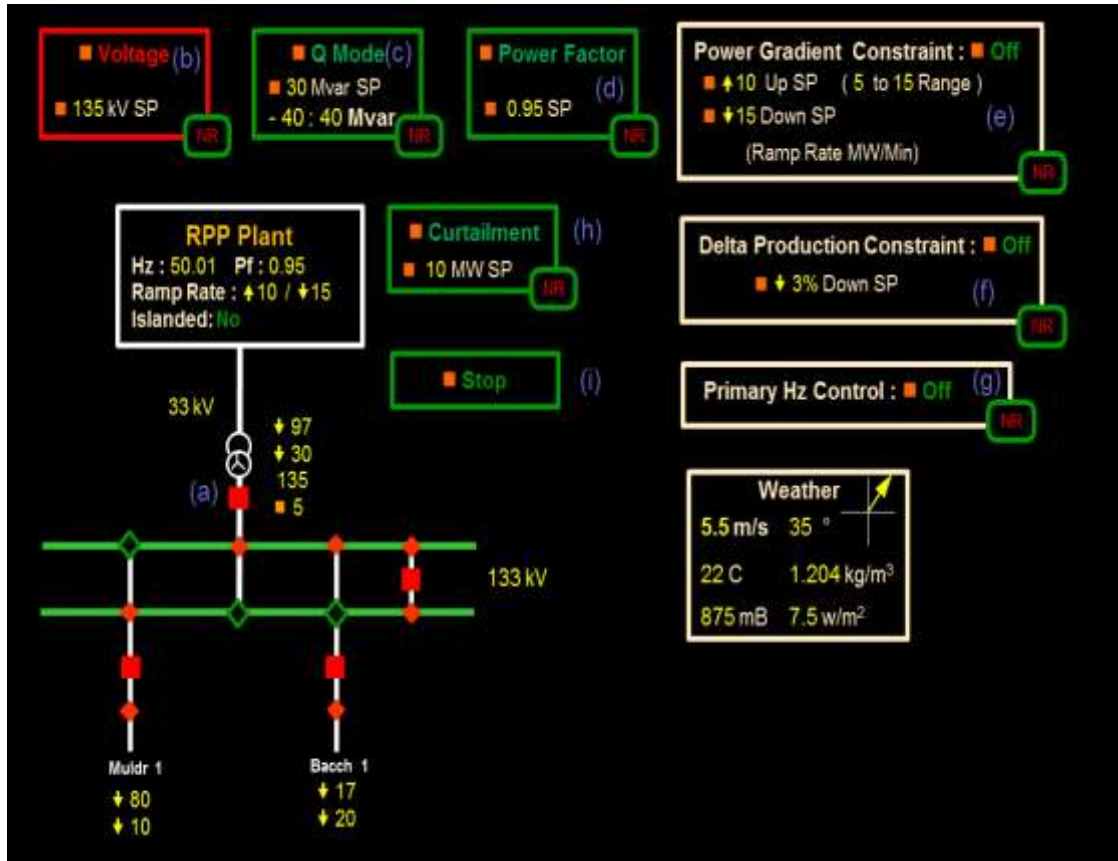


Figure 16: Example of one line Human Machine Interface layout

14. Testing and Compliance Monitoring

- (1) All *RPP generators* shall demonstrate compliance to all applicable requirements specified in this grid connection code and any other applicable code or standard approved by NERSA, as applicable, before being allowed to connect to the *DS* or the *TS* and operate commercially.
- (2) The *RPP generator* shall review, and confirm to the *SO and NERSA*, compliance by the *RPP* with every requirements of this code.
- (3) The *RPP generator* shall conduct tests or studies to demonstrate that the *RPP* complies with each of the requirements of this code.
- (4) The *RPP generator* shall continuously monitor its compliance in all material respects with all the connection conditions of this code.
- (5) Each *RPP generator* shall submit to the *SO* a detailed test procedure, emphasising system impact, for each relevant part of this code prior to every test.

(6) If *RPP generator* determines, from tests or otherwise, that the *RPP* is not complying with one or more sections of this code, then the *RPP Generator* shall (within 1 hour of being aware):

- (a) notify the *SO* of that fact,
- (b) advise the *SO* of the remedial steps it proposes to take to ensure that the relevant *RPP* can comply with this code and the proposed timetable for implementing those steps,
- (c) diligently take such remedial action to ensure that the relevant *RPP* can comply with this code; the *RPP generator* shall regularly report in writing to the *SO* on its progress in implementing the remedial action, and
- (d) after taking remedial action as described above, demonstrate to the reasonable satisfaction of the *SO* that the relevant *RPP* is then complying with this code.

(7) The *SO* may issue an instruction requiring the *RPP generator* to carry out a test to demonstrate that the relevant *RPP* complies with the code requirements. A *RPP generator* may not refuse such an instruction, provided it is issued timeously and there are reasonable grounds for suspecting non-compliance.

(8) The *RPP generator* shall keep records relating to the compliance of the *RPP* with each section of this grid connection code, or any other *code* applicable to that *RPP*, setting out such information that the *SO* reasonably requires for assessing power system performance, including actual *RPP* performance during abnormal conditions. Records shall be kept for a minimum of 5 years (unless otherwise specified in the *code*) commencing from the date the information was created.

15. Reporting to NERSA

(1) The *RPP generator* shall design the system and maintain records so that the following information can be provided to the *NERSA* on a monthly basis in an electronic spread sheet format:

- (a) Non-renewable/supplementary fuel used by the power plant as outlined under Supplementary Fuel Specification schedule of the PPA during the month.
- (b) Day ahead forecast output energy to the grid and hourly availability as specified in 13.4 and 13.5 above.
- (c) Actual hourly availability and output energy to the grid that occurred and the average primary resource for that hour (i.e. Wind speed for wind generators and solar radiation for solar generation)
- (d) Actual hourly electricity imports from all sources as applicable.
- (e) Direct monthly emissions per unit of electricity generated by the *RPP* (tCO₂/kWh).
- (f) Any curtailed energy during the month.

(2) These reports are to be submitted before the 15th of the following month to IPSreports@nersa.org.za

(3) These reports should also include details of incidents relating any unavailability of the network which prevented the RPP from generating and any incidents where their right to self-dispatch was impinged upon where the PPA gives them a right to self-dispatch.

16. Provision of Data and Electrical Dynamic Simulation Models

(1) The *SO*, *Distributors* and *TNSPs* require suitable and accurate dynamic models, in the template specified by the requesting party applying for a connection to the *DS* or *TS*, in order to assess reliably the impact of the *RPP* proposed installation on the dynamic performance and security and stability of the power system.

(2) The required dynamic models must operate under *RMS* and *EMT* simulation to replicate the performance of the *RPP* facility or individual units for analysis of the following network aspects:

- (a) *RPP* impact on network voltage stability
- (b) *RPP* impact on *QOS* at *POC*
- (c) *RPP* switching transients impact on network performance
- (d) *RPP* impact on breakers *TRV* (Transient Recovery Voltage)
- (e) *RPP* impact on network insulation co-ordination requirements
- (f) *RPP* impact on network protection co-ordination
- (g) *RPP FRT* (Fault Ride Through) capability for different types of faults and positions
- (h) *RPP* response to various system phenomena such as:
 - (i) switching on the network
 - (ii) power swings
 - (iii) small signal instabilities

(3) Generic instead of type tested *EMT* models can be accepted on condition that they represent *RPP* performance with frequency spectrum from 0 to 1kHz with accuracy level as indicated in Appendix 14.

(4) *EMT* models must include all parameters required for *EMT* simulations such as positive, negative and zero sequence impedances for all elements, magnetising curves, losses and tap changer data for transformers as well as positions of surge arresters and their *V/I* characteristics. *EMT* models must also include all protection and control functions of the plant.

(5) *RPP* data exchange shall be a time-based process:

(a) **First stage** (during the application for connection)

(i) The following information shall be submitted by the *RPP generator* to the *SO and Distributor or TNSP*, as applicable:

- Physical location of the *RPP* (including the GPS coordinates)
- Site Plan
- Number of wind turbines or *units* to be connected
- MW output per turbine or *unit*
- Initial phase MW value
- Final phase MW value and timelines
- Any other information that the service provider may reasonably require

(ii) For the detailed *RPP* design, the *NSP* shall make available to the *RPP* generator or its agent at least the following information:

- *Point of Connection* and the *Point of Common Coupling* including the nominal voltages,
- Expected fault levels
- The *network service provider's* connection between the *Point of connection* and the *RPP*,
- The busbar layout of the *PCC* and *POC* substations,
- The portion of the network service provider's grid that will allow accurate and sufficient studies to design the *RPP* to meet the Grid Code. This information shall include:
 - Positive and zero sequence parameters of the relevant network service provider's transmission and distribution, transformers, reactors, capacitors and other relevant equipment,
 - The connection of the various lines transformers, reactors and capacitors etc.

(b) **Second stage** (after detailed *RPP* designs have been completed but before commissioning the *RPP*).

(i) During this stage, the *RPP generator* shall provide information on:

- Selected *RPP* technology data.
- Fault ride through capability and harmonic studies test report
- Generic test model and dynamic modelling data per wind turbine or unit as from the type approval and tests result

(c) **Third stage** (after commissioning and optimisation of the *RPP*)

(i) During this stage, the *RPP generator* is compelled to provide information on:

- A validated *RPP* electrical dynamic simulation model using commissioning test data and measurements
- Test measurement data in the format agreed between the *RPP generator* and the *Distributor*, *NTC* or *SO*, as applicable.

(6) The dynamic modelling data shall be provided in a format as may be agreed between the *RPP generator* and the *Distributor*, *NTC* or *SO*, as applicable.

(7) In addition, the *RPP Generator* shall provide the *SO* with operational data as prescribed in **Appendix 8**.

Appendices

Appendix 1 – Wind

A1.1 High Wind Curtailment

(1) It shall be possible to continuously downward regulate the active power supplied by the *RPP* to an arbitrary value in the interval from 100% to at least 40% of the rated power. When downward regulation is performed, the shutting-down of individual wind turbine *generator* systems is allowed so that the load characteristic is followed as well as possible.

(2) The wind power plant shall stay connected to the TS and DS at average wind speeds below a predefined cut-out wind speed. The wind power plant shall declare the cut-out wind speed to the SO, based on the wind speed measured as an average value over a 10-minute period. To prevent instability in the TS and DS, the wind power plant shall be equipped with an automatic downward regulation function making it possible to avoid a temporary interruption of the active power production at wind speeds close to the cut-out wind speed.

(3) Downward regulation shall be performed as continuous or discrete regulation. Discrete regulation shall have a step size of maximum 25% of the rated power within the hatched area shown in Figure A1.1. When downward regulation is being performed, the shutting-down of individual wind turbine generator systems is allowed. The downward regulation band shall be agreed with the NSP upon commissioning of the wind power plant.

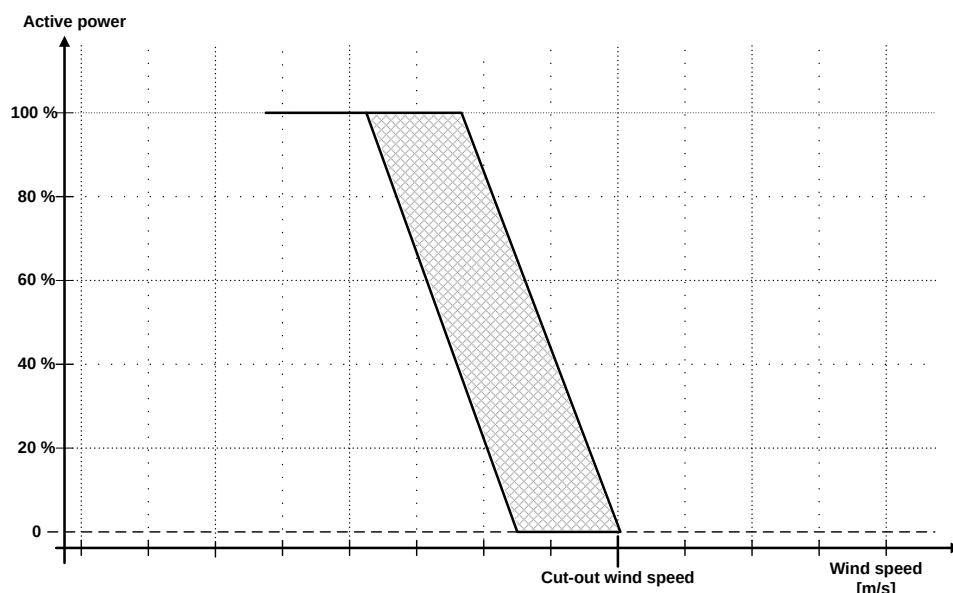


Figure A1-1: Downward regulation of active power at high renewable speeds

Appendix 2 – Photovoltaic

No special requirements for solar PV except the general requirement specified in this code.

Appendix 3 – Concentrated Solar Power

Table A3.1: Special requirements for solar CSP on Active Power Constraint Functions

Response Time [s]	Maximum 10 sec
Load(Upward) Ramp Rate [MW/min] - minimum	1.5% of P _{max} /min
De-load (Downward Ramp Rate [MW/min] - minimum	2.0% of P _{max} /min
Achievable accuracy limit	±3% of P _{max}

Appendix 4 – Small Hydro

Table A4.1: Special requirements for small Hydro on Active Power Constraint Functions

Response Time [s]	Maximum 10 sec
Load(Upward) Ramp Rate [MW/min] - minimum	10% of Pmax/min
De-load (Downward Ramp Rate [MW/min] - minimum	10% of Pmax/min
Achievable accuracy limit	±7.5% of Pmax

Appendix 5 – Landfill Gas

Table A5.1: Special requirements for Landfill Gas on Active Power Constraint Functions

Response Time [s]	Maximum 10 sec
Load(Upward) Ramp Rate [MW/min] - minimum	10% of Pmax/min
De-load (Downward Ramp Rate [MW/min] - minimum	10% of Pmax/min
Achievable accuracy limit	±8% of Pmax

Appendix 6 – Biomass

Table A6.1: Special requirements for Biomass on Active Power Constraint Functions

Response Time [s]	Maximum 10 sec
Load(Upward) Ramp Rate [MW/min] - minimum	10% of Pmax/min
De-load (Downward Ramp Rate [MW/min] - minimum	10% of Pmax/min
Achievable accuracy limit	±8% of Pmax

Appendix 7 – Biogas

Table A7.1: Special requirements for Biogas on Active Power Constraint Functions

Response Time [s]	Maximum 10 sec
Load(Upward) Ramp Rate [MW/min] - minimum	10% of P _{max} /min
De-load (Downward Ramp Rate [MW/min] - minimum	10% of P _{max} /min
Achievable accuracy limit	±8% of P _{max}

Appendix 8 – Documentation

A8.1 Master Data

Description	Text
Identification:	
Name of <i>electricity supply undertaking</i>	
Plant name	
ID number	
Planned commissioning	
Technical data:	
Manufacturer	
Type designation (model)	
Type approval	
Approval authority	
Installed kW (<i>rated power</i>)	
Cos ϕ (<i>rated power</i>)	
Cos ϕ (<i>20% rated power</i>)	
Cos ϕ (<i>no load</i>)	
3-phase short-circuit current immediately in front of the <i>power plant</i> (RMS)	
<i>Point of connection</i>	
Voltage level	

Description	Text
Plant address:	
Contact person (technical)	
Address1	
House number	
Letter	
Postal code	
BBR municipality	
X/Y coordinates	
Title number	
Owners' association on titled land	
Owner:	
C ID number	
Company name	
Contact person (administrative)	
Address1	
House number	
Letter	
Floor	
To the right/left	
Postal code	
Email address	

A8.2 Technical Documentation

A8.2.1 Step-Up Transformer

Description	Value
Make	
Type	
Comments	

Description	Symbol	Unit	Value
Nominal apparent power (1 p.u.)	S_n	MVA	
Nominal primary voltage (1 p.u.)	U_p	kV	
Nominal secondary voltage	U_s	kV	
Coupling designation, eg Dyn11	-	-	
Step switch location	-	-	Primary side ☐ Secondary side ☐
Step switch, additional voltage per step	du_{tp}	%/trin	
Step switch, phase angle of additional voltage per step:	$\phi_{i_{tp}}$	degree/step	
Step switch, lowest position	n_{tpmin}	-	
Step switch, highest position	n_{tpmax}	-	
Step switch, neutral position	n_{tp0}	-	
Short-circuit voltage, synchronous	u_k	%	
Copper loss	P_{cu}	kW	
Short-circuit voltage, zero system	u_{k0}	%	
Resistive short-circuit voltage, zero-sequence system	u_{kr0}	%	
No-load current	I_0	%	
No-load loss	P_0	%	

A8.2.2 Single Line Diagram Representation

(1) This applies to all *RPPs* of category B and C. The *SO*, *NSP* or local *network operator* may request that a single-line diagram representation be provided for *RPPs* of category A.

(2) A single-line diagram representation of the plant shall be created, with indication of *POC*, metering points, including settlement metering, limits of ownership and operational supervisor limits/limits of liability. In addition, the type designation for the switchgear used shall be stated so as to make it possible to identify the correct connection terminals.

(3) In instances when a single-line diagram representation is included in the grid use agreement between *RPP Generator* and *SO*, the *grid connection agreement* can be enclosed as documentation.

A8.2.3 PQ Diagram

(1) This applies to all *RPPs* of category B and C. The *SO*, *NSP* or local *network operator* may also request that a PQ diagram representation be provided for *RPPs* of category A.

A8.2.4 Short-circuit data

(1) **Application:** This applies to all *RPPs* of category B and C.

(2) For the purposes of static calculations, the *RPP generator* shall provide short-circuit data at different voltage drops in the *TS* and *DS*, using the requirements in section 0 as starting point. Voltage drops in connection with faults shall be stated with a short-circuit time of 150ms.

(3) The fault sequence is logged through simulation in the 0-500ms time interval. Short-circuit data shall be provided in the following tables.

(4) Assumptions for the calculation of short-circuit data:

- All *generator* in the *RPP* are connected
- The *RPP* produces *rated power*
- Current values are calculated in the *POC*
- Symmetrical voltage drop is indicated as a percentage (dU) of the output voltage
- The *RPP's* protection functions/settings are included
- The *short-circuit power* in the *POC* is set to $10 \times P_n$ with an X/R of 10
- 50 Hz component of the active current, I_{active}
- 50 Hz component of the reactive current, $I_{reactive}$
- Total current incl. DC component and harmonics, I_{peak}

dU=20%			
Time [ms]	I_{active} [A]	$I_{reactive}$ [A]	I_{peak} [A]
0			
5			
10			
20			
50			
100			
150			
200			
300			
500			

dU=30%			
Time [ms]	I_{active} [A]	$I_{reactive}$ [A]	I_{peak} [A]
0			
5			

10			
20			
50			
100			
150			
200			
300			
500			

dU=50%			
Time [ms]	I_{active} [A]	I_{reactive} [A]	I_{peak} [A]
0			
5			
10			
20			
50			
100			
150			
200			
300			
500			

dU=80%			
Time [ms]	I_{active} [A]	I_{reactive} [A]	I_{peak} [A]
0			
5			
10			
20			
50			
100			
150			
200			
300			
500			

Appendix 9 – Compliance test specifications

A9.1 Introduction

This section specifies the procedures to be followed in carrying out testing to verify compliance with this *Code*.

A9.2 Test procedures

A9.2.1 - RPP protection function verification		
Parameter	Reference	Description
Protection function and settings	Section 10	APPLICABILITY AND FREQUENCY All new <i>RPPs</i> coming on line or at which major refurbishment or upgrades of protection systems have taken place. Routine review: All <i>generators</i> to confirm compliance every six years. PURPOSE To ensure that the relevant protection functions in the <i>RPP</i> are coordinated and aligned with the system requirements. PROCEDURE 1. Establish the system protection function and associated trip level requirements from the <i>SO</i> or relevant <i>NSP</i>. 2. Derive protection functions and settings that match the <i>RPP</i> and system requirements. 3. Confirm the stability of each protection function for all relevant system conditions. 4. Document the details of the trip levels and stability calculations for each protection function. 5. Convert protection tripping levels for each protection function into a per <i>unit</i> base. 6. Consolidate all settings in a per <i>unit</i> base for all protection functions in one document. 7. Derive actual relay dial setting details and document the relay setting sheet for all protection functions. 8. Document the position of each protection function on one single line diagram of the generating <i>unit</i> and associated connections. 9. Document the tripping functions for each tripping function on one tripping logic diagram. 10. Consolidate detail setting calculations, per unit setting sheets, relay setting sheets, plant base information on which the settings are based, tripping logic diagram, protection function single line diagram and relevant protection relay manufacturers' information into one document. 11. Submit to the <i>SO</i> or relevant <i>NSP</i> for its acceptance and update. Review: 1. Review Items 1 to 10 above. 2. Submit to the <i>SO</i> or relevant <i>NSP</i> for its acceptance and update. 3. Provide the <i>SO</i> or relevant <i>NSP</i> with one original master copy and one working copy. ACCEPTANCE CRITERIA All protection functions are set to meet the necessary protection

		requirements of the <i>RPP</i> with a minimal margin, optimal fault clearing times and maximum plant availability. Submit a report to the <i>SO</i> or relevant <i>NSP</i> one month after commissioning and six-yearly for routine tests.
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A.9.2.2 - RPP protection integrity verification		
Parameter	Reference	Description
Protection integrity	Section 10	APPLICABILITY AND FREQUENCY All new <i>RPPs</i> coming on line and all other <i>power stations</i> after major works of refurbishment of protection or related plant. Also, when modification or work has been done to the protection, items 2 to 5 must be carried out. This may, however, be limited to the areas worked on or modified. Routine review: All <i>RPPs</i> on: item 1 below: Review and confirm every 6 years Item 2, and 3 below: at least every 12 years. PURPOSE To confirm that the protection has been wired and functions according to the specifications. PROCEDURE 1. Apply final settings as per agreed documentation to all protection functions. 2. With the <i>unit</i> off load and de-energized, inject appropriate signals into every protection function and confirm correct operation and correct calibration. Document all protection function operations. 3. Carry out trip testing of all protection functions, from origin (e.g. Buchholz relay) to all tripping output devices (e.g. HV breaker). Document all trip test responses. 4. Apply short-circuits at all relevant protection zones and with <i>generator</i> at nominal speed excite <i>generator</i> slowly, record currents at all relevant protection functions and confirm correct operation of all relevant protection functions. Document all readings and responses. Remove all short-circuits. 5. With the <i>RPP</i> at nominal production. Confirm correct operation and correct calibration of all protection functions. Document all readings and responses. Review: Submit to the <i>SO</i> or relevant <i>NSP</i> for its acceptance and update. ACCEPTANCE CRITERIA All protection functions are fully operational and operate to required levels within the relay <i>OEM</i> allowable tolerances. Measuring instrumentation used shall be sufficiently accurate and calibrated to a traceable standard. Submit a report to the <i>SO</i> or relevant <i>NSP</i> one month after test.

A9.2.3 - RPP active power control capability verification		
Parameter	Reference	Description
Active power control function and operational range	Section 11 depending on category	APPLICABILITY All new <i>RPPs</i> coming on line and after major modifications or refurbishment of related plant components or functionality. Routine test/reviews: Confirm compliance every 6 years. PURPOSE To confirm that the active power control capability specified is met. PROCEDURE The following tests shall be performed within an active power level range of at least 0.2p.u.or higher 1. The <i>RPP</i> will be required to regulate the active power to a set of specific setpoints within the design margins. 2. The <i>RPP</i> will be required to obtain a set of active power setpoints within the design margins with minimum two different gradients for ramping up and two different gradients for ramping down. 3. The <i>RPP</i> will be required to maintain as a minimum two different set levels of spinning reserve within the design margins. 4. The <i>RPP</i> will be required to operate as a minimum to limit active power output according to two different absolute power constraint set levels within the design margins. 5. The <i>RPP</i> will be required to verify operation according to as a minimum two different parameter sets for a frequency response curve within the design margins. ACCEPTANCE CRITERIA 1. The <i>RPP</i> shall maintain the set output level within $\pm 2\%$ of the capability registered with the <i>SO, NSP or another network operator</i> for at least one hour. 2. The <i>RPP</i> shall demonstrate ramp rates with precision within $\pm 2\%$ of the capability registered with the <i>SO, NSP or another network operator</i> for ramp up and down. 3. The <i>RPP</i> shall maintain a spinning reserve set level within $\pm 2\%$ of the capability registered with the <i>SO, NSP or another network operator</i> for at least one hour. 4. The <i>RPP</i> shall maintain an absolute power constraint set level within $\pm 2\%$ of the capability registered with the <i>System Operator</i> for at least one hour. 5. The <i>RPP</i> shall demonstrate that the requested frequency response curves can be obtained. Submit a report to the <i>SO, NSP or another network operator</i> one month after the test.

A9.2.4 - RPP reactive power control capability verification		
Parameter	Reference	Description
Reactive power control function and operational range	Sections 7 and 8 depending on category	APPLICABILITY All new <i>RPPs</i> coming on line and after major modifications or refurbishment of related plant components or functionality. Routine test/reviews: Confirm compliance every 6 years. PURPOSE To confirm that the reactive power control capability specified is met. PROCEDURE The following tests shall be performed within a minimum active power level range of at least 0.2 p.u. or higher 1. The <i>RPP</i> will be required to regulate the voltage at the PCC to a set level within the design margins. 2. The <i>RPP</i> will be required to provide a fixed Q to a set level within the design margins. 3. The <i>RPP</i> will be required to obtain a fixed PF within the design margins. ACCEPTANCE CRITERIA 1. The <i>RPP</i> shall maintain the set voltage within $\pm 5\%$ of the capability registered with the <i>SO</i>, <i>NSP</i> or another <i>network operator</i> for at least one hour. 2. The <i>RPP</i> shall maintain the set Q within $\pm 2\%$ of the capability registered with the <i>SO</i>, <i>NSP</i> or another <i>network operator</i> for at least one hour. 3. The <i>RPP</i> shall maintain the set PF within $\pm 2\%$ of the capability registered with the <i>SO</i>, <i>NSP</i> or another <i>network operator</i> for at least one hour. Submit a report to the <i>SO</i>, <i>NSP</i> or another <i>network operator</i> one month after the test.

A9.2.5 - RPP power quality calculations		
Parameter	Reference	Description
Power quality calculations for: 1. Rapid voltage changes 2. Flicker 3. Harmonics 4. Inter-harmonics 5. High frequency disturbances	Section 9 depending on category	APPLICABILITY All new <i>RPPs</i> coming on line and after major modifications or refurbishment of related plant components or functionality. Routine test/reviews: Confirm compliance every 6 years. PURPOSE To confirm that the limits for all power quality parameters specified is met. PROCEDURE The following tests shall be calculated within a minimum active power level range from 0.2p.u. to 1.0p.u. 1. Calculate the levels for rapid voltage changes are within the limits specified over the full operational range. 2. Calculate the flicker levels are within the limits specified over the full operational range. 3. Calculate the harmonics are within the limits specified over the full operational range. 4. Calculate the interharmonics are within the limits specified over the full operational range. 5. Calculate the disturbances higher than 2 Hz are within the limits specified over the full operational range. ACCEPTANCE CRITERIA 1. The calculations shall demonstrate that the levels for rapid voltage changes are within the limits specified over the full operational range. 2. The calculations shall demonstrate that the flicker levels are within the limits specified over the full operational range. 3. The calculations shall demonstrate that the harmonics are within the limits specified over the full operational range. 4. The calculations shall demonstrate that the interharmonics are within the limits specified over the full operational range. 5. The calculations shall demonstrate that the disturbances higher than 2 Hz are within the limits specified over the full operational range Submit a report to the <i>System Operator</i> one month after the test.

A.9.2.6 - RPP fault ride through simulations		
Parameter	Reference	Description
Simulations of fault ride through voltage droops and peaks.	Section 5.2.1 for category B and C	APPLICABILITY All new <i>RPPs</i> coming on line and after major modifications or refurbishment of related plant components or functionality. Routine test/reviews: None. PURPOSE To confirm that the limits for all power quality parameters specified is met. PROCEDURE By applying the electrical simulation model for the entire RPP it shall be demonstrated that the RPP performs to the specifications. 1. Area A - the RPP shall stay connected to the network and uphold normal production. 2. Area B - the RPP shall stay connected to the network. The RPP shall provide maximum voltage support by supplying a controlled amount of reactive power within the design framework offered by the technology, see Figure 5. 3. Area C - the RPP is allowed to disconnect. 4. Area D - the RPP shall stay connected. The RPP shall provide maximum voltage support by absorbing a controlled amount of reactive power within the design framework offered by the technology, see Figure 5. ACCEPTANCE CRITERIA 1. The dynamic simulations shall demonstrate that the RPP fulfils the requirements specified. Submit a report to the <i>SO, NSP or another network operator</i> three month after the commission.

Appendix 10 – Signal List Description

(1) This appendix provides detail on the signals listed in Chapter 13.

A10.1 The Concept of a Bay

(1) Figure A10-1 is an example of a typical Transmission connected *RPP* electrical interface diagram.

(2) In this figure the Breakers are shown as squares and the line isolators and busbar isolators are shown as diamonds. In this case a solid red symbol indicates that the device is closed and a hollow green symbol indicates that the device is open or tripped.

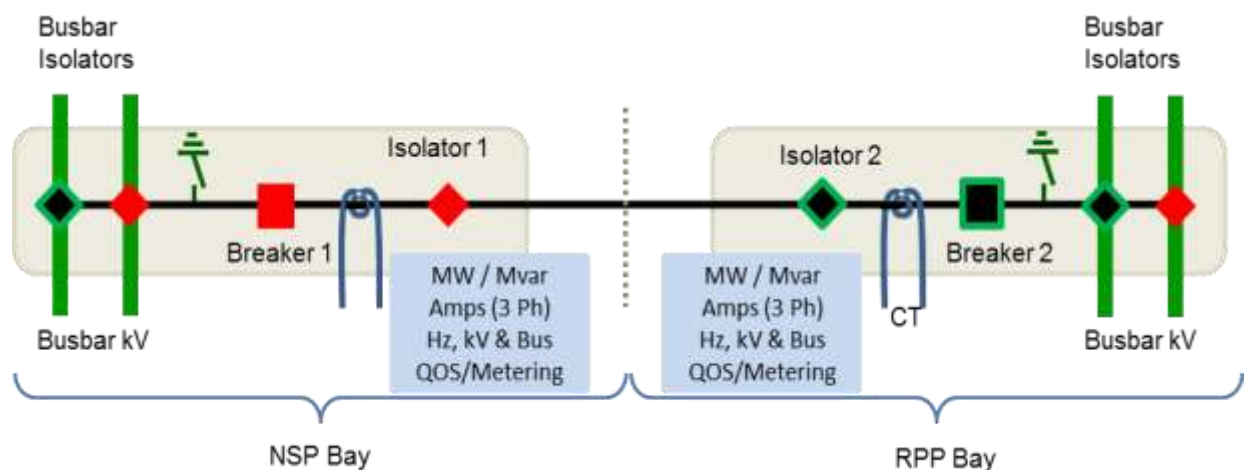


Figure A10-1: Simplified View of an *NSP-RPP* Electrical Interface

(3) The indications shown in Table A10-2 below refer to the Bay as a unit and the alarms associated with it.

(4) In all cases, each Bay shall have a Supervisory Switch associated with it. All operating on the Bay under these conditions shall be performed locally from the local control panel by the substation personnel.

(5) The *RPP* Breaker (Breaker 2 in Figure A10-1) shall not normally be tripped directly by the *NSP* or *SO*. The *NSP/SO* will use the 'Stop' command to reduce the plant output to zero when required.

(6) In the case of alarms, the alarm indications shall be reset when the plant returns to normal operating conditions.

A10.2 Double-bit Indications

(1) Breakers, isolators, and earth switches shall be indicated by means of two sensors/contacts, one bit that indicates all poles fully opened and another bit that indicates all poles fully closed.

(2) The reporting of double-bit indications is protocol dependent.

(3) Double-bit indications being sent to the *NSP* or *SO* are required to adhere to the convention described in the table below.

Table A10-1: Double-bit indications

Bit values per index number n+1 n	Meaning for Breaker and Isolator states	Meaning for Supervisory Switch states	Meaning for Start/Stop function states
0 0	In transit(Intermediate)	In transit(Intermediate)	In transit(Intermediate)
0 1	Opened	Off (inactive)	Stopped
1 0	Closed	On (active)	Started
1 1	Invalid(bad)	Invalid(bad)	Invalid(bad)

Description of table column headings used in this section:

Index	Input	State	Command	Action	Type	Explanation	Cross Reference to control/input
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Index: Index number of digital inputs, digital outputs, analogue inputs and analogue outputs

Input: Digital or analogue input as referenced from the SCADA system

State: Applicable to digital signals. Each binary status point can be mapped to one or two binary bits. In the case of a breaker or isolator, the state is reported via two bits. In the case of single-bit alarm points, only one bit is used to report the state

Command: Digital or analogue output as referenced from the SCADA system

Action: Action associated with a digital or analogue command

Type: Indicates the number of bits used to report the state of the point in question

Explanation: Explanation of the signal

Cross Reference to control/input: Digital and analogue controls are cross referenced to the corresponding digital and analogue indications and vice-versa

A10.3 Digital Input Signals From the RPP

A10.3.1 Bay wide binary indications

(1) The bay-wide digital indications for each bay as required by the *NSP/SO* are as per Table A10-2 below. All Breakers and Isolators between the HV side of the RPP transformer and the *POC* should be telemetered.

Table A10-2: Bay-wide binary indications

Index	Input	State	Type	Explanation	Cross reference to control
DI 1	Supervisory Switch	In transit-00 Off-01 On-10 Invalid-11	Double-bit	When Off this switch prevents supervisory controls from being transferred from the Gateway to the Bay devices.	
DI 2	Earth switch	In transit-00 Opened-01 Closed-10 Invalid-11	Double-bit	Status of the earth switch	
DI 3	Earth Applied	Yes-1 No-0	Single-bit	This indication must be set to 0 only when all earth switches in the bay are open.	
DI 4	Plant tripped on loss of grid protection or islanding protection	Yes-1 No-0	Single-bit	This indication must be set to 1 when the RPP detects loss of grid. Not required from a plant which is allowed to island by the <i>NSP/SO</i>	
DI 5	Plant islanded	Yes-1 No-0	Single-bit	This indication must be set to 1 when the RPP is islanded (Required from a plant which is allowed to island by the <i>NSP/SO</i>)	

DI 6	Plant Shutdown	Yes-1 No-0	Single-bit	This indication must be set to 1 when the RPP initiates a shutdown and stays set to 1 as long as the RPP is shutdown. Not set to 1 when a stop or trip command is issued from the NSP/SO	
DI 7	Breaker State	In transit-00 Opened-01 Closed-10 Invalid-11	Double-bit	Circuit Breaker State	DO 1
DI 8	Isolator State	In transit-00 Opened-01 Closed-10 Invalid-11	Double-bit	Isolator State	

A10.3.2 Breaker and auxiliary alarms

(1) If there is a supervisory command associated with a breaker, the following alarms should be provided.

Table A10-3(a): Breaker health and Auxiliary alarms

Index	Input	State	Type	Explanation
DI 9	Breaker Fail	Operated-1 Reset-0	Single-bit	Inability of the breaker to trip on operation of protection system
DI 10	Breaker Unhealthy	Operated-1 Reset-0	Single-bit	Status indicating Breaker's inability to operate and requires Operator investigation e.g. SF6 pressure low, springs not wound, limit switches not closed, etc.
DI 11	Breaker Fail Trip	Operated-1 Reset-0	Single-bit	Bus Strip is a consequence of breaker failure operation where the breaker fail output is routed via the buszone protection to trip all breakers connected to the relevant bus zone.
DI 12	Protection Operation	Operated-1 Reset-0	Single-bit	If any bay protection, main or back-up has operated then this indication should be set to '1'.

DI 13	Protection Unhealthy	Operated-1 Reset-0	Single-bit	Any condition implying that the protection system is not capable of performing its intended function. This is triggered by: 1) failed indication from any IED self-monitoring internal alarm (i.e. watchdog); 2) any D.C. fail alarm; 3) Teleprotection isolate switch (TPIS) selected to "OFF"; 4) Test Normal Switch (TNS) not selected to "Normal"; 5) VT supply fail;
DI 14	Pole discrepancy	Operated-1 Reset-0	Single-bit	One of the three Breaker poles is not in the same state as the other two.
DI 15	Breaker SF6 Gas Urgent	Operated-1 Reset-0	Single-bit	The Sulphur Hexafluoride insulation gas pressure is too low to allow the Breaker to operate safely. It blocks any operations – trip or close. In case of primary fault on the system, the breaker fail will operate and initiate Bus Strip.
DI 16	Breaker SF6 Gas Non-Urgent	Operated-1 Reset-0	Single-bit	The Sulphur Hexafluoride insulation gas pressure is below normal, the breaker can still interrupt fault current but it may be prevented from closing.
DI 17	Scheme AC Supply Fail	Operated-1 Reset-0	Single-bit	Loss of AC supply for the particular breaker
DI 18	Scheme DC Supply Fail	Operated-1 Reset-0	Single-bit	Loss of DC supply for the particular breaker

Table A10-3(b): Station auxiliary alarms

Index	Input	State	Type	Explanation
DI 19	Charger Mains Fail	Operated-1 Reset-0	Single-bit	Loss of station AC supply
DI 20	Station DC Fail	Operated-1 Reset-0	Single-bit	Loss of station DC supply. Battery Voltage Low, Battery Open Circuit, Battery Fuse Blown, Load Voltage Low

A10.4 Analogue Input Signals

(1) The analogue indications for each Bay are listed below:

Table A10-4 Analogue indications for each Bay

Index	Input	Explanation
AI 1	Active power sent out	Measured summated three phase active power export or import at the POC Export/Produce (+) Import/Absorb (-)
AI 2	Reactive power sent out	Measured summated three phase Reactive Power sent out at the POC Export/Produce (+) Import/Absorb (-)
AI 3	Current sent out	Red, White and Blue phase currents at the POC
AI 4	Actual ramp rate	Active up Power Ramp rate of the entire facility.
AI 5	Power Factor	Power Factor of the RPP
AI 6	Voltage sent out	Voltage at the POC
AI 7	Frequency	Frequency of the generated energy (only required where Islanding is allowed)

A10.5 Meteorological Data

Table A10-5: RPP meteorological inputs

Index	Input	Explanation
AI 9	Wind Speed	Within 75% of the hub height) – measured signal in meters/second (for WPP only)
AI 10	Wind Direction	Within 75% of the hub height) – measured signal in degrees from true north (0-359) (for WPP only)
AI 11	Air temperature	Measured signal in degrees centigrade (-20.0 to 50.0);
AI 12	Air pressure	Measured signal in millibar (800 to 1400).
AI 13	Air density	Measured signal in kg/m ³ (for WPP only)
AI 14	Direct Solar radiation	Measured signal in watts/m ² (for solar plants)
AI 15	Diffuse solar radiation	Measured signal in watts/m ² (for solar plants)
AI 16	Humidity	Measured signal in Percentage

A10.6 Command Function Requirements

- (1) There are typically three types of controls:
- (a) Normal device state change – Trip/Close or On/Off or Stop/Start etc.
 - (b) Setpoint controls – Analogue Output commands.
 - (c) Digital Raise/Lower commands

A10.6.1 Breaker trip

- (1) The NSP/SO shall be able to send a trip signal to the RPP Breaker/s at the HV side of the POC.

Table A10-6: Breaker Command Signal

Index	Command	Action	Explanation	Cross Reference to indication
DO 1	Breaker Trip	Trip (no Close)	Isolate the RPP from the Grid.	DI 7

A10.6.2 Frequency Response System Settings

- (1) Primary frequency control services are only provided if the RPP generator has entered into an ancillary services agreement with the SO.
- (2) The RPP generator shall make the following primary frequency control signals available via the Gateway:

Table A10-7: Primary frequency response command

Index	Command	Action	Explanation	Cross Reference to indication
DO 2	Frequency Control mode state	On/Off	Activate or deactivate Frequency Control Mode as requested by the NSP or SO	DI 21

Table A10-8: Primary frequency response indications

Index	Input	State	Type	Explanation	Cross Reference to control
DI 21	Frequency Control Mode State	On-1 Off-0	Single-bit	Will report a '1' state when active and '0' state when not active	DO 2
DI 22	Frequency Control mode	Not ready-1 Ready-0	Single-bit	This indication shall be set to 1 if frequency control cannot be done	

A10.6.3 Active Power Constraints

- (1) The following discussion pertains to the following constraint areas:
- (a) Absolute Production Constraint (Curtailment)

- (b) Delta Production Constraint (P-delta)
- (c) Power Gradient Constraint (Power Gradient)

A10.6.3.1 Absolute Production constraint

- (a) Curtailment to a set power output

- (1) The *NSP* or *SO* will send a 'Curtailment mode ON' command and then the setpoint.
- (2) Once the Curtailment setpoint is received, the *RPP* shall limit the total MW output to the value that is defined by the Curtailment setpoint. The *RPP* shall set the Curtailment in progress to 'ON' when it is moving to the requested setpoint.
- (3) When conditions in the power system allow, the *NSP/SO* will reset the 'Curtailment mode' state to 'OFF'. When the *RPP* detects the 'Curtailment mode' reset, it may resume its planned MW output.
- (4) It is essential that when Curtailment is initiated and cancelled, the times and the Curtailment setpoint values are captured and logged to ensure that disputes are minimised.

Table A10-9: Curtailment commands

Index	Command	Action	Explanation	Cross Reference to indication/analogue
DO 3	Curtailment mode state	On/Off	Activate or deactivate production curtailment in the event of system constraints.(also called absolute production constraint)	DI 23
AO 1	Curtailment setpoint	Setpoint command	Setpoint command to change the active power setpoint of the RPP	AI 17

Table A10-10: Curtailment indications

Index	Digital indications	State	Type	Explanation	Cross Reference to control
DI 23	Curtailment mode status	On-1 Off-0	Single-bit	Will report a '1' state when active and '0' state when not active	DO 3
DI 24	Curtailment in progress	Yes-1 No-0	Single-bit	Will be set to '1' while the facility is moving from the current value	

				to the curtailed value. Once the facility reaches the curtailment value, this bit will be reset.	
DI 25	Curtailment mode	Not ready-1 Ready-0	Single-bit	Will be set to '1' in the event of conditions at the plant preventing the plant from being curtailed. In the case of any not ready indication being detected, it is up to the RPP to correct the problem as soon as possible.	

Table A10-11: Curtailment analogue

Index	Input	Explanation	Cross Reference to control
AI 17	Curtailment Setpoint feedback	RPP echo response to a new Power setpoint issued by the NSP or SO.	AO 1

(b) Reduction to 0 MW (Stop command)

(1) Under extreme system conditions and when the curtailment option will take too long, provision has been made to send a 'Stop' command to the *RPP* to initiate a controlled shutdown of the installation at the negative (down) ramp rate setting.

(2) When this 'Stop' command is received by the *RPP* it should set the 'Curtailment Setpoint feedback' to zero, the 'Curtailment Mode' and 'Curtailment in Progress' indications to 'ON' and should ramp the installation to zero output at the negative ramp rate setting. Once generation has stopped, the 'Curtailment in Progress' indication should be set 'Off'.

(3) It is not expected that this command action would open the *RPP* Breaker but only reduce the output to zero. This will maintain supply to auxiliaries as required.

(4) When the 'START' command is issued, the *RPP* shall set the 'Curtailment Mode' to 'Off' and may ramp up generation to the scheduled generation output but no faster than the positive (up) ramp rate setting.

Table A10-12: Generation Stop/Start command

Index	Command	Action	Explanation	Cross Reference to indication
DO 4	Generation State	Start/Stop	By sending a START command, the NSP/SO should be able to start generation of the RPP and by sending a STOP command, the NSP/SO should be able to bring the RPP to a non-generating mode, but do not open the Breaker.	DI 26

Table A10-13: Generation Stop/Start indication

Index	Input	State	Type	Explanation	Cross Reference to
DI 26	Generation State	Shutdown-01 Generating-10	Double-bit	Will report a '10' state on receipt of a START command and '01' state for STOP command	DO 4

A10.6.3.2 Delta Production Constraint

(1) To activate this function the NSP/SO will send the 'ON' command to the 'P-Delta Mode' address.

(2) The NSP/SO will set the P-delta setpoint (percentage) as a 'setpoint' command. The RPP shall decrease the output by P-delta providing reserves for frequency control. The NSP/SO will reset the 'Delta Production Mode' to 'OFF' if the reserve functionality is not required any further.

(3) The 'P-delta Constraint mode not ready' shall be set to 1 whenever the Delta Production Constraint facility is not available.

Table A10-14: Delta production commands

Index	Command	Action	Explanation	Cross reference to indication / analogue
DO 5	P-delta constraint mode state	On/Off	Activate or deactivate delta production constraint. The RPP shall decrease its output by the set percentage of available active power to provide reserve for frequency control	DI 27
AO 2	P-delta Setpoint	Setpoint command	Setpoint command to change the P-delta setpoint (expressed in percentage).	AI 18

Table A10-15: Delta production binary indications

Index	Input	Start	Type	Explanation	Cross Reference to control
DI 27	P-delta constraint mode state	On-1 Off-0	Single-bit	Will report a '1' state when active and '0' state when not active	DO 5
DI 28	P-delta mode	Not ready-1 Ready-0	Single-bit	Will be set to '1' in the event of conditions at the plant preventing the plant from going into Delta Production Mode.	

Table A10-16: Delta production analogues

Index	Input	Explanation	Cross Reference to control
AI 18	P-delta setpoint feedback	RPP echo response to a new P-delta setpoint (percentage) issued by the NSP or SO	AO 2

A10.6.3.3 Power Gradient Constraint

(1) A Power Gradient Constraint is used to limit the maximum ramp rates by which the active power can be changed in the event of changes in primary renewable energy supply of the RPP.

(2) To implement the Power Gradient Constraint, the *NSP/SO* will send an 'ON' command to the 'Power gradient constraint mode' address and send setpoint commands to change ramp rate setpoints in MW/min of the *RPP* to new values within the limits specified by the *RPP*.

(3) The 'Power gradient mode not ready' shall be set to 1 by the *RPP* whenever ramp rate modifications cannot be done.

Table A10-17: Power gradient commands

Index	Command	Action	Explanation	Cross Reference to indication / analogue
DO 6	Power gradient constraint mode state	On/Off	Activate or deactivate power gradient constraint in the event of system constraints.	DI 29
AO 3	Positive Ramp Rate Setpoint	Setpoint command	Setpoint command to change the up ramp rate of the <i>RPP</i>	AI 19
AO 4	Negative Ramp Rate Setpoint	Setpoint command	Setpoint command to change the down ramp rate of the <i>RPP</i>	AI 20

Table A10-18: Power gradient indications

Index	Input	State	Type	Explanation	Cross Reference to control
DI 29	Power gradient constraint mode state	On-1 Off-0	Single-bit	Will report a '1' state when active and '0' state when not active	DO 6
DI 30	Power gradient mode	Not ready-1 Ready-0	Single-bit	Will report a '1' state when 'not ready' and '0' state when 'ready'	

Table A10-19: Power gradient analogues

Index	Input	Explanation	Cross Reference to control
AI 19	Up Ramp Rate setpoint feedback	<i>RPP</i> echo response to a new 'up' ramp rate setpoint issued by the <i>NSP</i> or <i>SO</i>	AO 3
AI 20	Down Ramp Rate setpoint feedback	<i>RPP</i> echo response to a new 'down' ramp rate setpoint issued by the <i>NSP</i> or <i>SO</i>	AO 4

A10.6.4 Reactive Power, Power factor and Voltage Control Functions

(1) The reactive power, power factor and voltage control functions are mutually exclusive, which means that only one of the three functions mentioned below can be active at a time. At least one of these functions must be active.

- (a) Reactive Power Control (Q control)
- (b) Power Factor control (PF Control)
- (c) Voltage control (V Control)

(2) The *NSP* or *SO* will select the relevant mode and change the setpoint. The *RPP* shall deactivate the other two modes. The setpoint feedback shall be updated within 2 seconds. The generator shall respond to the new setpoint within 30 seconds after receipt of an order to change to the new setpoint

A10.6.4.1 Reactive Power Control Mode

Table A10-20: Q mode commands

Index	Command	Action	Explanation	Cross Reference to indication/analogue
DO 7	Q control mode state	On (no OFF)	Activate Reactive Power Control Mode as requested by the NSP or SO	DI 31
AO 5	Q control setpoint	Setpoint command	Command to change the kvar or Mvar setpoint of the RPP. Producing vars (+), Absorbing vars (-)	AI 23
DO 8	Q control setpoint raise/lower	Raise-01 Lower-10	Raise/lower command can be used instead of setpoint command (AO 5)	AI 23

Table A10-21: Q mode binary indications

Index	Input	State	Type	Explanation	Cross Reference to control
DI 31	Q control mode status	On-1 (no OFF)	Single-bit	Will report a '1' state when active and '0' state when not active	DO 7
DI 32	Q control mode	Not ready-1 Ready-0	Single-bit	Will be set to '1' in the event of conditions at the plant preventing the plant from going into Q mode.	

Table A10-22: Q mode analogues

Index	Input	Type	Explanation	Cross Reference to control
AI 21	Q inductive Limit	Analogue	This value is the maximum inductive limit (-) based on plant available.	
AI 22	Q capacitive Limit	Analogue	This value is the maximum capacitive limit (+) based on plant available	
AI 23	Q Setpoint feedback	Analogue	This value is an indication of the Reactive Power setpoint issued by the NSP or SO	AO 5 or DO 8

A10.6.4.2 Power Factor Control Mode

Table A10-23: Power factor commands

Index	Command	Action	Explanation	Cross Reference to input
DO 9	PF control mode state	On (no OFF)	Activate Power Factor Mode as requested by the NSP or SO	DI 33
AO 6	PF control setpoint	Setpoint command	Setpoint command to change the power factor of the RPP. Producing vars (+), Absorbing vars (-)	AI 24
DO 10	PF control setpoint raise/lower	Raise-01 Lower-10	Raise/lower command can be used instead of setpoint command (AO 6)	AI 24

Table A10-24: Power factor binary indications

Index	Input	State	Type	Explanation	Cross Reference to control
DI 33	PF control mode state	On-1 (No Off)	Single-bit	Will report a '1' state when active and '0' state when not active	DO 9
DI 34	PF control mode	Not ready-1 Ready-0	Single-bit	Will be set to '1' in the event of conditions at the plant preventing the plant from going into this mode.	

Table A10-25: Power factor analogue

Index	Input	Type	Explanation	Cross Reference to control
AI 24	PF setpoint feedback	Analogue	Echo response to a new Power Factor setpoint issued by the NSP or SO. Producing vars (+), Absorbing vars (-)	AO 6 or DO 10

A10.6.4.3 Voltage Control mode**Table A10-26: Voltage mode commands**

Index	Command	Action	Reason	Cross Reference to indication/analogue
DO 11	Voltage control mode state	On (no Off)	Activate Voltage control Mode as requested by the NSP or SO	DI 35
AO 7	Voltage control setpoint	Setpoint command	Setpoint command to change the voltage	AI 25
DO 12	Voltage control setpoint raise/lower	Raise-01 Lower-10	Raise/lower command can be used instead of setpoint command(AO 7)	AI 25

Table A10-27: Voltage mode binary indications

Index	Input	State	Type	Explanation	Cross Reference
DI 35	Voltage control Mode state	On-1 Off-0	Single-bit	Will report a '1' state when active and '0' state when not active	DO 11
DI 36	Voltage mode	Not ready-1 Ready-0	Single-bit	Will be set to '1' in the event of conditions at the plant preventing the plant from going into the Voltage mode.	

Table A10-28: Voltage mode analogue

Index	Input	Type	Explanation	Cross Reference
AI 25	Voltage setpoint feedback	Analogue	RPP echo response to a new voltage setpoint issued by the NSP or SO	AO 7 or DO 12

A10.7 Provision of the Signal Data and Substation Layout Information to NSP or SO

(1) When the *RPP* generator is ready to commence with commissioning of their plant, the following data should be provided to *NSP* or *SO* commissioning personnel at least 8 weeks prior to the planned commissioning date.

- (a) A spread sheet containing the following signal sets – each in its own work sheet.
 - (i) Digital inputs - including cross links to the associated commands
 - (ii) Analogue inputs – including scaling information e.g. Low Engineering Value, Low Transmitted Value, High Engineering Value, High transmitted value, associated Setpoint command if any etc.
 - (iii) Digital commands – including cross links to the associated digital inputs.
 - (iv) Setpoint commands – including scaling information e.g. Low Engineering Value, Low Transmitted Value, High Engineering Value, High transmitted value, associated analogue input etc.
 - (v) Revision History
 - (vi) Each row on the spread sheet should list the point address, description in full, and space to record the commissioning tests.
- (b) Station Electric/operating diagram showing all bays conforming to the *NSP* or *SO* requirements,
- (c) An example spreadsheet is available from the *NSP* or *SO* on request.

Appendix 11 – Test Procedures for Gateway Factory Acceptance Tests (FAT), Site Acceptance Tests (SAT) and Commissioning

A11.1 Pre-FAT

(1) *NSP* or *SO* will provide the generic Factory Acceptance Test (FAT) procedures which shall be updated by the *RPP generator*. The final test procedure shall be agreed upon by the parties. The *RPP generator* shall submit the pre-factory acceptance tests results to *SO* or *NSP* at least two months prior to Grid connection date.

A11.2 FAT

(1) Factory Acceptance tests shall be done by the *RPP generator* where *NSP* or *SO* reserves the right to witness.

A11.3 Pre-SAT

(1) *NSP* or *SO* will provide generic Site Acceptance Test (SAT) procedures which shall be updated by the *RPP generator*. The final test procedure shall be agreed upon by the parties. Pre-site acceptance tests shall be done jointly by the parties using the *NSP* or *SO*'s applicable development systems to ensure inter-operability.

A11.4 SAT

(1) Site acceptance tests shall be done to the *NSP* or *SO*'s production systems.

A11.5 Commissioning

(1) Commissioning is the point to point testing of all specified signals to ensure that they are correctly configured and effect the correct operation.

(2) *NSP* or *SO* will provide test procedures for both plant online and offline testing. These procedures shall be used to verify the correct response to the issued commands.

Appendix 12 – Provision of Technical Network / Grid Data to RPP Generators

A12.1 Introduction

(1) *Generators* that want to develop conventional or renewable power plants in South Africa have to comply with the conditions set out in the applicable Codes which are approved by the *National Energy Regulator of South Africa (NERSA)*. The *NERSA* Codes define what technical grid information should be provided by the *Network Service Provider (NSP)* to the generators, and, the generators to the *NSP* for planning and design purposes. The aim of this appendix is to regulate provision of network/grid data from the *NSP* to the generators. This document is applicable to all generators connected to the medium voltage level and above. It is not applicable to generators less than 1 MW.

A12.2 Definitions

Term	Definition
Lumped Load	This refers to the aggregated sum of individual loads connected at a specific busbar
Lumped Generation	This refers to the aggregated sum of individual generation units connected at a specific busbar
Equivalent Network	This is a reduced network consisting of a source infeed/s, boundary bus(es), interconnecting transformer(s) and surrounding network that represents the behaviour of the external system as seen from its boundary bus
Interconnecting Transformers	This is the network transformer(s) that connect the surrounding network to the boundary bus representing the external network.
Surrounding network	This is the network at the same voltage level as the PoC (e.g. surrounding 132kV network if POC is at 132kV or 33kV network if PoC is at 33kV). For extensive surrounding networks, parts of the network may be represented by lumped sources and/or lumped loads.
Boundary	Network boundary is one voltage level above the POC, e.g. if POC is 132kV, then information shall be at 400kV, 275kV or 220kV, It is typically one voltage level higher than PoC,
External network	The rest of the network which is represented by the boundary bus and source infeed(s).
Fault level(s)	Known as Short circuit level(s) – Provides information on the strength of the network and equivalent source impedance under defined conditions.

Normal short circuit level	Normal short circuit level must be calculated on basis of an operational scenario with maximum generation and equipment (such as line, transformers etc) in service (in the sourcing grid using a high load case) as per the Transmission Development Plan (TDP).
Maximum design short circuit level	The generating plant shall be designed to withstand the maximum recommended IEC fault level rating for the specific voltage level at the POC.
Minimum short circuit level	Minimum short circuit level must be calculated on basis of an operational scenario with minimum generation (in the sourcing grid) in service (typically minimum load case) and contingency scenario that gives the minimum fault level; provided the network is within an operable range. • Note that the calculation method for both the maximum and minimum short circuit levels shall be stated/included with the specific values i.e. IEC 60909 or the complete method. • Worst case contingency scenario: Credible contingency scenarios (contingencies having a probability that is still in a range that makes it likely to be observed over a lifetime of e.g. 20 years) leading to minimum short circuit level/maximum network impedance at the POC

A12.3 Information Exchange

A12.3.1 General

(1) The *NSPs* shall make every endeavour to provide the latest up to date information for the purposes required, however the *NSPs* do not guarantee the accuracy of the information. The information that shall be provided shall be the *NSPs* best available information at the time of request as stipulated in the *Code*.

A12.3.2 Simulation Studies Requirements for Generation Integration

A12.3.2.1 Studies to be performed by the NSP

(1) The *Network Service Provider (NSP)* is responsible to conduct generation grid impact studies. It must be noted that some of these studies are done in stages i.e. some are conducted during the feasibility (depending on generator technology types); while more detailed studies are only done during the design phases due to the overall grid impacts of all approved *Generators*.

(2) These studies may include:

- (a) Load flow studies, contingency analysis (verification of thermal loading and voltages),
- (b) Short Circuit studies (verification of short circuit rating of existing equipment, impact on protection),
- (c) Stability Studies:
 - (i) Transient stability studies (impact on critical fault clearing times, stability constraints and transfer limits of the entire connected system)
 - (ii) Oscillatory stability studies
 - (iii) Frequency stability studies
 - (iv) Voltage stability studies (long-term and short-term)
- (d) Power Quality studies (The network harmonic impedance for network normal and a reasonable range of contingencies at the proposed point of connection).

A12.3.2.2 Studies to be performed by the generator(s)

(1) The *generator* shall design his/her power plant and conduct grid code compliance studies. The *renewable energy power plant* technology can either be synchronous generator (for example, Concentrated Solar Power or CSP, biomass, landfill gas etc.) or inverter-based and/or asynchronous machines (such as Solar PV or wind farms).

a. Inverter based generators and/or asynchronous generators

(1) The studies to be completed by the *generator* for inverter-based and asynchronous generating facilities (e.g. wind or PV farms) are listed as follows:

- (a) Design studies
 - (i) Load flow and short circuit studies for design purposes (equipment rating)
 - (ii) Quality of supply (QoS) studies:
 - to meet allowed QoS emission limits
 - limit the resonance caused by the installation
 - anticipated voltage variations due to internal switching actions.
- (b) Grid Code Compliance Studies
 - (i) Reactive power capability studies (based on load flows)
 - (ii) LVRT studies (wind farms/PV farms)
 - (iii) HVRT studies (wind farms/ PV farms)
 - (iv) Power quality assessment (harmonics, voltage unbalance, rapid voltage changes and voltage flicker)

b. Synchronous generators

(1) The following studies shall be conducted if the generating facility is a synchronous machine (such as CSP, biomass, hydro, landfill plants):

(a) Design studies

- (i) Load flow and short circuit studies for design purposes (equipment rating)
- (ii) Quality of supply (QoS) studies:
 - to meet allowed QoS emission limits
 - limit the resonance caused by the installation
 - anticipated voltage variations due to internal switching actions.

(b) Grid Code Compliance Studies

- (i) Reactive power capability studies (based on load flows)
- (ii) Transient Stability Studies, subject to the information provided as per the document
- (iii) Excitation system requirements as per GCR3 of the current South African Grid Code – The Network Code.
- (iv) Power quality assessment (harmonics, voltage unbalance, rapid voltage changes and voltage flicker)

A12.3.3 Information Exchange between the NSP and the RPP generator

(1) Generator data exchange shall be a time-based process as follows.

(a) **First stage** (during the application for connection; also referred to as feasibility stage)

- (i) The following information shall be submitted by the *RPP generator* to the *NSP*:
 - Physical location of the generating plant (including the GPS coordinates)
 - Site Plan
 - Type of the generating power plant (e.g. wind, Solar PV, CSP, land-fill, biomass etc)
 - Total generating plant MW output
 - Total capacity (MW) that will be exported into the grid of the NSP
 - Initial phase MW value
 - Final phase MW value and timelines
 - Any other information that the service provider may reasonably require
- (ii) The *NSP* shall make available to the generator or its duly appointed agent at least the following information:

- Define the *Point of Connection* and the *Point of Common Coupling* including the nominal voltages
- Expected maximum fault levels at the point of connection
- Expected minimum fault levels at the point of connection

(b) Second stage (during the design stage; for the *generator* to design his plant and conduct the grid code compliance studies).

- (i) During this stage, the *NSP* shall provide information on the equivalent network as described in A10.3.3.1 below.
- (ii) It must be noted that in certain cases, the development of an equivalent network will be time-consuming due to the numerous in-feeds and voltage levels that must be represented. This may also lead to instances where the accuracy of the equivalent network may be in question. Where no agreement can be reached on the equivalent network model developed by the *NSP*, then a full network model shall be utilized via joint studies by both the *NSP* and the applicant (*generator*).

A12.3.4 Data provisions by the NSP for the Equivalent Network, for the generating power plant design and grid code compliance studies (excluding harmonic studies)

(a) Source infeed

- (i) The source infeed(s) at the boundary bus(es) which represents the external network. This is typically one voltage level higher than the *POC* voltage.
- (ii) Transformer(s) connecting the boundary bus(es) to the surrounding network e.g. 400/132kV, 275/132kV, 132/33kV etc.
- (iii) Expected operating maximum and minimum fault levels [before and after connection, using the year of connection as the reference] at the boundary bus(es) (i.e. 400kV, 275kV, 132kV), including X/R ratios to model the correct network source in-feed(s) of the external network

(b) Fault levels and X/R ratios will be provided for the following network conditions:

- (i) Normal short circuit level,
- (ii) Maximum design short circuit level based on the planning horizon,
- (iii) Minimum short circuit level. Minimum short circuit level must be calculated on basis of an operational scenario with minimum generation (in the sourcing grid) in service

(typically minimum load case) and contingency scenario that gives the minimum fault level; provided the network is within an operable state.

Note: the calculation method for both the maximum and minimum short circuit levels shall be stated/included with the specific values i.e. IEC 60909.

- (iv) Worst case contingency scenario: Credible contingency scenarios (contingencies having a probability that is still in a range that makes it likely to be observed over a lifetime of e.g. 20 years) leading to minimum short circuit level/maximum network impedance at the POC.

(c) Surrounding network

- (i) Surrounding network at the same voltage level as the *POC*,
- (ii) The lumped peak load and minimum load (based on metering data, where available),
- (iii) If there is any generation within the surrounding network, such generation shall be represented by generic models,
- (iv) Any loads and/or generation at lower voltage levels, fed from the surrounding network will be lumped at the surrounding network voltage level,
- (v) Information on the available present network interruption performance and Quality of Supply (QOS) levels at the *PCC* or the closest node in the network; including the expected background harmonic distortion of the grid before connection of the generation facility. This shall include:
 - Indicative frequency sweeps at the *POC*
 - Voltage unbalance
 - Voltage dips
 - Network interruption performance data
 - Voltage Total Harmonic Distortion (THD) levels (where available)
 - The system normal and minimum fault levels at *POC*

Note: voltage unbalance, voltage dips, network interruption performance data, voltage harmonics is for information purposes only.

(d) The equivalent network

- (i) All lines (or cables) of the surrounding network. Note that the surrounding network is the same voltage of the *POC*.
- (ii) All synchronous power plants feeding into the surrounding network, using an aggregated generic model of the generating plant.
- (iii) All other non-synchronous power plants (e.g. wind farms, PV-inverters) using an aggregated, generic model (e.g. IEC-type model, IEEE standard models) at the POC of each of the farms.

- (iv) Loads shall be aggregated and modelled by one equivalent load connected to the surrounding network.
- (v) Feeders having load and generation shall be reduced and represented by one aggregated load and one equivalent generator connected to the surrounding network.
- (vi) Transformer(s) connecting to MTS of the surrounding network, e.g. 400kV/132kV, 275kV/132kV, 132kV/33kV
- (vii) Shunt compensation devices
- (viii) Source in-feed(s)

d1. Lines and cables data

- (i) Lines:
 - Length, positive and zero sequence impedance per length ($R1', X1' / R0', X0'$), positive and zero sequence capacitance (or susceptance per length) ($C1' / C0'$).
- (ii) Cables:
 - Length, positive and zero sequence impedance per length ($R1', X1' / R0', X0'$), positive and zero sequence capacitance (or susceptance per length) ($C1' / C0'$).

d2. Synchronous power plant data

- (i) Synchronous power plants having connection point in surrounding network:
 - Generator transformer (Rated capacity, short circuit impedance ($R1', X1', R0', X0'$), no-load impedance (iron losses/magnetizing reactance)
 - Synchronous generator (Rated capacity, voltage level and synchronous machine parameters)
 - Generic models for Automatic Voltage Regulator (AVR) and Power System Stabilizers (PSS)

d3. Non-synchronous power plant data

- (i) Other, Non-Synchronous power plants (e.g. wind farms, PV-farms) having connection point in surrounding network:
 - Aggregated generator transformer (Rated capacity, short circuit impedance ($R1', X1', R0', X0'$), no-load impedance (iron losses/magnetizing reactance)
 - Main MV bus-bar of generator installation
 - Aggregated, generic model of non-synchronous generator at main MV- bus bar of the windfarm/PV farm, e.g. IEC-type model/IEEE standard model.

- (ii) At a minimum, the following data/properties shall be represented:
- Rating of the generation installation
 - Control mode during normal operation (farm controller, e.g. voltage, power factor etc.)
 - Reactive power capacity (using grid code characteristic)
 - Dynamic model with main focus on the generator's behaviour during LVRT-events.
 - If additional dynamic components are used (e.g. STATCOM, Capacitors, SVCs) must also be represented

d4. Feeders

- (i) Feeder loads:
- Lumped, aggregated load model (without feeder transformer).
 - Maximum load with applicable power factor.
- (ii) Feeders with embedded generation:
- Parallel connection of equivalent Feeder-Load and equivalent, generic generator.
 - Maximum load with applicable power factor.
 - Maximum generation with operating power factor

d5. Transformers and shunts

- (i) MTS Transformer (e.g. 400/132kV, 275/132kV, 132/33kV):
- Transformer rating
 - Rated voltages of transformer
 - Short circuit impedance (R, X)
 - No-load current and iron losses
 - Information about tap changer (number of taps, tap steps, max./min. taps, on-load/off-load, controlled node, control mode)
- (ii) Shunt compensation devices
- Generic models of the shunt compensation devices
 - All the voltage switching ranges
 - Control set point

A12.3.5 Examples

(a) Connection to a sub-transmission network with a single source in-feed:

(i) Proposed generating plant *POC* is at 132kV as shown in 0.

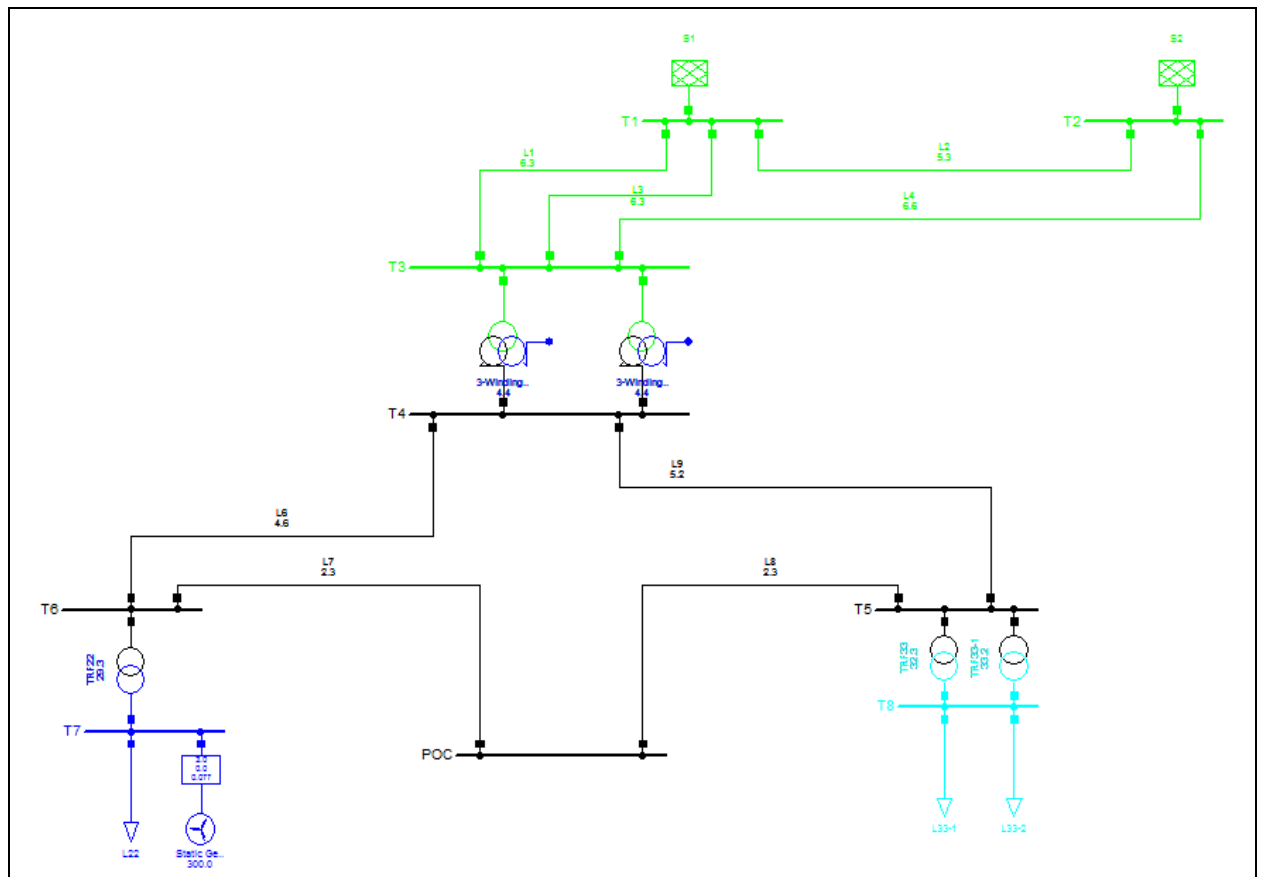


Figure A12-1: Network showing a single source in-feed and the proposed connection
PoC is at 132kV

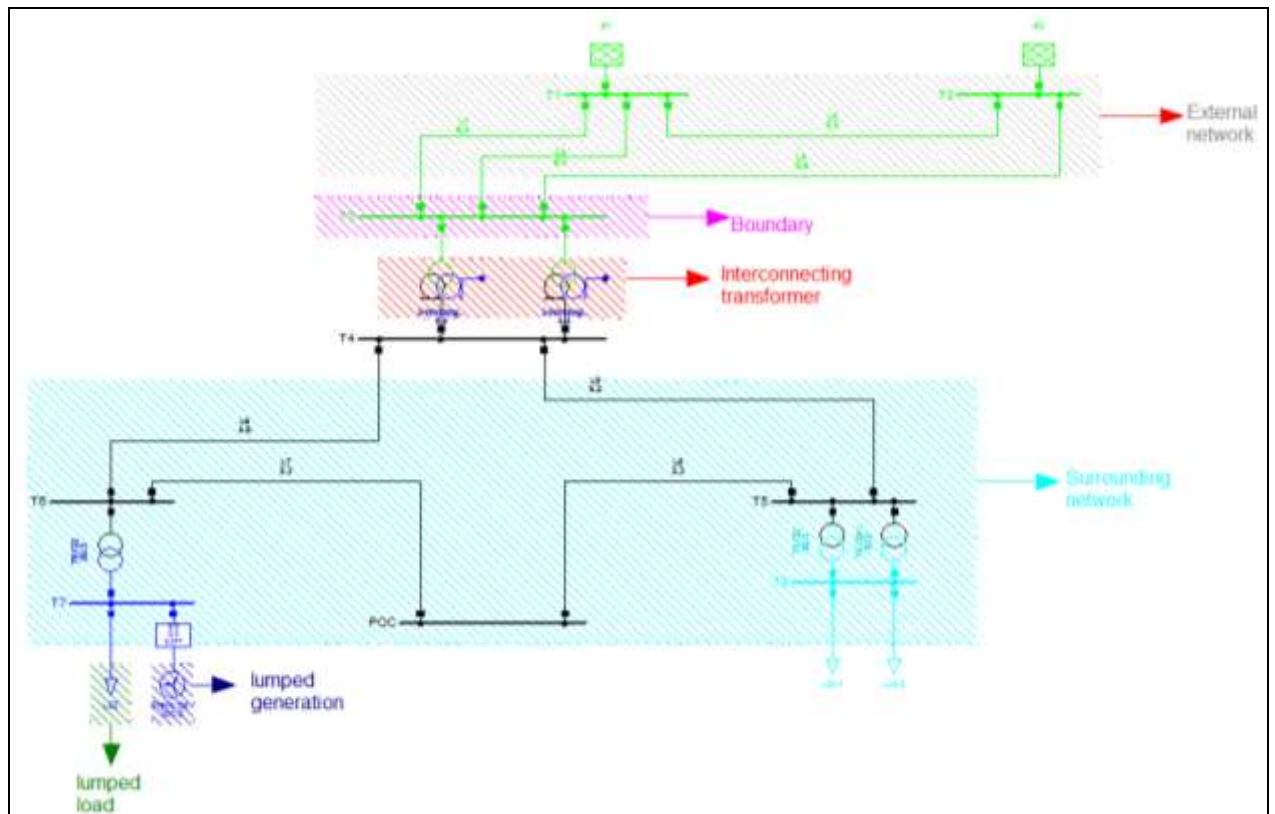


Figure A12-2: Single source in-feed network illustrating some definitions

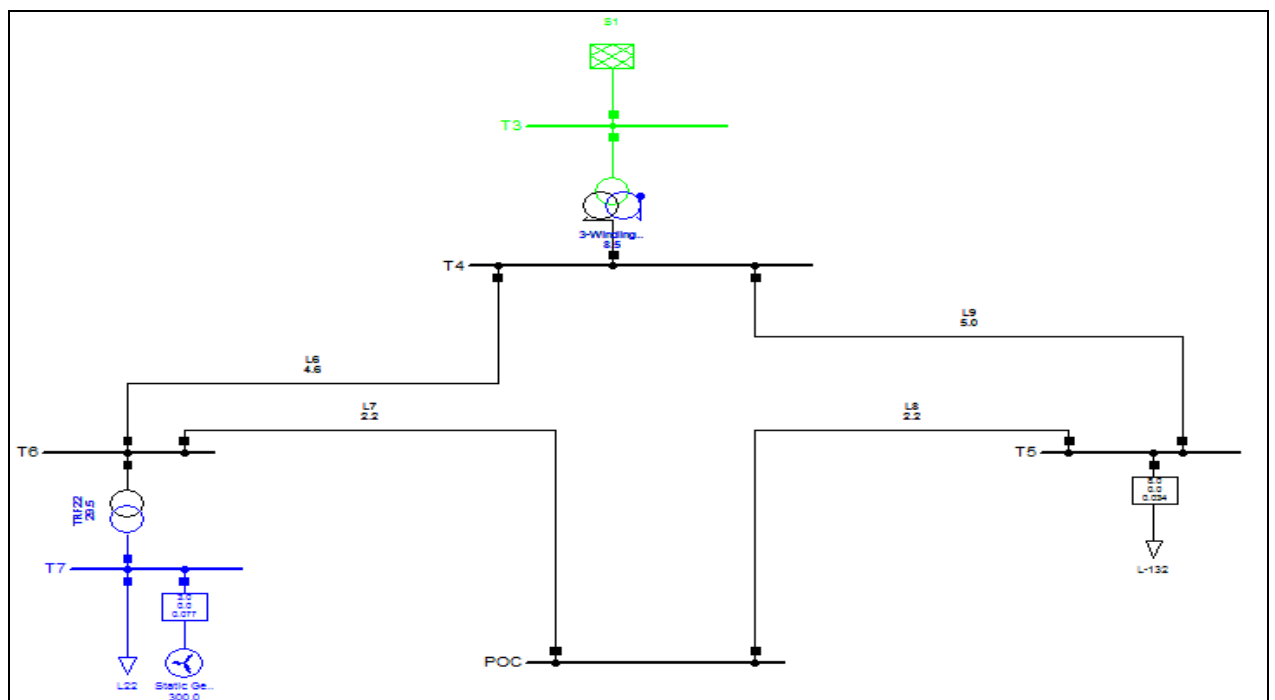


Figure A12-3: Equivalent network with single source in feed

(b) Connection to a sub-transmission network with two source in-feeds:

(i) Proposed generating plant is at 132kV as shown in 0.

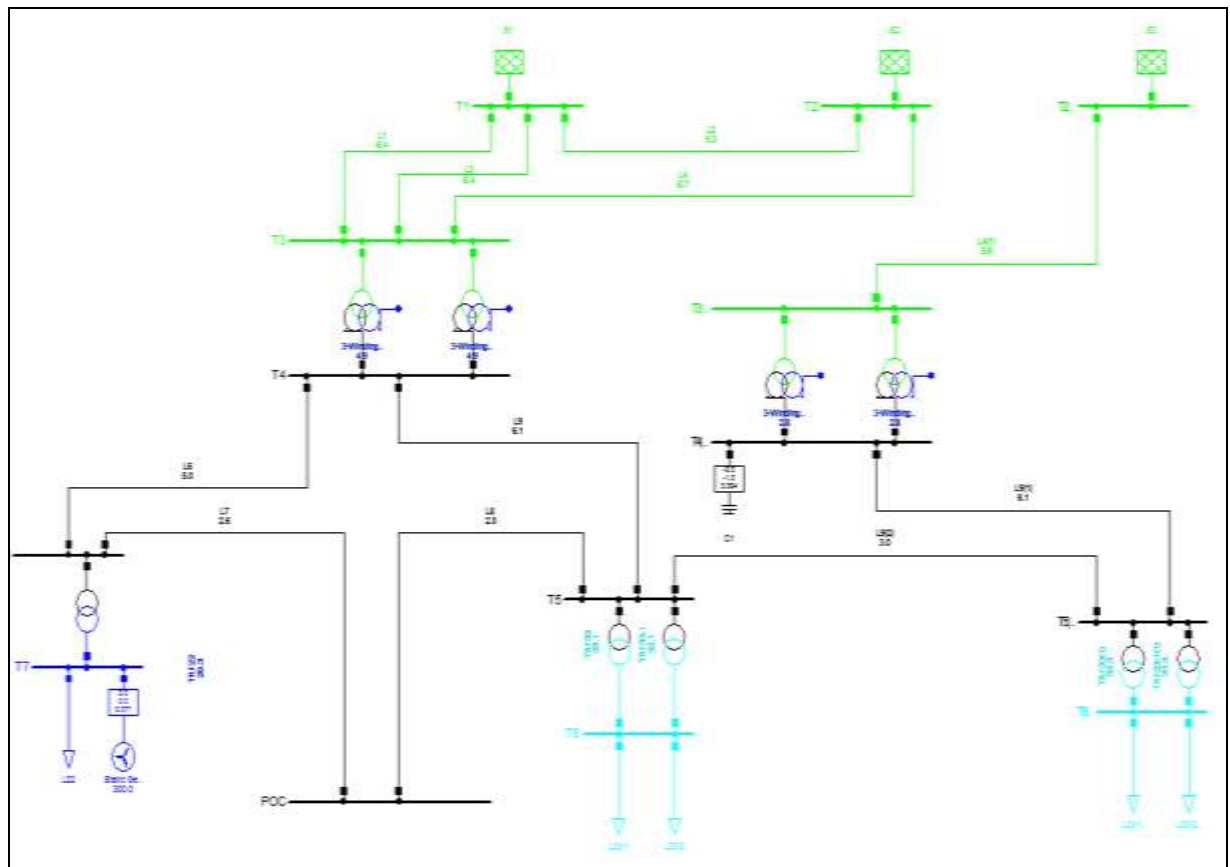


Figure A12-4: Network with two source in-feeds showing the POC

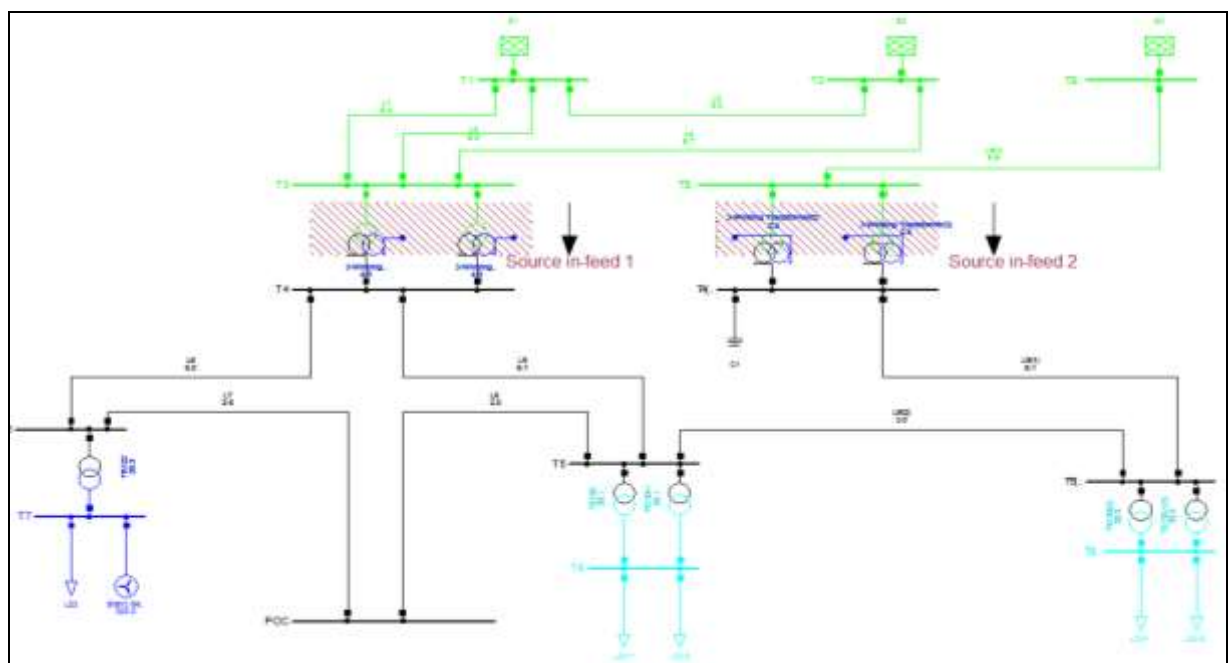


Figure A12-5: Two source in-feeds network illustrating some definitions

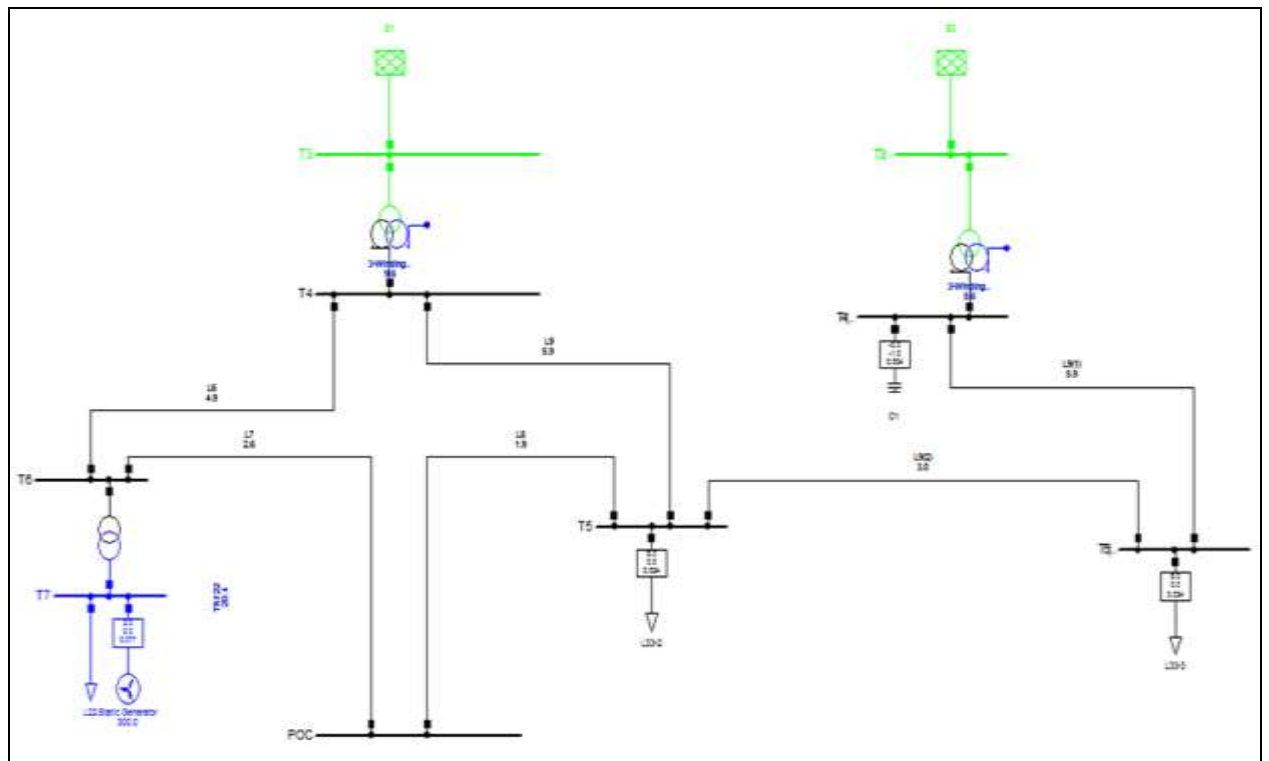


Figure A12-6: Equivalent network of two source in-feeds

Appendix 13 – RPP Power Quality Compliance Guideline

A13.1 Abbreviations

A13.1 Introduction

- (1) This document provides requirements for the compliance process to enable *RPP Owner(s)* to meet the Grid Code requirements with respect to power quality (PQ) as stipulated by the *NSP*.

A13.1.1 Definition of Emission Level:

- (1) The definitions as stipulated in the IEC 61000-3-6 document for harmonic emission level is interpreted as the magnitude of the vector that is caused by the installation at the point of evaluation (*POC* for *RPP*). The concept is illustrated in Figure A13-1 for a voltage harmonic phasor.

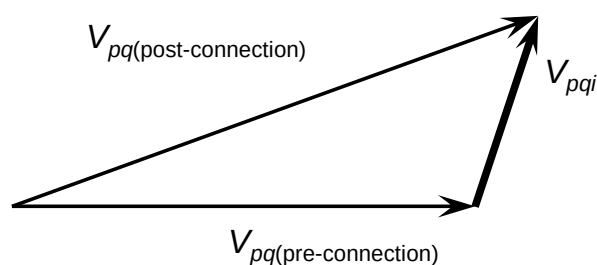


Figure A13-1: Definition of emission level.

- (2) A similar definition is provided for voltage unbalance in IEC 61000-3-13. Whilst the definition is not as clear for voltage flicker in IEC 61000-3-7, the principle is similar, since voltage flicker would not add linearly between different installations unless there is an intentional link.

A13.2 Power Quality Emission Limits

- (1) *NSP's* shall calculate the specific emission limits as guided by NRS048:4 or IEC 61000-3-6.

NOTE: NRS048:2 do not specify emission limits but rather compatibility levels that the *NSP* must account for. Compatibility levels as stated in NRS048:2 shall not be issued to Generators as emission levels.

A13.3 NSP and Reference Data

(1) In line with the provision of technical data guideline, the following information is relevant to the *RPP Owner* for power quality purposes (where available):

(i) If available, historical performance of power quality background levels at the PoC or closest available point; as plots of the 95th percentile weekly values (10 minute values) for the voltage trends for at least a year or the period for which data is available.

(ii) A range of impedance frequency sweep data at the PoC, including network healthy and a realistic range of network contingencies.

NOTE: These contingencies are based on network operational requirements in consultation between the NSP PQ engineer and NSP Network Operational personnel.

(iii) A range of impedance frequency sweep data at the PoC for the future network. This data may be used to assess the future harmonic impact due to varying impedance, however it would not be any guarantee that the network would evolve as provided.

NOTE Frequency sweep data should be provided as the complex network impedance vs. frequency, i.e. R and X, or, Z and phi,

(2) Reference data includes compatibility levels for all power quality parameters as provided in NRS048-2, or as specified in the connection agreement.

(3) It is important to note the interaction between the various power quality parameters; e.g. background harmonics and voltage unbalance that often impact the harmonic emissions from equipment.

NB: Background levels/total levels after connection of all power quality parameters up to the compatibility levels should be taken into account when calculating emission levels as well as immunity of equipment. The historical background data is, therefore, not critical for any power quality assessment.

A13.4 Compliance Assessment

(1) The main compliance criteria for harmonics will be harmonic current measurements at the PoC (Although both voltage and current harmonic emission levels are prescribed by the NSP).

(2) Compliance for voltage flicker shall be based on measured voltage values at the PoC.

(3) Compliance for voltage unbalance shall be based on measured voltage or current (negative sequence current) values at the PoC.

- (4) Grid code compliance must be proven by means of measurements under normal operating conditions (i.e. *RPP* operating) for a period of no less than 7 consecutive days (24 hours per day). Measurements should be taken once the Generator is connected to the grid but is not yet commercially operational. The generation profile must be representative of the *RPP* final commercial operations, i.e. with all generation capacity in service.
- (5) Harmonic emission simulations will not be required for grid code compliance, however should the;
- (i) NPS experience a violation of the PQ levels within NRS048:2 due to the *RPP* connection, the NSP must request that the *RPP Owner* to reduce its output within 7 days so as to manage the PQ levels to within NRS048:2 compatibility levels.
 - (ii) *RPP Owner* become aware or should be reasonably being aware of a violation of the PQ levels within NRS048:2, the *RPP Owner* must reduce its output within 7 days to manage the PQ levels within NRS048:2 compatibility levels.

A13.4.1 Assessment Process

A13.4.1.1 Generators larger than 5 MVA (Category B >5 MVA and larger categories)

- (1) The following assessment process is applicable to all *RPP* larger than 5 MVA:

STEP 1: *RPP Owner* to obtain voltage and current emission limits from the NSP.

STEP 2: *RPP Owner* to measure and report power quality assessment, in line with prescribed methods in section A6.5, to the NSP within 6 months of grid connection (note that the plant generation pattern must be representative as if the plant is in full commercial operation). Measurements must include a period (at least 24 hours continuous) where the *RPP* is not connected (connecting breaker open) for the harmonic, unbalance and flicker assessment, these measurements must be taken at the PoC prior to the plants.

STEP 3: *RPP Owner* to engage with the NSP and discuss the results of the emission assessment. Where violations of the respective harmonic emission limits are evident, agreement must be reached on which harmonics are emitted and which are absorbed/sunk by the *RPP*. The same holds true for voltage unbalance and voltage flicker.

Where there are no emission violations then the *RPP* will be considered to be Grid Code compliant for PQ.

Where there are violations of the emission levels and if agreement is obtained between the NSP and *RPP* on the harmonic emission violations i.e. absorbed vs emitted, then the NSP will support an appropriate conditional exemption for the *RPP* to mitigate the emission violations.

NOTE: Clause A13.4 (5) applies during this exemption period.

The above mentioned process holds true for voltage unbalance and voltage flicker.

Step 4: *RPP Owner* must redo their compliance assessment after mitigation, within exemption period granted and present to NSP for final grid code approval. (The 24 hour period of disconnection may not be required if all emission levels are below the allocated emission limits.)

Figure A13. 2 shows a flows diagram of the assessment process described above.

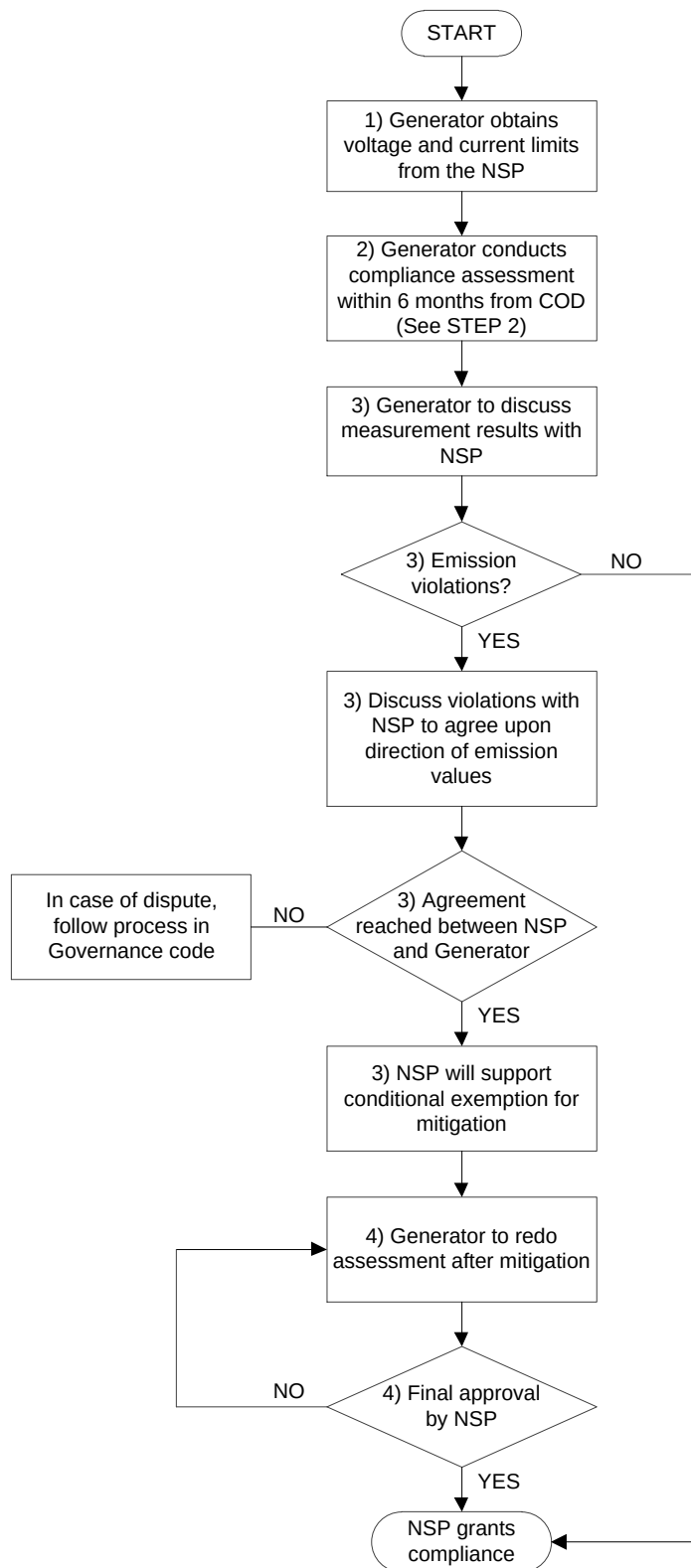


Figure 13.2: Flow diagram of assessment process

A13.4.2 Resonance Screening Process

- (1) The basic screening process involves checking whether the connection of the *RPP* will cause excessive resonance. Resonance is regarded as excessive when the impedance envelope ($3 \times Z_h$) is exceeded at any frequency.
- (2) *RPP Owner(s)* are encouraged to engage with the *NSP* for the resonance screening. *RPP Owner(s)* are to integrate their plant models into the *NSP* utility model (DigSilent simulation tool) to evaluate the combined impact on the resonance screening. Only the results (frequency sweeps) of such integration will be shared with the *RPP*. Full *NSP* models will not be shared.

A13.4.3 Emission Requirements for Inverters used in Category A and B (≤ 5 MVA) Generator

A13.4.3.1 Harmonics

- (1) The individual harmonic currents of the inverters shall not exceed the limits specified IEC61727. These values are specified in Table A13.4.3-1 and the total harmonic distortion (THD) of the current (to the 50th harmonic) shall be less than 5%.

Table A13.4.3-1: Inverter harmonic emission requirements

Harmonic order (h)*	$3 \leq h < 9$	$11 \leq h < 15$	$17 \leq h < 21$	$23 \leq h < 33$	$33 \leq h \leq 50$	Even	THD
Percentage of rated current (Odd harmonics)	<4,0	<2,0	<1,5	<0,6	<0,3	25% of odd harmonics	<5,0

A13.4.3.2 Flicker

- (1) The inverter shall conform to the voltage fluctuation and flicker limits as per SANS61000-3-3 for equipment rated less than or equal to 16 A per phase and SANS61000-3-5 for equipment rated greater than 16 A per phase. Compliance shall be determined by type testing in accordance with the appropriate Standard.

A13.4.3.3 Voltage Unbalance

- (1) For *RPP* connected to multi-phase supplies (two- or three-phase connection at the POC), the difference in installed capacity between phases may not exceed 4.6 kVA per phase

A13.5 Grid Code Compliance: Measurement

A13.5.1 Measurement Requirements

- (1) All *RPP* above >5MVA are required to demonstrate compliance at installation level via power quality measurements. All measurements for compliance purposes will be done at

the PoC, unless not practically feasible. Voltage and Current measurements shall be taken.

- (2) The following requirements apply for these measurements:
 - (a) The PQ instrument(s) shall comply with IEC 61000-4-30, class A Edition 3 (or latest version).
 - (b) The instrument(s) shall be set-up to record the 10-minute aggregated values as a minimum. Where possible phasor measurements (including aggregated phasors) may be measured and applied.
 - (c) These instruments shall be set-up to record all current and voltage harmonics up to the 50th harmonic.
 - (d) Three-phase measurements shall be done:
 - (i) For a PoC that is connected in Wye with a solidly earthed neutral, the phase-to-neutral voltages and the phase currents shall be monitored;
 - (ii) For all other systems, the phase-to-phase voltages and the phase currents shall be monitored.
 - (iii) Any deviation shall be motivated and confirmed with the NSP when required.
 - (e) Where the PoC has two or more feeders to the RPP, a physical current summation method shall be implemented.
- (3) Measurements used for the analysis of the emissions shall be taken for a minimum of one week, starting at midnight on day 1 lasting until midnight after 7 days. The duration of the measurement shall be representative of the plant's expected normal production cycle.
- (4) Simultaneous, GPS synchronised measurements must be taken when more than one PQ instrument is used in the assessment of Grid Code Compliance.
- (5) A measurement uncertainty assessment must be completed for the measurement circuit utilised by the Generator, these include all instrument transformers, transducers, cable lengths and the measurement instrument. The JCGM 100 and JCGM 106 documents may be used to calculate the measurement uncertainty as well as to apply the uncertainty to the measured parameters. Accuracy levels for the equipment utilised are specified in the equipment specification sheets. Generic accuracy levels for instrument transformers may also be obtained from the IEC 61000-4-7 and from IEC 61869-103. The measurement uncertainty assessment shall be included in the assessment report to the NSP.

A13.5.2 Harmonic Emission Evaluation/Assessment for category B (>5MVA) and C Generator's

- (1) There are cases where the current flowing through the PoC will exceed the limits even though the emissions of the plant itself do not exceed these limits. This will be true where the plant presents low impedance at a certain frequency and therefore absorbs harmonic currents from the network and acts as a harmonic filter. For these cases, and others, a number of different methods (as listed below) can be used to prove that the *RPP* is absorbing the harmonic currents measured and not emitting them. The *RPP* must remove the measured values deemed to be absorption and prove compliance based on the remaining dataset (considered to be emission only). The 95th percentile value of the emission dataset must be compared to the respective limit to prove compliance.
- (2) The different methods include the following:
- (a) The impedance slope method, as described in Cigré Technical Report 468: Review of Disturbance Emission Assessment Techniques, section 4.2.2: Harmonic Emissions Level Compliance Assessment;
 - a. Use only if the linear trend line of the 10 minute measurements has a positive slope
 - b. Impedance slope must reflect the network at the time of measurement
 - c. The load impedance slope must represent the full Generator internal network.
 - (b) Consistent reduction background voltage harmonic emissions coinciding with the increase of measured harmonic currents during power generation by the Generator.
 - (c) Measurement and assessment of harmonic emission using harmonic phasors, (including aggregated harmonic phasors).
- (3) Triplen harmonics, in a Delta/Star or Star/Delta transformer system, may be equated to zero for balanced systems (i.e. zero negative phase sequence measured) only. For all other unbalanced systems the following equation will apply to cater for the penetration of triplen harmonics through the Delta windings.

$$\%I_h = 0.95h * V_{\%UB}$$

Where $\%I_h$ is the percentage of harmonic current flowing through the star-delta transformer, h the harmonic order and $V_{\%UB}$ the percentage voltage unbalance measured.

A13.5.2.1 Relaxation of Limits: Harmonic Current Emission Evaluation for Category B (>5MVA) and C Generators

- (1) With respect to the requirements for meeting harmonic emission limits provided by the *NSP*, the *RPP Owner* shall be allowed to exceed individual current harmonic emission limits by up to 50% (e.g. if the 5th harmonic limit is 1A, the *RPP* may emit up to 1.5A) provided that the Harmonic Distortion (HD) band limit is met for the following specified bands: $2 \leq h \leq 13$, $14 \leq h \leq 25$, $26 \leq h \leq 39$, $40 \leq h \leq 50$.

The THD bands are defined as follows:

$2 \leq h \leq 13$:

$$HD_{IlimitG1} = \sqrt{\sum_{h=2}^{13} I_{limit}(h)^2} \quad HD_{I_{measured}G1} = \sqrt{\sum_{h=2}^{13} I_{measured}(h)^2}$$

Grid Code compliance requirement: $HD_{I_{measured}G1} < HD_{IlimitG1}$

For $14 \leq h < 25$:

$$HD_{IlimitG2} = \sqrt{\sum_{h=14}^{25} I_{limit}(h)^2} \quad HD_{I_{measured}G2} = \sqrt{\sum_{h=14}^{25} I_{measured}(h)^2}$$

Grid Code compliance requirement: $HD_{I_{measured}G2} < THD_{IlimitG2}$

For $26 \leq h \leq 39$:

$$HD_{IlimitG3} = \sqrt{\sum_{h=26}^{39} I_{limit}(h)^2} \quad HD_{I_{measured}G3} = \sqrt{\sum_{h=26}^{39} I_{measured}(h)^2}$$

Grid Code compliance requirement: $HD_{I_{measured}G3} < THD_{IlimitG3}$

For $40 \leq h \leq 50$:

$$HD_{IlimitG4} = \sqrt{\sum_{h=40}^{50} I_{limit}(h)^2} \quad HD_{I_{measured}G4} = \sqrt{\sum_{h=40}^{50} I_{measured}(h)^2}$$

Grid Code compliance requirement: $HD_{I_{measured}G4} < HD_{IlimitG4}$

- (2) Where it has been shown that a plant absorbs a specific harmonic order, this harmonic order may be reduced to the allowable limit (as allocated by the *NSP*) in the Group HD band calculation.

- (3) The HD Band requirement is illustrated in figure A13.5.2-1 for clarity.

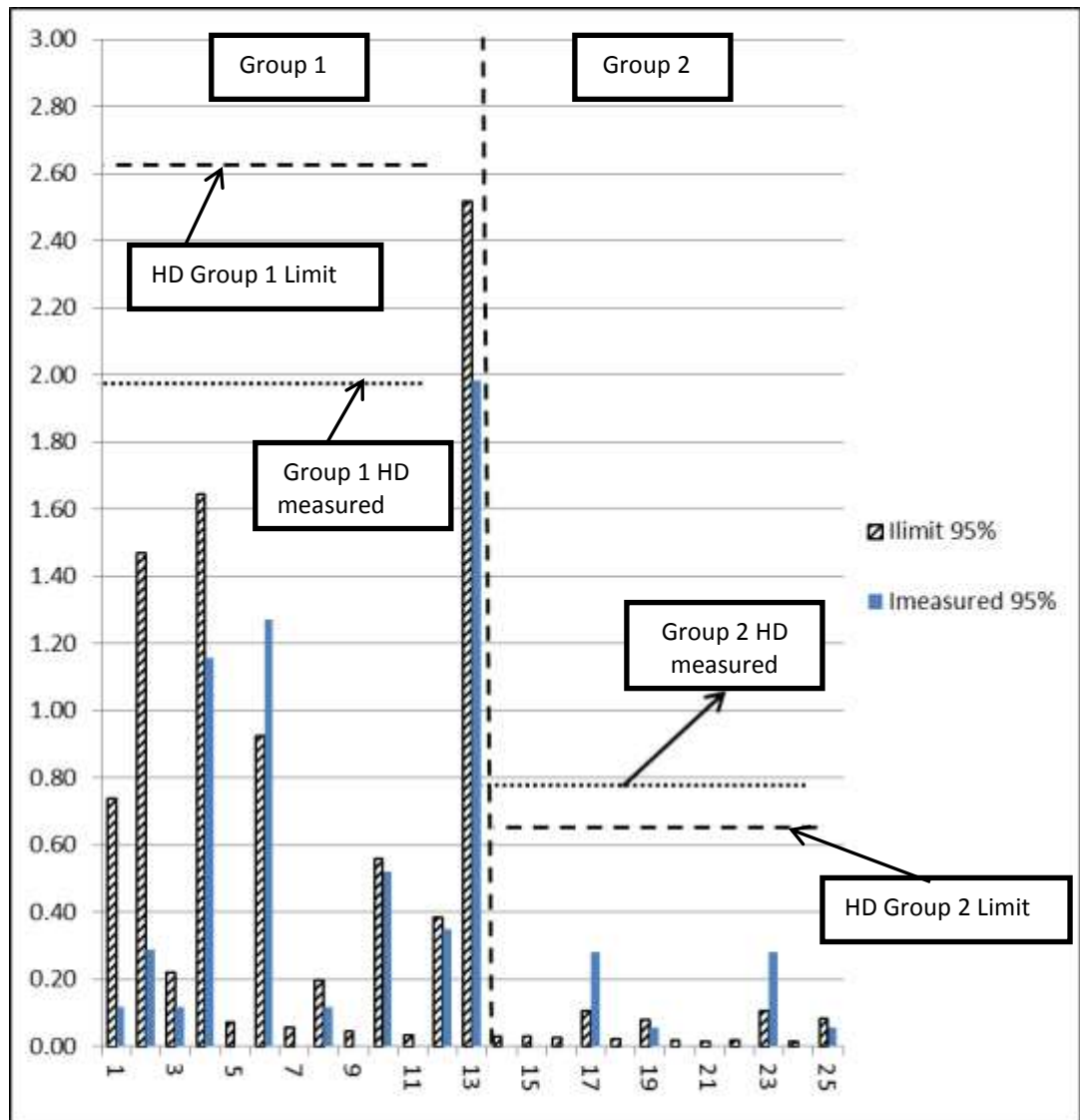


Figure A13.5.2-1: Group HD Emissions Assessment

- (4) Figure A13.5.2-1 shows the individual current harmonic measurements and limits as well as the HD bands measurements and limits for bands $2 \leq h \leq 13$ (Group 1) and $14 \leq h \leq 25$ (Group 2).
- (5) For HD band $2 \leq h \leq 13$, it can be observed that the measured current for the 7th harmonic exceeds the limit, however the Group 1 HD measured is less than the limit set for the HD Group. The *RPP* is deemed to meet the harmonic emissions requirements and does not need to take any actions.
- (6) For HD band $14 \leq h \leq 25$, it can be observed that the measured harmonic currents for the 17th and the 23rd harmonic exceed their limits. Additionally, the Group 2 HD measured exceeds the Group 2 HD limit. In this instance the *RPP* is required to take actions to ensure that the Group 2 HD harmonic emissions are reduced to below the limits.

- (7) The weekly sliding 95% values at each harmonic current shall be assessed against the emission limits specified by the *NSP* and the values which were calculated.

A13.5.3 Flicker Emission Evaluation

- (1) The following method is recommended for the assessment of flicker.
- (2) This assessment method for flicker is based on the method from the Cigré Technical Report 468: Review of Disturbance Emission Assessment Techniques, section 5.3.2 - Simplified Approach. The method has been modified to reflect the requirements for Plt as specified in South African standards.
- (3) However, any of the other methods listed in section 5.3 of Cigré Technical Report 468 may also be used by the Generator to assess flicker.
- (4) The typical configuration for individual flicker emission level assessment is given in Figure A13.5.3-1. The installation under consideration is connected at point A. This point is connected to the *POC* (point B) through impedance Z_2 (in this case the transformer connecting to the *NSP*). Other fluctuating loads are possibly connected at B.
- (5) Typically, when other fluctuating loads are operating in the electrical vicinity, the background flicker at the *POC* (point B) cannot be neglected. However, on the secondary side of the transformer (A) the dominance of the emission Plt emission (A) of the investigated installation in the global flicker level Plt (A) increases and the influence of other sources can often be neglected, especially when the 95th on 99th percentile of Plt are considered (IEC 61000-3-7).

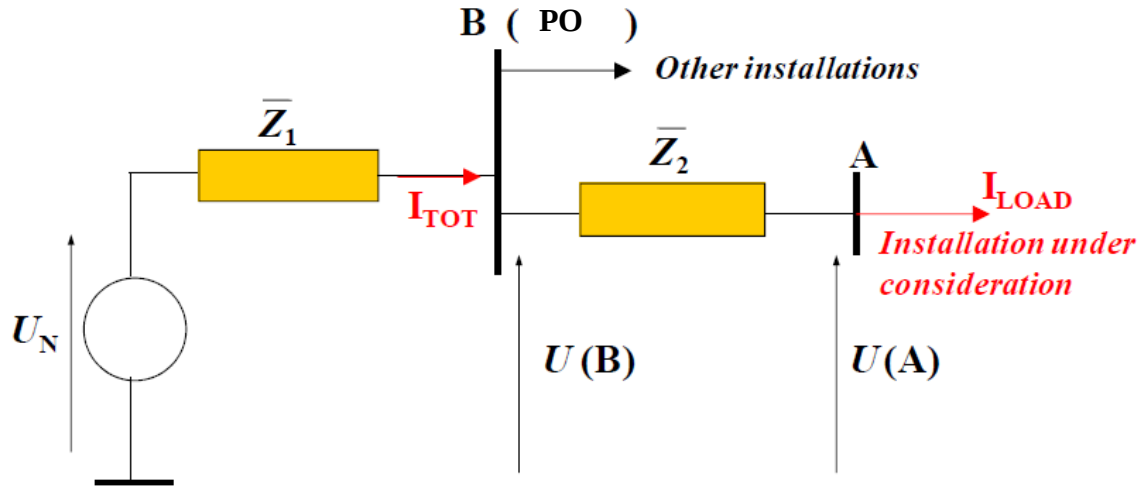


Figure A13.5.3-1: Network configuration for the application of the "simplified approach"

- (6) The simplified approach consists of measuring the flicker level on the secondary side(A) of the transformer and to transpose it to the primary side (B), by simply using the impedance ratio below.

$$P_{lt \text{ emission}}(A) \cong P_{lt}(A)$$

$$P_{lt \text{ emission}}(B) \cong \left| \frac{\bar{Z}_1}{\bar{Z}_1 + \bar{Z}_2} \right| P_{lt \text{ emission}}(A) = \frac{S_{SC,A}}{S_{SC,B}} P_{lt \text{ emission}}(A)$$

The following definitions apply:

- (f) Z_1 is the network impedance,
- (g) Z_2 is the transformer impedance,
- (h) $P_{lt}(A)$ is the global flicker level at A,
- (i) $P_{lt \text{ emission}}(A)$ is the flicker emission level of the installation under consideration, at A,
- (j) $P_{lt \text{ emission}}(B)$ is the flicker emission level of the installation under consideration, at B (POC),
- (k) $S_{SC,A}$ is the short-circuit power at A and
- (l) $S_{SC,B}$ is the short-circuit power at B (POC).

- (7) The short circuit power at the POC, SSC, B shall be provided by the NSP.

- (8) All flicker measurements used shall be the Weekly 95th percentile of the voltage flicker Plt and Pst

A13.5.4 Voltage Unbalance Emission Evaluation/Assessment

- (1) The following method is recommended for the assessment of voltage unbalance:
- (i) The assessment for voltage unbalance is taken from the Cigré Technical Report 468: Review of Disturbance Emission Assessment Techniques, section 6.4.3.3 – Emission level Assessment based on Current Measurement.
- (2) However, any of the other methods listed in section 6.4.3 of Cigré Technical Report 468 may also be used by the Generator to assess voltage unbalance.

A13.5.4.1 Total emission level assessment based on measurement before and after connection

- (1) Connection of an installation to the system can lead to an increase or decrease (see Figure A13.5.4-1b) of the resultant unbalance level (i.e. $U_{2,i}$). This can be investigated by measuring the unbalance levels at the POE before (i.e. the unbalance level = $U_{2,oc}$ and labelled as $U_{2,pre-connection}$) and after (i.e. the unbalance level = U_2 and labelled as $U_{2,post-connection}$) the connection. The increase or decrease of the resultant unbalance level is determined by the phase angle of the emission $U_{2,i}$ caused by the connection of the load.

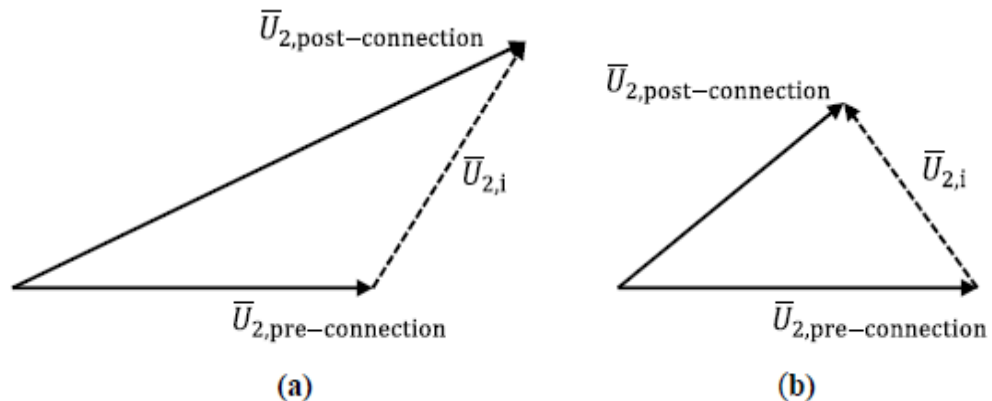


Figure A13.5.4-1: Comparison of the voltage unbalance level before and after the connection of an installation

- (2) This is the first step which leads to one of the following procedures:
- (a) If a decrease of the resultant unbalance level is caused by the connection of the installation, no emission limit is allocated to the installation and no further action is necessary.

(b) If an increase of the resultant unbalance level is caused by the connection of the installation, the fraction of the emission level which the installation is responsible for (i.e. $U_{2,i-load}$) has to be determined for the purpose of comparison against the emission limit allocated based

(3) In the case of measuring instruments providing the phase angle information and when it is known that the vectors $U_{2,pre-connection}$ and $U_{2,post-connection}$ are likely to be in-phase, the total emission $U_{2,i}$ caused by the connection of the load can be determined by:

$$|\bar{U}_{2,i}| = |\bar{U}_{2,post-connection} - \bar{U}_{2,pre-connection}|$$

(4) In the case where the above information is unavailable, as directed in IEC/TR 61000-3-13, the summation law ($\alpha = 1.4$) can be applied to determine the emission $U_{2,i} = |U_{2,i}|$ as shown in:

$$|\bar{U}_{2,i}| = \left(|\bar{U}_{2,post-connection}|^\alpha - |\bar{U}_{2,pre-connection}|^\alpha \right)^{1/\alpha}$$

(5) All unbalance measurements used shall be Weekly 95th percentile of the unbalance measured on a 10-min basis.

Appendix 14 – Calculation of Errors for Validation of EMT Models

A14.1 Introduction

(1) The accuracy of EMT model representation shall be assessed during model validation. The validation shall provide comparison of recorded oscillographies of voltages and currents at POC to simulated values of the same voltages and currents during fault event external to the RPP.

(2) Evaluated currents shall represent sole contribution of the RPP to the fault on the NSP network.

(3) The severity of the event shall be such that the voltage depression at POC shall exceed 20% of U_n .

A14.2 Definitions

Term	Definition
Sample-by-sample error	Percentage error calculated as a ratio of difference the between recorded and simulated values of synchronized samples referred to maximum peak value of recorded voltage or current in any phase.
Average error	Average of errors over a certain period.
Average absolute error	Average of absolute values of errors over a certain period.
90% error band	Error band calculated in such a way that 90% of sample-by-sample errors are within this band. 10% of samples may have errors higher than the band.

A14.3 Validation Criteria

(1) The following requirements shall be met for successful validation of the model:

1. Voltage validation requirements:
 - a. Average absolute error of sample-by-sample evaluation shall not exceed 5% in all 3 phases;
 - b. The 90% error band in sample-by-sample evaluation shall not exceed $\pm 10\%$ in all 3 phases. Where deviations occur, these shall be explained based on sound engineering principles.
2. Current validation requirements:
 - a. Average absolute error of sample-by-sample evaluation shall not exceed 15% in all 3 phases;
 - b. The 90% error band in sample-by-sample evaluation must not exceed $\pm 20\%$ in all 3 phases. Where deviations occur, these Transgressions shall be explained based on sound engineering principles;
 - c. Average error of 1 cycle RMS values shall not exceed 15% in all 3 phases;
 - d. Average error of Symmetrical Components values shall not exceed 20% for all 3 Symmetrical Components (positive, negative and zero sequence). Calculations shall be done using 1 cycle Fourier Transform.

- (2) Calculations of the above indices shall be done over the fault duration and include one 50Hz cycle after the fault was cleared.
- (3) For RMS and Sequence Components calculations the front of the single cycle window shall start one cycle after fault inception and end 2 cycles after the fault was cleared.
- (4) Comparisons for voltages and currents as well as calculated errors shall be presented graphically, on time axis, to reflect pre-fault conditions, fault duration and post-fault conditions.
- (5) Calculation spreadsheets shall be submitted in support of verification of results.

A14.3 Examples of Graphical Presentation

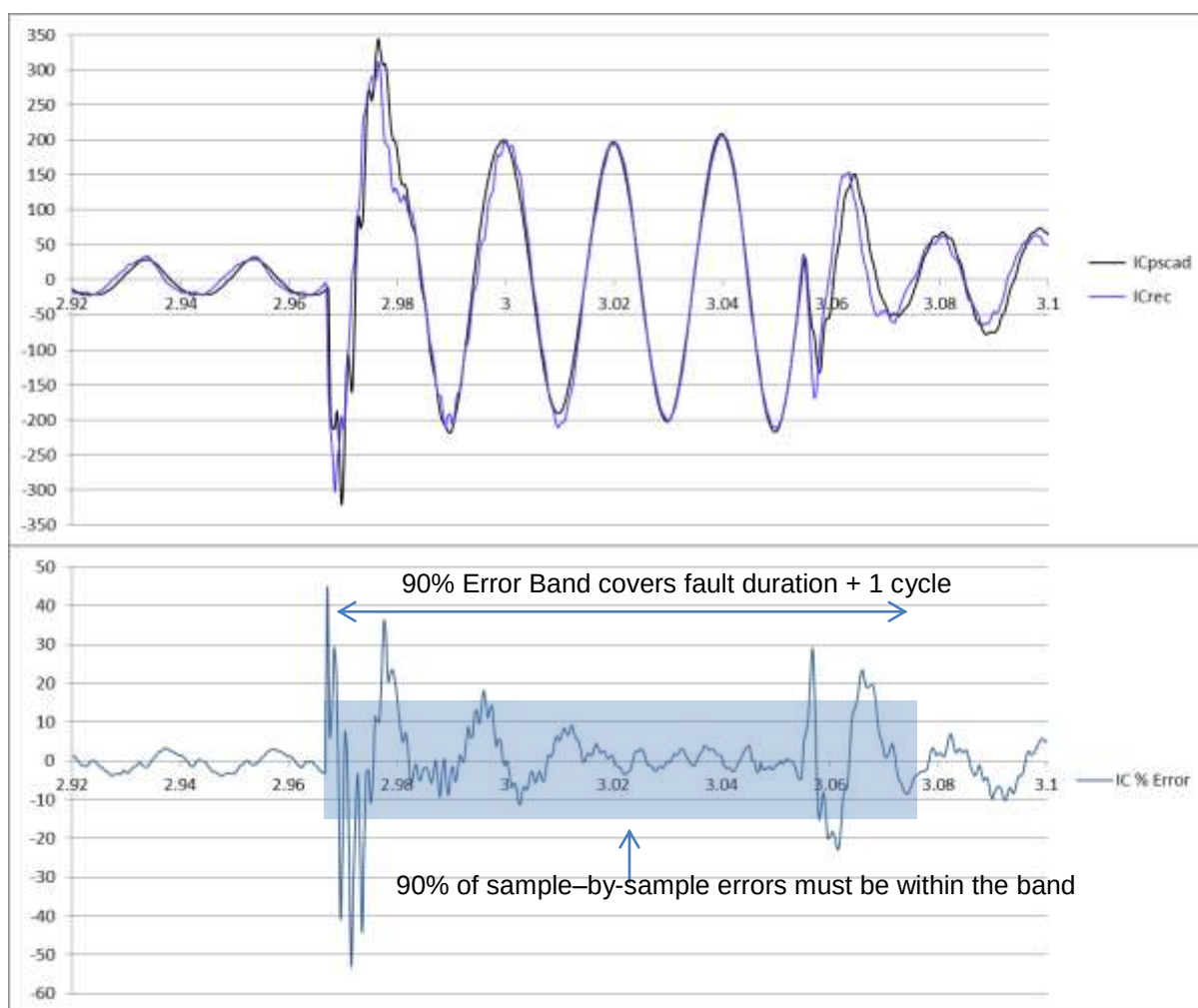


Figure A14.3.1 An example of superimposed simulated and recorded currents and sample-by-sample errors with error band over fault duration plus 1 cycle that ensures that 90% of samples are within this band.

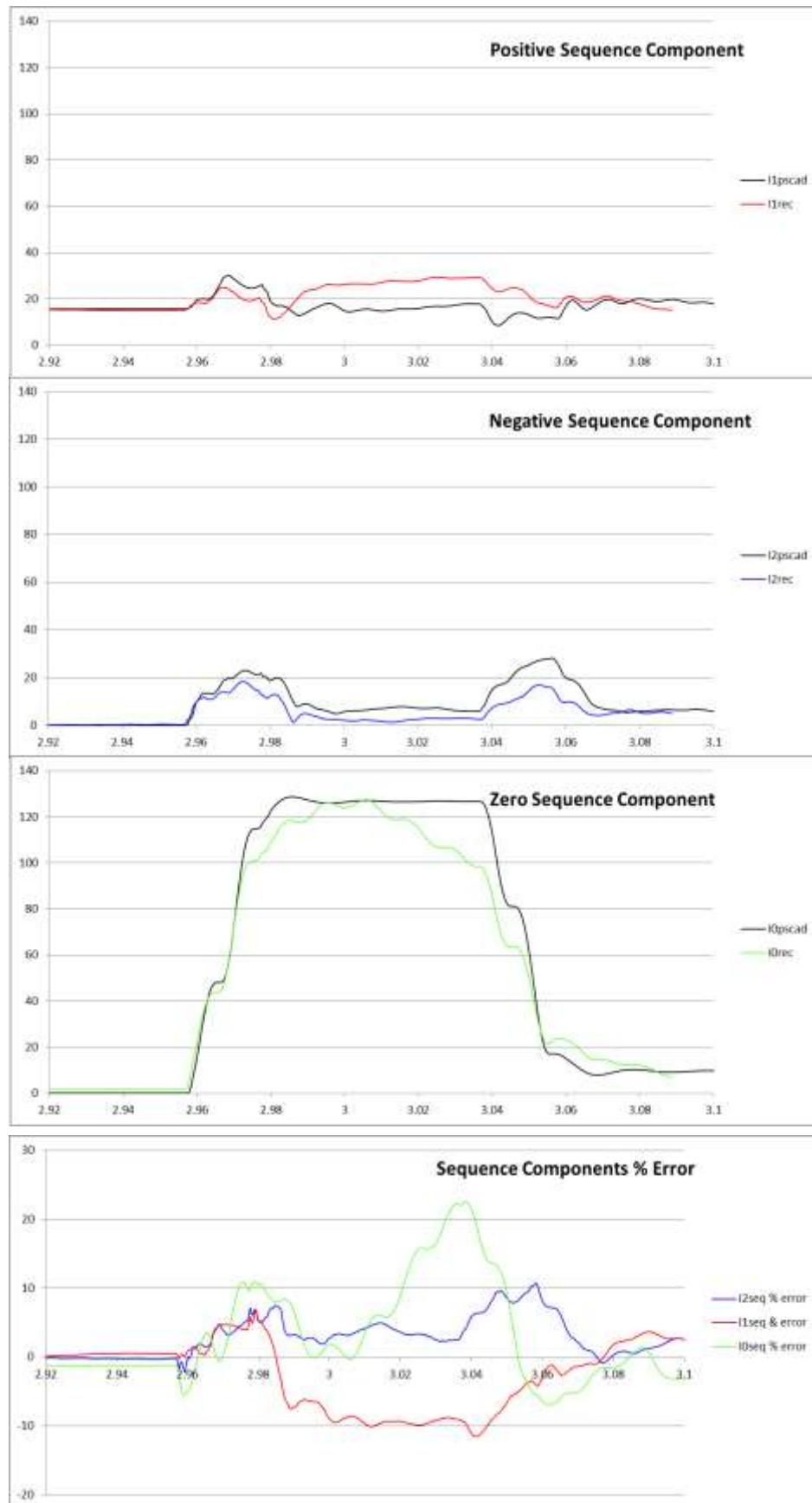


Figure A14.3.2 An example of Symmetrical Components comparison and errors calculated with single cycle moving window.